

Project Data Page:





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DESIGN CALCULATION

In Accordance with ASME Section VIII Division 1

ASME Code Version : 2025

Analysis Performed by : SL User

Job File : ...\\Syngas Cooler-AK.pvdb

Date of Analysis : Sep 3, 2026 3:29pm

PV Elite 28, January 2026

Vessel Design Summary:

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 780.000 mm. ID
 Vessel Design Length, Tangent to Tangent 5600.97 mm.
 Specified Datum Line Distance 0.00 mm.
 Shell Side Design Temperature 93 °C
 Channel Side Design Temperature 93 °C
 Shell Side Design Pressure 191.000 bars
 Channel Side Design Pressure 147.000 bars
 Shop Shell Side Test Pressure 250.008 bars
 Field Shell Side Test Pressure 261.149 bars
 Shop Channel Side Test Pressure 191.275 bars
 Field Channel Side Test Pressure 199.570 bars
 Wind Design Code ASCE/SEI 7-22
 Earthquake Design Code ASCE/SEI 7-22

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #	Material Treatment Options
Shell	SA-516 60	...	...	K02100	
Head	SA-516 60	...	...	K02100	
Flange	SA-266 2	...	...	K03506	
Nozzle	SA-266 2	...	...	K03506	
Nozzle Flg	SA-105	...	...	K03504	
Tubes	SA-179	...	...	K01200	
Tubesheet	SA-266 2	...	...	K03506	
Flg Bolting	SA-193 B7	...	<= 64	G41400	
Hrz Bolting	SA-193 B7	...	<= 64	G41400	
Saddles	SA-516 60	...	...	K02100	

+FG - Fine Grained, +R - rolled, +T - Tempered, +N - Normalized
 +Q - quenched, +CBV - Cb & V added, +IT - Impact tested

Element Pressures and MAWP (bars & mm.):

Element Description or Type	Design Pressure + Stat. head	Ext. Press.	Element M.A.W.P	Total Corrosion Allowance	Str. Flg. Gov.	In Creep Range
Blind Flange	147.148	0.00	151.000	3.0000	N/A	No
Ch Flg	147.076	0.00	147.000	3.0000	N/A	No
Ch Shell	147.077	0.00	147.000	3.0000	N/A	No
Main Shell	191.004	0.00	192.000	3.0000	N/A	No
Dish End	191.004	1.03	194.000	3.0000	No	No

Liquid Level: 1510.00 mm. Dens.: 0.000 kg./cm³ Sp. Gr.: 0.050

Element Types and Properties:

Element Type	"To" Elev mm.	Element Length mm.	Nominal Thickness mm.	Finished Thickness mm.	Reqd Thk Internal mm.	Reqd Thk External mm.	Long Eff	Circ Eff
Body Flg	263.2	260.0	268.0	260.0	256.4	...	1.00	1.00
Body Flg	620.2	357.0	253.0	245.0	238.0	...	1.00	1.00
Cylinder	1520.2	900.0	56.0	56.0	55.9	...	1.00	1.00
Cylinder	5550.2	3850.0	74.0	74.0	73.5	...	1.00	1.00
Ellipse	5601.0	50.8	80.0	68.0	67.0	...	1.00	1.00



External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	260	213.503	...	No Calc
20	30	245	238.049	...	No Calc
30	Tsht	56	No Calc	...	97.2
Tsht	40	56	No Calc	...	No Calc
40	50	74	No Calc	...	114
50	60	68	5.41026	1.03419	90.1

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm^4	Ring Inertia Available cm^4
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	1256.97	No Calc	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	50	3965.83	No Calc	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc

Local Stress Analysis Results:

Description	Analysis Type	Max Stress Ratio	High Stress Location	Pass Fail
T1-Tube Inlet	WRC-297	0.983	Shell	Passed
T2-Tube outlet	WRC-297	0.983	Shell	Passed
S1-Shell Inlet	WRC-297	0.983	Shell	Passed
S2-Shell Outle	WRC-297	0.983	Shell	Passed

Foundation/Support Summary

Saddle Parameters:

Saddle Width	200.000 mm.
Saddle Bearing Angle	120.000 deg.
Centerline Dimension	800.000 mm.
Wear Pad Width	480.000 mm.
Wear Pad Thickness	10.000 mm.
Wear Pad Bearing Angle	132.000 deg.
Distance from Saddle to Tangent	800.000 mm.
Baseplate Length	850.000 mm.
Baseplate Thickness	24.000 mm.
Baseplate Width	254.000 mm.
Number of Ribs (including outside ribs)	3
Rib Thickness	10.000 mm.
Web Thickness	10.000 mm.
Height of Center Web	306.000 mm.
Number of Bolts in Baseplate	4

Baseplate Sketch

Maximum Tensile Bolt Load 25. Kg

Summary of Maximum Saddle Loads, Operating Case, Un-Factored:

Maximum Vertical Saddle Load	29214.31 Kg
Maximum Transverse Saddle Shear Load	1078.39 Kg
Maximum Longitudinal Saddle Shear Load	2156.78 Kg

Summary of Maximum Saddle Loads, Pressure Test Case:

Maximum Vertical Saddle Load	2551.67	Kgf
Maximum Transverse Saddle Shear Load	99.76	Kgf
Maximum Longitudinal Saddle Shear Load	57.22	Kgf

Weights:

Fabricated - Bare w/o removable internals	21606.4	kg.
Shop Test - Fabricated + water (full)	23828.7	kg.
Shipping - Fab. + rem. intls.+ shipping app.	21733.1	kg.
Erected - Fab. + rem. intls.+ insul. (etc)	21733.1	kg.
Empty - Fab. + intls. + details + wghts.	21733.1	kg.
Operating - Empty + operating liquid (no ca)	22694.3	kg.
Field Test - Empty weight + test medium (full)	23955.4	kg.

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Nozzle Calculation Summary (bars & mm.):

Description	MAWP	Ext	MAPNC	UG-45	[tr]	Weld Path	Areas or Stresses
T1-Tube Inlet	147	...	...	OK	15.63	OK	Passed
T2-Tube outlet	147	...	...	OK	15.63	OK	Passed
S1-Shell Inlet	192	...	...	OK	22.87	OK	Passed
S2-Shell Outlet	192	...	...	OK	22.87	OK	Passed

Nozzle MAWP Summary:

Minimum MAWP Nozzles : 192 Nozzle : S1-Shell Inlet [Shellside]
 Minimum MAWP Nozzles : 147 Nozzle : T1-Tube Inlet [Tubeside]

Note: MAWPs (Internal Case) shown above are at the High Point.

Multiple output lines for the same nozzle indicates required Code calculations in both the longitudinal and circumferential planes of reinforcement where applicable.

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm.	Layout Angle deg	Dia. Limit mm.
30	T1-Tube Inlet	1070.175	270.000	443.800
30	T2-Tube outlet	1070.175	90.000	443.800
40	S1-Shell Inlet	2300.175	90.000	526.401
40	S2-Shell Outlet	4900.175	270.000	526.401

The nozzle spacing is computed by the following:

$$= \sqrt{ll^2 + lc^2} \text{ where}$$

ll - Arc length along the inside vessel surface in the long. direction.
 lc - Arc length along the inside vessel surface in the circ. direction

If any interferences/violations are found, they will be noted below.

No interference violations have been detected!

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Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia mm	Wall Thk mm	Reinforcing Pad Diameter	Pad Thk mm	Cut Length mm	Flg Class
T1-Tube Inlet	250 mm	160	RTJWn	273.0	28.575	...	...	309	1500
T2-Tube outlet	250 mm	160	RTJWn	273.0	28.575	...	...	309	1500
S1-Shell Inlet	300 mm	160	RTJWn	323.9	33.325	...	...	355	1500
S2-Shell Outlet	300 mm	160	RTJWn	323.9	33.325	...	...	355	1500

General Notes for the above table:

The cut length is the outside projection + inside projection + drop + in-plane shell thickness. This value does not include weld gaps, nor does it account for shrinkage.

In the case of oblique nozzles, the outside diameter must be increased. The re-pad width around the nozzle is calculated as follows:

Width of Pad = (Pad Outside Dia. (per above) - Nozzle Outside Dia.)/2

For hub nozzles, the thickness and diameter shown are those of the smaller and thinner section.

Nozzle Material and Weld Fillet Leg Size Details (mm.):

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
T1-Tube Inl	SA-266 2	56.000	10.000	...	...	...
T2-Tube out	SA-266 2	56.000	10.000	...	...	...
S1-Shell In	SA-266 2	74.000	10.000	...	...	...
S2-Shell Ou	SA-266 2	74.000	10.000	...	...	...

Note: The Outside projections below do not include the flange thickness.

Nozzle Miscellaneous Data:

Description	Elev/Distance From Datum mm.	Layout Angle deg	Proj Outside mm.	Proj Inside mm.	Installed in Component
T1-Tube Inlet	1070.175	270.0	200.00	0.00	Ch Shell
T2-Tube outlet	1070.175	90.0	200.00	0.00	Ch Shell
S1-Shell Inlet	2300.175	90.0	200.00	0.00	Main Shell
S2-Shell Outlet	4900.175	270.0	200.00	0.00	Main Shell

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Bill of Materials:

QTY	DESCRIPTION	MATERIAL
1	BLIND FLANGE: 268.0mm. THK X 1510.0mm. OD	SA-266 2
1	WELD NECK FLANGE: 253.0mm. THK X 1510.0mm. OD	SA-266 2
1	CYLINDER: 56.0mm. THK X 780.0mm. ID X 900.0mm.	SA-516 60
1	CYLINDER: 74.0mm. THK X 780.0mm. ID X 3850.0mm.	SA-516 60
1	ELLIPTICAL HEAD: 2.0 X 1, 80.0mm. THK X 780.0mm. ID X 50.8mm.	SA-516 60
10	BAFFLES: 720.0mm DIA X 12.0mm THK	SA-516 70
6	TIE RODS: 19mm DIA X 3600mm	SA-193 B7M
1	TUBESHEET: 928.0mm X 180.0mm THK	SA-266 2
520	TUBES: 3658.0mm X 19.0mm DIA X 3.0mm THK	SA-179
2	SADDLE: 200.0mm X 120 DEG	SA-516 60
1	INSULATION: 3850mm X 50mm THK	Hot Insulation
1	INSULATION: 245mm X 50mm THK	Hot Insulation
1	GASKET: 838.0mm. OD X 812.0mm. ID	...
40	BODY FLANGE BOLTS: 48.0mm. DIA	SA-193 B7
80	NUTS FOR BODY FLANGE BOLTS: 48.0mm. DIA	...
1	NAMEPLATE	...



Minimum Design Metal Temperature Results Summary (°C):

Description	Notes	Curve	Basic MDMT	Reduced MDMT	UG-20(f) MDMT	Thickness ratio	Gov Thk mm.	E*	PWHT reqd
Main Shell	[8]	C	4	4		1.000	74.000	1.00	Yes
Dish End	[10]	C	3	2		0.992	68.000	1.00	Yes
Dish End	[7]	C	6	2		0.922	80.000	1.00	Yes
Left Saddle	[24]	C	-48	-48		1.000	10.000	1.00	Yes
Sdl 2 Fr40	[24]	C	-48	-48		1.000	10.000	1.00	Yes
S1-Shell Inlet	[1]	B	25	25		0.992	74.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-32					
S2-Shell Outlet	[1]	B	25	25		0.992	74.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-32					
Tubesheet: SS	[13]	B	9	-2		0.814	74.000	1.00	Yes
Warmest MDMT:			25	25					
Blind Flange	[11]	B	23	20		0.958	65.000	1.00	No
Ch Flg	[11]	B	3	0		0.958	56.000	1.00	Yes
Ch Shell	[8]	C	-2	-2		0.999	56.000	1.00	Yes
T1-Tube Inlet	[1]	B	19	19		0.999	56.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-42					
T2-Tube outlet	[1]	B	19	19		0.998	56.000	1.00	Yes
Nozzle Flg	[5]	A	-18	-42					
Tubesheet: CS	[14]	B	3	-19		0.611	56.000	1.00	No
Bolting	[21]		-48						
Warmest MDMT:			23	20					

Exchanger Side	Computed MDMT	Required MDMT	Pass/Fail
Shell	25.0	-10.0	Fail
Channel/Tube	20.0	0.0	Fail

Notes:

- [!] - This was an impact tested material.
- [1] - Governing Nozzle Weld.
- [4] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-c).
- [5] - ASME Flange MDMT Calcs; Thickness ratio per UCS-66(b)(1)(-b).
- [6] - MDMT Calculations at the Shell/Head Joint.
- [7] - MDMT Calculations for the Straight Flange.
- [8] - Cylinder/Cone/Flange Junction MDMT.
- [9] - Calculations in the Spherical Portion of the Head.
- [10] - Calculations in the Knuckle Portion of the Head.
- [11] - Calculated (Body Flange) Flange MDMT.
- [12] - Calculated Flat Head MDMT per UCS-66.3
- [13] - Tubesheet MDMT, shell side, if applicable
- [14] - Tubesheet MDMT, tube side, if applicable
- [15] - Nozzle Material
- [16] - Shell or Head Material
- [17] - Impact Testing required
- [18] - Impact Testing not required, see UCS-66(b)(3)
- [20] - Cylinder/Cone Junction MDMT based on Longitudinal Stress considerations
- [21] - Body Flange Bolting Material
- [22] - Nozzle Flange Bolting Material
- [23] - Stiffening Ring to Shell Weld
- [24] - Saddle to Shell Weld

UG-84(b)(2) was not considered.
 UCS-66(g) was not considered.
 UCS-66(i) was not considered.

Notes:

Impact test temps were not entered in and not considered in the analysis.
 UCS-66(i) applies to impact tested materials not by specification and
 UCS-66(g) applies to materials impact tested per UG-84.1 General Note (c).



The Basic MDMT includes the (30F) PWHT credit if applicable.

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Class From To : Basic Element Checks.

Class From To: Check of Additional Element Data

There were no geometry errors or warnings.

PV Elite performs all calculations internally in Imperial Units to remain compliant with the ASME Code and any built in assumptions in the ASME Code formulas. The finalized results are reflected to show the set of selected units for this analysis.

Information Regarding Nozzle Loads

Please note that nozzle loadings, if included, are assumed to be local in nature and will not contribute to or create a net section bending moment. Therefore, the addition of nozzle loads will not affect the support load calculation. If you wish to create a load on the support(s) from nozzle loads, you can enable that feature in tools->configuration->Job Specific Settings.



The ASME Code Section VIII Division 1 does not explicitly require stresses on elements to be computed under test conditions. Please see interpretation VIII-1-10-12 for more information. PV Elite also computes weight due to test fluid and uses the weight in various calculations such as saddle and leg loads. PV Elite also computes static pressure due to operating and test liquids. The static pressure is used in conjunction with the design and test pressures to determine a total pressure in the determination of required thickness, MAWP etc. This ensures compliance with UG-22 requirements.

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Units used in this Analysis (bars):

Name	System Unit	Constant	User Unit
Length	Feet	304.8000	mm.
Force	Pounds	0.4536	Kgf
Mass	Pounds	0.4536	kg.
Area	sq. inches	6.4516	cm ²
Moment	ft. lbs.	0.1383	Kg-m.
Stress	lbs./sq.in.	0.0069	N./mm ²
Temperature	Degrees F	0.5556	°C
Pressure	psig	0.0689	bars
Elast. Modulus	lbs./sq.in.	6.8946	KPa.
Pipe Density	lbs./cu.in.	0.0277	kg./cm ³
Ins. Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Fluid Density	lbs./cu.ft.	0.1600E-04	kg./cm ³
Wind Speed	miles/hr	1.6093	Km/hr
Tray Weight	lbs./sq.ft.	0.0005	Kgs/cm ²
Inertia	in.4	41.6231	cm4
G Load	G's	1.0000	g's
Wind Load	lbs./sq.ft.	4.8825	Kgs/m ²
Elevation	Feet	304.8000	mm.
Volume	in.3	16.3900	cm ³
Diameter	inches	25.4000	mm.
Thickness	inches	25.4000	mm.

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	191	bars
Channel Side Design Pressure	147	bars
Shell Side Design Temperature	93.3	°C
Channel Side Design Temperature	93.3	°C
Radiography, Shell Side	RT-1	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	Lethal	
Service Type, Channel Side	Air/Water/Steam	
MDMT (CET), Shell Side	-10.0	°C
MDMT (CET), Tube Side	0.0	°C
User defined MAWP, Shell Side	0	bars
User defined MAWP, Channel Side	0	bars
User defined MAPnc, Shell Side	0	bars
User defined MAPnc, Channel Side	0	bars
User defined Test Pres., Shell Side	0	bars
User defined Test Pres., Channel Side	0	bars

Projection of Nozzle from Vessel Top	0	mm.
Projection of Nozzle from Vessel Bottom	0	mm.
Type of Construction	Welded	
Use Higher Longitudinal Stresses (Flag)	Y	
Select t for Internal Pressure (Flag)	N	
Select t for External Pressure (Flag)	N	
Select t for Axial Stress (Flag)	N	
Select Location for Stiff. Rings (Flag)	N	
Consider Vortex Shedding	N	

Shop Pressure Test:

Type of Pressure Test	UG-99(b)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	No

Field Pressure Test:

Type of Pressure Test	UG-99(c)
Pressure Test Position	Horizontal
Test Performed in Corroded Condition	Yes



Load Case 1	NP+EW+WI+FW+BW
Load Case 2	NP+EW+EE+FS+BS
Load Case 3	NP+OW+WI+FW+BW
Load Case 4	NP+OW+EQ+FS+BS
Load Case 5	NP+HW+HI
Load Case 6	NP+HW+HE
Load Case 7	IP+OW+WI+FW+BW
Load Case 8	IP+OW+EQ+FS+BS
Load Case 9	EP+OW+WI+FW+BW
Load Case 10	EP+OW+EQ+FS+BS
Load Case 11	HP+HW+HI
Load Case 12	HP+HW+HE
Load Case 13	IP+WE+EW
Load Case 14	IP+WF+CW
Load Case 15	IP+VO+OW
Load Case 16	IP+VE+EW
Load Case 17	NP+VO+OW
Load Case 18	FS+BS+IP+OW
Load Case 19	FS+BS+EP+OW
Load Case 20	BL+IP+OW

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 Km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	Moderately Smooth
Base Elevation	0 mm.
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment	H or Hh 0 mm.
Distance Upwind of Crest	Lh 0 mm.
Distance from Crest to the Vessel	x 0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Seismic Design Code	ASCE/SEI 7-22
Importance Factor	1.500
Force Modification Factor R	3.000
Site Class	D
Component Elevation Ratio	z/h 0.000
Amplification Factor	Ap 0.000
Force Factor	0.000
Consider Vertical Acceleration	Yes
Minimum Acceleration Multiplier	0.000
User Value of Sds (used if > 0)	0.000
User Value of Sd1 (used if > 0)	0.000
Component resonance duct. fact.	[CAR] 0.000
Component strength factor	[Rpo] 0.000
Overstrength factor	[Omega0] 0.000
Response Acceleration Factor	[SMS] 0.326
Response Acceleration Factor	[SM1] 0.139
Building Importance Factor	[Ie] 1.000
Site Latitude	0.000 deg
Site Longitude	0.000 deg

Design Pressure + Static Head	Y
Consider MAP New and Cold in Noz. Design	N
Consider External Loads for Nozzle Des.	Y
Use ASME VIII-1 Appendix 1-9	N

Perform Blast Load Analysis	No
Material Database Year	Current w/Addenda or Code Year

Configuration Directives:

g0 is taken as min(g0, attached shell thk)	Yes
--	-----

Do not use Nozzle MDMT Interpretation VIII-1 01-37	No
Use Table G instead of exact equation for "A"	Yes
Shell Head Joints are Tapered	Yes
Compute "K" in corroded condition	Yes
Use Code Case 2286	No
Use the MAWP to compute the MDMT	Yes
For thickness ratios <= 0.35, MDMT will be -155F (-104C)	Yes
For PWHT & P1 Materials the MDMT can be < -55F (-48C)	No
Using Metric Material Databases, ASME II D	Yes
Calculate B31.3 type stress for Nozzles with Loads	Yes
Reduce the MDMT due to lower membrane stress	Yes
Consider Longitudinal Stress in MDMT Calculations	Yes

Complete Listing of Vessel Elements and Details:

Element#	1/5
Element From Node	10
Element To Node	20
Element Type	Flange
Description	Blind Flange
Distance "FROM" to "TO"	260 mm.
Flange Outside Diameter	1510 mm.
Element Thickness	260 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	268 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	147 bars
Design Temperature Internal Pressure	93 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Allowable Stress, Ambient	138 N./mm ²
Allowable Stress, Operating	138 N./mm ²
Allowable Stress, Hydrotest	225 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm.
Yield Stress, Operating	228.14 N./mm ²
UCS-66 Chart Curve Designation	B
External Pressure Chart Name	CS-2
UNS Number	K03506
Product Form	Forgings
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	10
Detail Type	Liquid
Detail ID	Liquid: 10
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	1510 mm.
Liquid Density	0.0009984 kg./cm ³

Element#	2/5
Element From Node	20
Element To Node	30
Element Type	Flange
Description	Ch Flg
Distance "FROM" to "TO"	357 mm.
Flange Inside Diameter	780 mm.
Element Thickness	245 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	253 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	147 bars
Design Temperature Internal Pressure	93 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C



Effective Diameter Multiplier	1.2
Material Name	SA-266 2
Perform Flange Stress Calculation (Y/N)	Y
Weld is pre-Heated	No
Element From Node	20
Detail Type	Liquid
Detail ID	Liquid: 20
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	780 mm.
Liquid Density	0.0009984 kg./cm ³

Element#	3/5
Element From Node	30
Element To Node	40
Element Type	Cylinder
Description	Ch Shell
Distance "FROM" to "TO"	900 mm.
Inside Diameter	780 mm.
Element Thickness	56 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	56 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	147 bars
Design Temperature Internal Pressure	93 °C
Design External Pressure	0 bars
Design Temperature External Pressure	85 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 60
Allowable Stress, Ambient	118 N./mm ²
Allowable Stress, Operating	118 N./mm ²
Allowable Stress, Hydrotest	198 N./mm ²
Material Density	0.00775 kg./cm ³
P Number Thickness	32.004 mm.
Yield Stress, Operating	202.33 N./mm ²
UCS-66 Chart Curve Designation	C
External Pressure Chart Name	CS-2
UNS Number	K02100
Product Form	Plate
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	30
Detail Type	Liquid
Detail ID	Liquid: 30
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Liquid	780 mm.
Liquid Density	0.0009984 kg./cm ³

Element From Node	30
Detail Type	Nozzle
Detail ID	T1-Tube Inlet
Dist. from "FROM" Node / Offset dist	450 mm.
Nozzle Diameter	250 mm
Nozzle Schedule	160
Nozzle Class	1500
Layout Angle	270.0
Blind Flange (Y/N)	N
Weight of Nozzle (Used if > 0)	330.27 Kg
Grade of Attached Flange	GR 1.1
Nozzle Matl	SA-266 2

Element From Node	30
Detail Type	Nozzle
Detail ID	T2-Tube outlet
Dist. from "FROM" Node / Offset dist	450 mm.
Nozzle Diameter	250 mm



Nozzle Schedule	160	
Nozzle Class	1500	
Layout Angle	90.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	330.27	Kgf
Grade of Attached Flange	GR 1.1	
Nozzle Matl	SA-266 2	

Element#	4/5
Element From Node	40
Element To Node	50
Element Type	Cylinder
Description	Main Shell
Distance "FROM" to "TO"	3850 mm.
Inside Diameter	780 mm.
Element Thickness	74 mm.
Internal Corrosion Allowance	3 mm.
Nominal Thickness	74 mm.
External Corrosion Allowance	0 mm.
Design Internal Pressure	191 bars
Design Temperature Internal Pressure	93 °C
Design External Pressure	0 bars
Design Temperature External Pressure	200 °C
Effective Diameter Multiplier	1.2
Material Name	SA-516 60
Efficiency, Longitudinal Seam	1.0
Efficiency, Circumferential Seam	1.0
Weld is pre-Heated	No

Element From Node	40
Detail Type	Saddle
Detail ID	Left Saddle
Dist. from "FROM" Node / Offset dist	800 mm.
Width of Saddle	200 mm.
Height of Saddle at Bottom	800 mm.
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm.
Width of Wear Plate	480 mm.
Thickness of Wear Plate	10 mm.
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	0.60000002
Moment Factor	3.0
Dimension E at base (optional)	0 mm.
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm.

Element From Node	40
Detail Type	Saddle
Detail ID	Sdl 2 Fr40
Dist. from "FROM" Node / Offset dist	2650 mm.
Width of Saddle	200 mm.
Height of Saddle at Bottom	800 mm.
Saddle Contact Angle	120.0
Height of Composite Ring Stiffener	0 mm.
Width of Wear Plate	480 mm.
Thickness of Wear Plate	10 mm.
Contact Angle, Wear Plate (degrees)	132.0
Friction coefficient	5.9999999e-002
Moment Factor	3.0
Dimension E at base (optional)	0 mm.
Circumferential Eff. over Saddle	1.0
Circumferential Eff. at Midspan	1.0
Tangent to Tangent dist. (optional)	0 mm.

Element From Node	40
Detail Type	Liquid
Detail ID	Liquid: 40



Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Liquid 780 mm.
 Liquid Density 0.49920E-04 kg./cm³

Element From Node 40
 Detail Type Insulation
 Detail ID Hot Insulation
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Insulation 3850 mm.
 Thickness of Insulation 50 mm.
 Density 0.0001918 kg./cm³

Element From Node 40
 Detail Type Nozzle
 Detail ID S1-Shell Inlet
 Dist. from "FROM" Node / Offset dist 600 mm.
 Nozzle Diameter 300 mm
 Nozzle Schedule 160
 Nozzle Class 1500
 Layout Angle 90.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 530.16 Kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-266 2

Element From Node 40
 Detail Type Nozzle
 Detail ID S2-Shell Outlet
 Dist. from "FROM" Node / Offset dist 3200 mm.
 Nozzle Diameter 300 mm
 Nozzle Schedule 160
 Nozzle Class 1500
 Layout Angle 270.0
 Blind Flange (Y/N) N
 Weight of Nozzle (Used if > 0) 530.16 Kgf
 Grade of Attached Flange GR 1.1
 Nozzle Matl SA-266 2

Element# 5/5
 Element From Node 50
 Element To Node 60
 Element Type Elliptical
 Description Dish End
 Distance "FROM" to "TO" 50.8 mm.
 Inside Diameter 780 mm.
 Element Thickness 68 mm.
 Internal Corrosion Allowance 3 mm.
 Nominal Thickness 80 mm.
 External Corrosion Allowance 0 mm.
 Design Internal Pressure 191 bars
 Design Temperature Internal Pressure 93 °C
 Design External Pressure 1.0342 bars
 Design Temperature External Pressure 200 °C
 Effective Diameter Multiplier 1.2
 Material Name SA-516 60
 Efficiency, Longitudinal Seam 1.0
 Efficiency, Circumferential Seam 1.0
 Elliptical Head Factor 2.0
 Weld is pre-Heated No

Element From Node 50
 Detail Type Liquid
 Detail ID Liquid: 50
 Dist. from "FROM" Node / Offset dist 0 mm.
 Height/Length of Liquid 780 mm.
 Liquid Density 0.49920E-04 kg./cm³

Element From Node 50

PV Elite 28 Licensee: SL User
FileName : Syngas Cooler-AK
Input Echo:

Step: 1 3:29pm Sep 3,2026



HEXAGON

Detail Type	Insulation
Detail ID	Hot Insulation
Dist. from "FROM" Node / Offset dist	0 mm.
Height/Length of Insulation	245.8 mm.
Thickness of Insulation	50 mm.
Density	0.0001918 kg./cm ³

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XY Coordinate Calculations:

From	To	X (Horiz.) mm.	Y (Vert.) mm.	DX (Horiz.) mm.	DY (Vert.) mm.
Blind Flange		263.175	...	260	...
Ch Flg		620.175	...	357	...
Ch Shell		1520.17	...	900	...
Main Shell		5550.17	...	3850	...
Dish End		5600.97	...	50.8	...

The calculated dimensions are based on the given element lengths. Due to variability in manufacturing (weld gaps etc.), the Tangent to Tangent length may not be exact. The length includes gasket thicknesses and raised face heights.

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Flange Input Data Values

Description: New Flange :

Blind Flange

Flange Type		Blind	
Design Pressure	P	147.15	bars
Design Temperature		93	°C
Internal Corrosion Allowance	ci	3.0000	mm.
External Corrosion Allowance	ce	0.0000	mm.
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Outside Diameter	A	1510.000	mm.
Flange Thickness	t	260.0000	mm.
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Perimeter along the Center of the Bolts	L	4398.230	mm.
Diameter of Bolt Circle	C	1400.000	mm.
Nominal Bolt Diameter	a	48.0000	mm.
Type of Threads		TEMA Metric	
Number of Bolts		40	
Flange Face Outside Diameter	Fod	849.000	mm.
Flange Face Inside Diameter	Fid	780.000	mm.
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	838.000	mm.
Gasket Inside Diameter	Gi	812.000	mm.
Gasket Factor	m	2.0000	
Gasket Design Seating Stress	y	17.24	N./mm ²
Gasket Thickness	tg	3.1750	mm.
Raised face depth	Fd	0.0000	mm.

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Gasket Contact Width,	$N = (Go - Gi)/2$	13.000 mm.
Basic Gasket Width,	$bo = N/2$	6.500 mm.
Effective Gasket Width,	$b = Cb \cdot \sqrt{bo}$	6.425 mm.
Gasket Reaction Diameter,	$G = Go - 2 \cdot b$	825.151 mm.

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:
 $= \pi/4 \cdot G^2 \cdot P$
 $= \pi/4 \cdot 825^2 \cdot 147$
 $= 802430.812 \text{ Kg}$

Contact Load on Gasket Surfaces [Hp or Wo,2]:
 $= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$
 $= 2 \cdot 6.42 \cdot \pi \cdot 825 \cdot 2 \cdot 147$
 $= 99962.672 \text{ Kg}$

Design Bolt Load for the Operating Condition [Wo]:
 $= \max(Wo,1 + Wo,2 + H'p, 0)$
 $= \max(802431 + 99963 + 0, 0)$
 $= 902393.438 \text{ Kg}$

Design Bolt Load for the Gasket Seating Condition [Wg]:



$$= y \cdot b \cdot \pi \cdot G + y_{Part} \cdot b_{Part} \cdot l_p$$

$$= 17.2 \cdot 6.42 \cdot \pi \cdot 825 + 0 \cdot 0 \cdot 0$$

$$= 29273.355 \text{ Kgf}$$

Required Bolt Area [Am]:

$$= \max((W_o + F_a + 4M_e/G) / S_b, W_g / S_a)$$

$$= \max((902393 + 0 + 4 \cdot 0 / 825) / 172, 29273 / 172)$$

$$= 514.522 \text{ cm}^2$$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	514.522	537.184	
Radial Distance between Bolts and Edge:	50.000	55.000	
Circ. Spacing between the Bolts:	109.300	109.843	...

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:

$$= A_b \cdot S_a / (y \cdot \pi (G_o + G_i))$$

$$= 537 \cdot 172 / (17.2 \cdot \pi (838 + 812))$$

$$= 103.404 \text{ mm. [Note: Exceeds actual gasket width, 13.000]}$$

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172 \cdot 537$$

$$= 942138.38 \text{ Kgf}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o$$

$$= 902393.44 \text{ Kgf}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1400 - 825) / 2$$

$$= 287.4246 \text{ mm.}$$

Flange Stresses

Tangential Flange Stress, Flat Head (4.6.2.3), Operating [STo]:

$$= 1.9 \cdot W_{m1} \cdot h_G \cdot B_{sc} / (t^2 \cdot G) + C \cdot Z \cdot P_{eq} \cdot G^2 / t^2$$

$$= 1.9 \cdot 902393 \cdot 287 \cdot 1 / (257^2 \cdot 825) + 0.3 \cdot 1 \cdot 147 \cdot 825^2 / 257^2$$

$$= 134.18 \text{ N./mm}^2$$

Tangential Flange Stress, Flat Head (4.6.2.4), Seating [STa]:

$$= 1.9 \cdot W \cdot h_G \cdot B_{sc} / (t^2 \cdot G)$$

$$= 1.9 \cdot 942138 \cdot 287 \cdot 1 / (257^2 \cdot 825)$$

$$= 92.58 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= W_{m1} / A_b$$

$$= 902393 / 537$$

$$= 164.74 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (W_{m2} / A_b)$$

$$= (29273 / 537)$$

$$= 5.34 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual	Gasket Seating Allowed
Tangential Flange	134.18	138.00	92.58	138.00
Bolting	164.74	172.00	5.34	172.00

Reqd. Blind Flange Thickness at Center

256.425 mm.



Reqd. Blind Flange Thickness at Gasket 213.503 mm.

Estimated M.A.W.P. (Operating) 151 bars
Estimated Finished Weight of Flange at given Thk. 3608.7 kg.
Estimated Unfinished Weight of Forging at given Thk 3608.7 kg.

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.96, Temperature Reduction per Fig. UCS 66.1 = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 23 °C
Min Metal Temp. at Required thickness (UCS 66.1) 20 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Operating		Gasket Seating	
	Actual	Allowed	Actual	Allowed
Tangential Flange	131.11	138.00	90.46	138.00
Bolting	164.74	172.00	5.34	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 253.441 mm.

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Flange Input Data Values

Description: FLANGE :

Ch Flg

Flange Type	Integral Weld Neck		
Design Pressure	P	147.08	bars
Design Temperature		93	°C
Internal Corrosion Allowance	ci	3.0000	mm.
External Corrosion Allowance	ce	0.0000	mm.
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	780.000	mm.
Flange Outside Diameter	A	1510.000	mm.
Flange Thickness	t	245.0000	mm.
Thickness of Hub at Small End	go	56.0000	mm.
Thickness of Hub at Large End	g1	84.0000	mm.
Length of Hub	h	112.0000	mm.
Flange Material		SA-266 2	
Flange Material UNS number		K03506	
Flange Allowable Stress At Temperature	Sfo	138.00	N./mm ²
Flange Allowable Stress At Ambient	Sfa	138.00	N./mm ²
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	172.00	N./mm ²
Bolt Allowable Stress At Ambient	Sa	172.00	N./mm ²
Diameter of Bolt Circle	C	1400.000	mm.
Nominal Bolt Diameter	a	48.0000	mm.
Type of Threads		TEMA Metric	
Number of Bolts		40	
Flange Face Outside Diameter	Fod	849.000	mm.
Flange Face Inside Diameter	Fid	780.000	mm.
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	838.000	mm.
Gasket Inside Diameter	Gi	812.000	mm.
Gasket Factor	m	2.0000	
Gasket Design Seating Stress	y	17.24	N./mm ²
Gasket Thickness	tg	3.1750	mm.
Raised face depth	Fd	0.0000	mm.

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Hub Small End Required Thickness due to Internal Pressure:

$$= (P \cdot (D/2 + Ca)) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (147 \cdot (780/2 + 3)) / (118 \cdot 1 - 0.6 \cdot 147) + Ca$$

$$= 55.9476 \text{ mm.}$$

Small End Hub MAWP:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118 \cdot 1 \cdot 53) / (393 + 0.6 \cdot 53)$$

$$= 147.211 \text{ bars}$$

Hub Small End Hub MAPnc:

$$= (Sa \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118 \cdot 1 \cdot 56) / (390 + 0.6 \cdot 56)$$

$$= 155.984 \text{ bars}$$

Corroded Flange Thickness, tc = T-ci	242.000	mm.
Corroded Flange ID, Bcor = B+2*ci	786.000	mm.
Corroded Large Hub, g1Cor = g1-ci	81.000	mm.
Corroded Small Hub, g0Cor = go-ci	53.000	mm.
Code R Dimension, R = ((C-Bcor)/2)-g1cor	226.000	mm.
Gasket Contact Width, N = (Go - Gi)/2	13.000	mm.

Basic Gasket Width, $bo = N/2$ 6.500 mm.
 Effective Gasket Width, $b = Cb \cdot \sqrt{bo}$ 6.425 mm.
 Gasket Reaction Diameter, $G = Go - 2 \cdot b$ 825.151 mm.

Basic Flange and Bolt Loads:

Hydrostatic End Load due to Pressure [H or Wo,1]:
 $= \pi/4 \cdot G^2 \cdot P$
 $= \pi/4 \cdot 825^2 \cdot 147$
 $= 802040.562 \text{ Kgf}$

Contact Load on Gasket Surfaces [Hp or Wo,2]:
 $= 2 \cdot b \cdot \pi \cdot G \cdot m \cdot P$
 $= 2 \cdot 6.42 \cdot \pi \cdot 825 \cdot 2 \cdot 147$
 $= 99914.062 \text{ Kgf}$

Hydrostatic End Load at Flange ID [Hd]:
 $= \pi/4 \cdot Bcor^2 \cdot P$
 $= \pi/4 \cdot 786^2 \cdot 147$
 $= 727737.500 \text{ Kgf}$

Pressure Force on Flange Face [Ht]:
 $= H - Hd$
 $= 802041 - 727738$
 $= 74303.078 \text{ Kgf}$

Design Bolt Load for the Operating Condition [Wo]:
 $= \max(Wo,1 + Wo,2 + H'p, 0)$
 $= \max(802041 + 99914 + 0, 0)$
 $= 901954.688 \text{ Kgf}$

Design Bolt Load for the Gasket Seating Condition [Wg]:
 $= y \cdot b \cdot \pi \cdot G + yPart \cdot bPart \cdot lp$
 $= 17.2 \cdot 6.42 \cdot \pi \cdot 825 + 0 \cdot 0$
 $= 29273.355 \text{ Kgf}$

Required Bolt Area [Am]:
 $= \max((Wo + Fa + 4Me/G)/Sb, Wg/Sa)$
 $= \max((901955 + 0 + 4 \cdot 0/825)/172, 29273/172)$
 $= 514.272 \text{ cm}^2$

ASME Maximum Circumferential Spacing between Bolts, Table 4.16.11 [Bsmax]:
 $= 2a + 6t/(m + 0.5)$
 $= 2 \cdot 48 + 6 \cdot 242/(2 + 0.5)$
 $= 676.800 \text{ mm.}$

Actual Circumferential Bolt Spacing [Bs]:
 $= C \cdot \sin(\pi / n)$
 $= 1400 \cdot \sin(\pi/40)$
 $= 109.843 \text{ mm.}$

ASME Moment Multiplier for Bolt Spacing per Table 4.16.11 [Bsc]:
 $= \max(\sqrt{ Bs/(2a + t) }, 1)$
 $= \max(\sqrt{ 110/(2 \cdot 48 + 242) }, 1)$
 $= 1.0000$

Bolting Information for TEMA Metric Thread Series (Non Mandatory):

	Minimum	Actual	Maximum
Bolt Area:	514.272	537.184	
Radial Distance between Hub and Bolts:	62.000	226.000	
Radial Distance between Bolts and Edge:	50.000	55.000	
Circ. Spacing between the Bolts:	109.300	109.843	676.800

Min. Gasket Contact Width (Brownell Young) [Not an ASME Calc] [Nmin]:
 $= Ab \cdot Sa/(y \cdot \pi(Go + Gi))$
 $= 537 \cdot 172/(17.2 \cdot \pi(838 + 812))$
 $= 103.404 \text{ mm. [Note: Exceeds actual gasket width, 13.000]}$

Recommended Minimum Gasket Contact Width per Table 4.16.2

Sheet Gaskets including laminated sheet gaskets with or without a metal core: 16 mm
 Preformed comp. gaskets incl. spiral wound, jacketed, & solid flat metal gaskets: 16 mm

Flange Design Bolt Load, Gasket Seating [W or Wgs]:

$$= S_a \cdot A_b$$

$$= 172 \cdot 537$$

$$= 942138.38 \text{ Kg}$$

Gasket Load for the Operating Condition [HG]:

$$= W_o - H$$

$$= 901955 - 802041$$

$$= 99914.07 \text{ Kg}$$

Moment Arm Calculations:

Distance to Gasket Load Reaction [hg]:

$$= (C - G) / 2$$

$$= (1400 - 825) / 2$$

$$= 287.4246 \text{ mm.}$$

Distance to Face Pressure Reaction [ht]:

$$= 0.5 ((C - B) / 2 + h_g)$$

$$= 0.5 ((1400 - 786) / 2 + 287)$$

$$= 297.2123 \text{ mm.}$$

Distance to End Pressure Reaction [hd]:

$$= (C - B - g_1) / 2$$

$$= (1400 - 786 - 81) / 2$$

$$= 266.5000 \text{ mm.}$$

Summary of Moments for Internal Pressure: (Kg-m.)

Loading	Force	Distance	Bolt Corr	Moment
End Pressure, Md	727738.	266.5000	1.0000	193946.
Face Pressure, Mt	74303.	297.2123	1.0000	22084.
Gasket Load, Mg	99914.	287.4246	1.0000	28718.
Gasket Seating, Matm	942138.	287.4246	1.0000	270799.

Total Moment for Operation, Mo 244748. Kg-m.
 Total Moment for Gasket seating, Mg 270799. Kg-m.

Effective Hub Length, $h_o = \sqrt{B_{cor} \cdot g_o \cdot C_{or}}$ 204.103 mm.
 Hub Ratio, $h/h_o = H_L / H_o$ 0.549
 Thickness Ratio, $g_1/g_o = (g_1 \cdot C_{or}) / (g_o \cdot C_{or})$ 1.528

Flange Factors for Integral Flange:

Factor F per 4.16.5 0.838
 Factor V per 4.16.5 0.297
 Factor f per 4.16.5 1.000

Stress factor from Table 4.16.4 [K]:

$$= A / B$$

$$= 1510 / 786$$

$$= 1.921$$

Stress factor from Table 4.16.4 [Y]:

$$= (0.66845 + (5.7169 (K^2 \cdot \log_{10}(K))) / (K^2 - 1)) / (K - 1)$$

$$= (0.66845 + (5.7169 (1.92^2 \cdot \log_{10}(1.92))) / (1.92^2 - 1)) / (1.92 - 1)$$

$$= 3.140$$

Stress factor from Table 4.16.4 [T]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / ((1.0472 + 1.9448 \cdot K^2) \cdot (K - 1))$$

$$= (1.92^2 (1 + 8.55246 \cdot \log_{10}(1.92)) - 1) / ((1.0472 + 1.9448 \cdot 1.92^2) \cdot (1.92 - 1))$$

$$= 1.537$$

Stress factor from Table 4.16.4 [U]:

$$= (K^2 (1 + 8.55246 \cdot \log_{10}(K)) - 1) / (1.36136 (K^2 - 1) \cdot (K - 1))$$

$$= (1.92^2(1 + 8.55246 \cdot \log_{10}(1.92)) - 1) / (1.36136(1.92^2 - 1) \cdot (1.92 - 1))$$
$$= 3.450$$

Stress factor from Table 4.16.4 [Z]:

$$= (K^2 + 1) / (K^2 - 1)$$
$$= (1.92^2 + 1) / (1.92^2 - 1)$$
$$= 1.743$$

Stress factor from Table 4.16.4 for integral flanges [d]:

$$= U \cdot g_0^2 \cdot h_0 / V$$
$$= 3.45 \cdot 53^2 \cdot 204 / 0.3$$
$$= 6654849.000 \text{ mm}^3$$

Stress factor from Table 4.16.4 for integral flanges [e]:

$$= F / h_0$$
$$= 0.84 / 204$$
$$= 0.004 \text{ 1/mm}$$

Stress factor from Table 4.16.4 [L]:

$$= (t \cdot e + 1) / T + t^3 / d$$
$$= (242 \cdot 0.0041 + 1) / 1.54 + 242^3 / 6654849$$
$$= 3.427$$

Flange Stresses

Longitudinal Hub Stress, Operating [SHo]:

$$= (f \cdot Mop / Bcor) / (L \cdot g_1^2)$$
$$= (1 \cdot 244748 / 786) / (3.43 \cdot 81^2)$$
$$= 135.81 \text{ N./mm}^2$$

Longitudinal Hub Stress, Seating [SHa]:

$$= (f \cdot Matm / Bcor) / (L \cdot g_1^2)$$
$$= (1 \cdot 270799 / 786) / (3.43 \cdot 81^2)$$
$$= 150.27 \text{ N./mm}^2$$

Radial Flange Stress, Operating [SRo]:

$$= (\beta \cdot Mop / Bcor) / (L \cdot t^2)$$
$$= (2.32 \cdot 244748 / 786) / (3.43 \cdot 242^2)$$
$$= 35.37 \text{ N./mm}^2$$

Radial Flange Stress, Seating [SRa]:

$$= (\beta \cdot Matm / Bcor) / (L \cdot t^2)$$
$$= (2.32 \cdot 270799 / 786) / (3.43 \cdot 242^2)$$
$$= 39.13 \text{ N./mm}^2$$

Tangential Flange Stress, Operating [STo]:

$$= (Y \cdot Mo / (t^2 \cdot Bcor)) - Z \cdot SRo$$
$$= (3.14 \cdot 244748 / (242^2 \cdot 786)) - 1.74 \cdot 35.4$$
$$= 102.05 \text{ N./mm}^2$$

Tangential Flange Stress, Seating [STa]:

$$= (y \cdot Matm / (t^2 \cdot Bcor)) - Z \cdot SRa$$
$$= (3.14 \cdot 270799 / (242^2 \cdot 786)) - 1.74 \cdot 39.1$$
$$= 112.91 \text{ N./mm}^2$$

Average Flange Stress, Operating [SAo]:

$$= (SHo + \max(SRo, STo)) / 2$$
$$= (136 + \max(35.4, 102)) / 2$$
$$= 118.93 \text{ N./mm}^2$$

Average Flange Stress, Seating [SAa]:

$$= (SHa + \max(SRa, STa)) / 2$$
$$= (150 + \max(39.1, 113)) / 2$$
$$= 131.59 \text{ N./mm}^2$$

Bolt Stress, Operating [BSo]:

$$= Wm1 / Ab$$
$$= 901955 / 537$$
$$= 164.66 \text{ N./mm}^2$$

Bolt Stress, Seating [BSa]:

$$= (Wm^2 / Ab)$$

$$= (29273/537)$$

$$= 5.34 \text{ N./mm}^2$$

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	135.81	207.00	150.27	207.00
Radial Flange	35.37	138.00	39.13	138.00
Tangential Flange	102.05	138.00	112.91	138.00
Maximum Average	118.93	138.00	131.59	138.00
Bolting	164.66	172.00	5.34	172.00

Minimum Required Flange Thickness 238.049 mm.

Estimated M.A.W.P. (Operating) 147 bars
 Estimated Finished Weight of Flange at given Thk. 2655.4 kg.
 Estimated Unfinished Weight of Forging at given Thk 3632.8 kg.

Flange rigidity based on required thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 270799 / 1 \cdot 9807 \cdot 0.3 / (3.23 \cdot 202713 \cdot 53^2 \cdot 204 \cdot 0.3)$
 $= 0.365 \text{ (should be } \leq 1)$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 244748 / 1 \cdot 9807 \cdot 0.3 / (3.23 \cdot 198576 \cdot 53^2 \cdot 204 \cdot 0.3)$
 $= 0.337 \text{ (should be } \leq 1)$

Flange rigidity based on given thickness [ASME]:

Flange Rigidity Criterion, Seating per Table 4.16.10 [Jg]:
 $= 52.14 \cdot Ma / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eamb \cdot go^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 270799 / 1 \cdot 9807 \cdot 0.3 / (3.43 \cdot 202713 \cdot 53^2 \cdot 204 \cdot 0.3)$
 $= 0.344 \text{ (should be } \leq 1)$

Flange Rigidity Criterion, Operating per Table 4.16.10 [Jo]:
 $= 52.14 \cdot Mo / Bsc \cdot Cnvfac \cdot V / (\lambda \cdot Eop \cdot goc^2 \cdot ho \cdot Ki)$
 $= 52.14 \cdot 244748 / 1 \cdot 9807 \cdot 0.3 / (3.43 \cdot 198576 \cdot 53^2 \cdot 204 \cdot 0.3)$
 $= 0.317 \text{ (should be } \leq 1)$

Minimum Design Metal Temperature Results:

Thickness Ratio = 0.96, Temperature Reduction per Fig. UCS 66.1 = 2 °C

Min Metal Temp. w/o impact per UCS-66, Curve B 19 °C
 Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit 3 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 0 °C

Note: UCS-66(b)(-c) was considered in the flange MDMT calculation.

Per ASME VIII-2, 4.16.2.3, Perform Calculations in the New Condition:

Flange Stress Analysis Results: N./mm²

	Actual	Operating Allowed	Gasket Seating Actual	Allowed
Longitudinal Hub	132.33	207.00	145.49	207.00
Radial Flange	36.02	138.00	39.60	138.00
Tangential Flange	97.91	138.00	107.66	138.00
Maximum Average	115.12	138.00	126.58	138.00
Bolting	164.66	172.00	5.34	172.00

Minimum Req'd Thickness, ambient, no Corrosion: 232.029 mm.

PV Elite 28 Licensee: SL User
FileName : Syngas Cooler-AK
Flg Calc [Int P]: FLANGE

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Internal Pressure Results Summary:

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd bars	Nominal Thickness mm.	Total Corr Allowance mm.	Element Diameter mm.	Allowable Stress(SE) N./mm ²	Element#
Blind Flange		147.15	268	3	1510	138	1
Ch Flg		147.08	253	3	780	138	2
Ch Shell		147.08	56	3	780	118	3
Main Shell		191	74	3	780	118	4
Dish End		191	80	3	780	118	5

Element Required Thickness and MAWP :

From	To	Design Pressure bars	M.A.W.P. Corroded bars	M.A.P. New & Cold bars	Minimum Thickness mm.	Required Thickness mm.	Element#
Blind Flange		147	151	154	260	256.425	1
Ch Flg		147	147	154	245	238.049	2
Ch Shell		147	147	156	56	55.9478	3
Main Shell		191	192	201	74	73.4627	4
Dish End		191	194	202	68	67.0149	5

Summary of Heat Exchanger Maximum Allowable Working Pressures :

Note:

For Exchanger designs, the following values include MAWPs that consider the tubesheet, tubes, tube/tubesheet joint etc. These values were determined by iteration. Review the tubesheet analysis report for more information.

Shell Side MAWP = 192 bars
 Shell Side MAPnc = 201 bars
 Channel Side MAWP = 147 bars
 Channel Side MAPnc = 154 bars

Elements Suitable for Design Internal Pressure.

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: Ch Shell

Cylindrical Shell Design Information:					
Design Pressure	P	147.077 bars	Int. Design Temperature		93 °C
Material		SA-516 60	Ext. Design Temperature		85 °C
Length		900.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02100
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	118.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	118.0 N./mm ²
Inside Diameter	Di	780.000 mm.	Actual Stress		117.9 N./mm ²
Spec. Min. Thk t _e , or s _a		56.0000 mm.	Spec. Nominal Thk.		56.000 mm.
Surface Area		25220.7 cm ²	SG of Contents		1.000
Element Volume		430129.7 cm ³	Weight of Contents		429.9 Kg
Empty Weight		1025.9 Kg	Operating Weight		1455.8 Kg
			Comp. All. Ope. Stress		141.6 N./mm ²

Radiography Information:			
User Defined			User Defined
Circ. Joint Efficiency	100 %		Long. Joint Eff. E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (147 \cdot 393) / (118 \cdot 1 - 0.6 \cdot 147)$$

$$= 52.9478 + 3.0000 = 55.9478 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.077 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118 \cdot 1 \cdot 53) / (393 + 0.6 \cdot 53)$$

$$= 147 - 0.077 = 147 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118 \cdot 1 \cdot 56) / (390 + 0.6 \cdot 56)$$

$$= 156 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (147 \cdot (393 + 0.6 \cdot 53)) / (1 \cdot 53)$$

$$= 117.890 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 6.699 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 56, t_r = 53, c = 3 mm., E* = 1
 Thickness Ratio = t_r * E* / (t_g - c) = 1, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve C -2 °C

Note:

Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:	
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PV Elite 28 Licensee: SL User
FileName : Syngas Cooler-AK
Element: Ch Shell

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MDMT Curve	C	Press. at MDMT	147.134 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	0 °C
Governing Thk.	56.000 mm.	Basic MDMT	-2 °C
Stress/Thk. Ratio	0.999	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	-2 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: Main Shell

Cylindrical Shell Design Information:					
Design Pressure	P	191.004 bars	Int. Design Temperature		93 °C
Material		SA-516 60	Ext. Design Temperature		200 °C
Length		3850.000 mm.	External Pressure Chart		CS-2
			UNS Number		K02100
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	118.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	118.0 N./mm ²
Inside Diameter	Di	780.000 mm.	Actual Stress		117.2 N./mm ²
Spec. Min. Thk t _e , or s _a		74.0000 mm.	Spec. Nominal Thk.		74.000 mm.
Surface Area		112242.8 cm ²	SG of Contents		0.050
Element Volume		1241944.1 cm ³	Weight of Contents		62.1 Kg
Empty Weight		5924.2 Kg	Operating Weight		5986.3 Kg
			Comp. All. Ope. Stress		141.6 N./mm ²

Radiography Information:			
User Defined			User Defined
Circ. Joint Efficiency	100 %		Long. Joint Eff. E or v 100 %

Required Thickness due to Internal Pressure [tr]:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= (191 \cdot 393) / (118 \cdot 1 - 0.6 \cdot 191)$$

$$= 70.4627 + 3.0000 = 73.4627 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.004 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118 \cdot 1 \cdot 71) / (393 + 0.6 \cdot 71)$$

$$= 192 - 0.0038 = 192 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118 \cdot 1 \cdot 74) / (390 + 0.6 \cdot 74)$$

$$= 201 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [Sact]:

$$= (P \cdot (R + 0.6 \cdot t)) / (E \cdot t)$$

$$= (191 \cdot (393 + 0.6 \cdot 71)) / (1 \cdot 71)$$

$$= 117.192 \text{ N./mm}^2$$

% Elongation per Table UG-79-1 (50 * t_{nom} / R_f * (1 - R_f / R_o)) 8.665 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

Minimum Design Metal Temperature Results:

Govrn. thk, t_g = 74, t_r = 71, c = 3 mm., E* = 1
 Thickness Ratio = t_r * E* / (t_g - c) = 1, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve C 4 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:	
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PV Elite 28 Licensee: SL User
FileName : Syngas Cooler-AK
Element: Main Shell

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MDMT Curve	C	Press. at MDMT	192.314 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	74.000 mm.	Basic MDMT	4 °C
Stress/Thk. Ratio	1.000	Temperature Difference	0 °C
UCS-66(c) red.	No	Computed Min. Temp.	4 °C
		Joint Efficiency E*	100 %

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Internal Pressure Calculation Results:

ASME Code, Section VIII Division 1, 2025

Internal Pressure Results for: Dish End

Elliptical Head Design Information:					
Design Pressure	P	191.004 bars	Int. Design Temperature		93 °C
Material		SA-516 60	Ext. Design Temperature		200 °C
Length		50.800 mm.	External Pressure Chart		CS-2
			UNS Number		K02100
Int. Corr. All.	c	3.0000 mm.	Allowable Stress (ope)	S	118.0 N./mm ²
Ext. Corr. All.	ce	0.0000 mm.	Allowable Stress (amb)	Sa	118.0 N./mm ²
Inside Diameter	Di	780.000 mm.	Actual Stress		116.2 N./mm ²
Spec. Min. Thk t _e , or s _a		68.0000 mm.	Spec. Nominal Thk.		80.000 mm.
Surface Area		11078.3 cm ²	SG of Contents		0.050
Element Volume		86408.3 cm ³	Weight of Contents		4.3 Kg _f
Empty Weight		634.0 Kg _f	Operating Weight		638.3 Kg _f
			Comp. All. Ope. Stress		141.6 N./mm ²
			Aspect Ratio		2.00

Radiography Information:			
User Defined			User Defined
Circ. Joint Efficiency	100 %	Long. Joint Eff.	E or v 100 %

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 9.750 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 4.875 mm.

Required Thickness due to Internal Pressure [t_r]:

$$= (P \cdot D \cdot K_{cor}) / (2 \cdot S \cdot E - 0.2 \cdot P) \text{ Appendix 1-4(c)}$$

$$= (191 \cdot 786 \cdot 0.99) / (2 \cdot 118 \cdot 1 - 0.2 \cdot 191)$$

$$= 64.0149 + 3.0000 = 67.0149 \text{ mm.}$$

Max. Allowable Working Pressure at given Thickness, corroded [MAWP]:

Less Operating Hydrostatic Head Pressure of 0.004 bars

$$= (2 \cdot S \cdot E \cdot t) / (K_{cor} \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 118 \cdot 1 \cdot 65) / (0.99 \cdot 786 + 0.2 \cdot 65)$$

$$= 194 - 0.0038 = 194 \text{ bars}$$

Maximum Allowable Pressure, New and Cold [MAPNC]:

$$= (2 \cdot S \cdot E \cdot t) / (K \cdot D + 0.2 \cdot t) \text{ per Appendix 1-4 (c)}$$

$$= (2 \cdot 118 \cdot 1 \cdot 68) / (1 \cdot 780 + 0.2 \cdot 68)$$

$$= 202 \text{ bars}$$

Actual stress at given pressure and thickness, corroded [S_{act}]:

$$= (P \cdot (K_{cor} \cdot D + 0.2 \cdot t)) / (2 \cdot E \cdot t)$$

$$= (191 \cdot (0.99 \cdot 786 + 0.2 \cdot 65)) / (2 \cdot 1 \cdot 65)$$

$$= 116.239 \text{ N./mm}^2$$

Straight Flange Required Thickness:

$$= (P \cdot R) / (S \cdot E - 0.6 \cdot P) + c \text{ per UG-27 (c)(1)}$$

$$= (191 \cdot 393) / (118 \cdot 1 - 0.6 \cdot 191) + 3$$

$$= 73.463 \text{ mm.}$$

Straight Flange Maximum Allowable Working Pressure:

Less Operating Hydrostatic Head Pressure of 0.004 bars

$$= (S \cdot E \cdot t) / (R + 0.6 \cdot t) \text{ per UG-27 (c)(1)}$$

$$= (118 \cdot 1 \cdot 77) / (393 + 0.6 \cdot 77)$$

$$= 207 - 0.0038 = 207 \text{ bars}$$

Factor K, corroded condition [K_{cor}]:

$$= (2 + (\text{Inside Diameter} / (2 \cdot \text{Inside Head Depth}))^2) / 6$$

$$= (2 + (786 / (2 \cdot 198))^2) / 6$$

$$= 0.989937$$

Percent Elong. per UCS-79, VIII-1-01-57 $(75 \cdot t_{nom} / R_f) \cdot (1 - R_f / R_o)$ 34.762 %

Note: Please Check Requirements of UCS-79 as Elongation is > 5%.

MDMT Calculations in the Knuckle Portion:

Govrn. thk, tg = 68, tr = 64.5, c = 3 mm., $E^* = 1$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.99$, Temp. Reduction = 0 °C

Min Metal Temp. w/o impact per UCS-66, Curve C 3 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 2 °C

MDMT Calculations in the Head Straight Flange:

Govrn. thk, tg = 80, tr = 71, c = 3 mm., $E^* = 1$
 Thickness Ratio = $tr \cdot E^* / (tg - c) = 0.92$, Temp. Reduction = 4 °C

Min Metal Temp. w/o impact per UCS-66, Curve C 6 °C
 Min Metal Temp. at Required thickness (UCS 66.1) 2 °C

Note:
 Post Weld Heat Treatment is required for this Element/Joint and it was specified as being heat treated.

MDMT Design Information:			
MDMT Curve	C	Press. at MDMT	192.314 bars
UCS-66(b) red.	Yes	Min. Des. Mtl. Temp.	-10 °C
Governing Thk.	80.000 mm.	Basic MDMT	6 °C
Stress/Thk. Ratio	0.922	Temperature Difference	4 °C
UCS-66(c) red.	No	Computed Min. Temp.	2 °C
		Joint Efficiency E^*	100 %

Note: Temperature values displayed above are for the location with the highest basic MDMT

Tolerance for Formed Heads per UG-81(a):

Head inner surface maximum deviation outside the specified shape, 1.25% of D: 9.750 mm.
 Head inner surface maximum deviation inside the specified shape, 0.625% of D: 4.875 mm.

EXTERNAL PRESSURE RESULTS for: Dish End

Elliptical Head From 50 to 60 Ext. Chart: CS-2 at 200 °C

Elastic Modulus from Chart: CS-2 at 200 °C: 193602528 KPa.

Results for Maximum Allowable Ext. Pressure				MAEP
Tca	Outer Dia	Do/t	Factor A	Factor B
65.000	916.00	14.09	0.0098557	114.22

$$MAEP = B / (K_0 \cdot Do / t) = 114 / (0.9 \cdot 14.1) = 90.1 \text{ bars}$$

Results for Required Thickness				Tca
Tca	Outer Dia	Do/t	Factor A	Factor B
2.410	916.00	380.04	0.0003655	35.38

$$MAEP = B / (K_0 \cdot Do / t) = 35.4 / (0.9 \cdot 380) = 1.03 \text{ bars}$$

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	250.008 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	248.300 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	261.149 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	211.545 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	273.130 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	273.130 bars
App 27-4	= M.A.W.P.	192.314 bars

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	191.275 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	191.100 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	199.570 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	161.848 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	210.210 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	210.210 bars
App 27-4	= M.A.W.P.	147.134 bars

Horizontal Test performed per: UG-99b

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Ch Shell	144.8	198.0	0.731	191.35
Main Shell	146.8	198.0	0.742	250.08
Dish End	145.9	198.0	0.737	250.08

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
T1-Tube Inlet	Nozzle	138.00	138.00	1.000
T2-Tube outlet	Nozzle	138.00	138.00	1.000
S1-Shell Inlet	Nozzle	138.00	138.00	1.000
S2-Shell Outlet	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Blind Flange	138.00	138.00	1.000
Ch Flg	138.00	138.00	1.000
Ch Shell	118.00	118.00	1.000
Main Shell	118.00	118.00	1.000
Dish End	118.00	118.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	92.40	92.40	1.000
Tubesheet Material	138.00	138.00	1.000

Minimum 1.000



Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
T1-Tube Inlet	83.77	225.00	0.372
T2-Tube outlet	83.77	225.00	0.372
S1-Shell Inlet	111.52	225.00	0.496
S2-Shell Outlet	111.52	225.00	0.496

Elements Suitable for Test Pressure.

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Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	250.008 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	248.300 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	261.149 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	211.545 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	273.130 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	273.130 bars
App 27-4	= M.A.W.P.	192.314 bars

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	191.275 bars
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	191.100 bars
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	199.570 bars
UG100	= 1.10 * M.A.W.P. * Sa/S	161.848 bars
PED	= max(1.43*DP, 1.25*DP*ratio)	210.210 bars
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	210.210 bars
App 27-4	= M.A.W.P.	147.134 bars

Horizontal Test performed per: UG-99c

Please note that Nozzle, Shell, Head, Flange, etc MAWPs are all considered when determining the hydrotest pressure for those test types that are based on the MAWP of the vessel.

Stresses on Elements due to Test Pressure (N./mm² & bars):

From To	Stress	Allowable	Ratio	Pressure
Ch Shell	160.0	198.0	0.808	199.65
Main Shell	160.3	198.0	0.809	261.23
Dish End	159.0	198.0	0.803	261.23

Stress ratios for Nozzle and Pad Materials (N./mm²):

Description	Pad/Nozzle	Ambient	Operating	Ratio
T1-Tube Inlet	Nozzle	138.00	138.00	1.000
T2-Tube outlet	Nozzle	138.00	138.00	1.000
S1-Shell Inlet	Nozzle	138.00	138.00	1.000
S2-Shell Outlet	Nozzle	138.00	138.00	1.000

Minimum 1.000

Stress ratios for Pressurized Vessel Elements (N./mm²):

Description	Ambient	Operating	Ratio
Blind Flange	138.00	138.00	1.000
Ch Flg	138.00	138.00	1.000
Ch Shell	118.00	118.00	1.000
Main Shell	118.00	118.00	1.000
Dish End	118.00	118.00	1.000

Minimum 1.000

Stress ratios for Exchanger Materials (N./mm²):

Description	Ambient	Operating	Ratio
Tube Material	92.40	92.40	1.000
Tubesheet Material	138.00	138.00	1.000

Minimum 1.000



Hoop Stress in Nozzle Wall during Pressure Test (N./mm²):

Description	Ambient	Operating	Ratio
T1-Tube Inlet	87.41	225.00	0.388
T2-Tube outlet	87.41	225.00	0.388
S1-Shell Inlet	116.49	225.00	0.518
S2-Shell Outlet	116.49	225.00	0.518

Elements Suitable for Test Pressure.

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Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main Shell

Shell Test Pressure		261.15 bars
Shell Thickness	ts	74.0000 mm.
Shell Internal Corrosion Allowance	cas	0.0000 mm.
Shell External Corrosion Allowance	casext	0.0000 mm.
Inside Diameter of Shell	Ds	780.000 mm.
Shell Temperature for Internal Pressure	Ts	21.11 °C
Shell Material		SA-516 60

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02100
Shell Allowable Stress at Temperature	Ss	118.00 N./mm ²
Shell Allowable Stress at Ambient		118.00 N./mm ²

Channel Data:

Channel Description: Ch Shell

Channel Type:		Cylinder
Channel Test Pressure		199.57 bars
Channel Thickness	tc	56.0000 mm.
Channel Internal Corrosion Allowance	cac	0.0000 mm.
Channel External Corrosion Allowance	cacext	0.0000 mm.
Inside Diameter of Channel	Dc	780.000 mm.
Channel Design Temperature	TEMPC	21.11 °C
Channel Flange Matl		SA-516 60

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02100
Channel Allowable Stress at Temperature	Sc	118.00 N./mm ²
Channel Allowable Stress at Ambient		118.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	520
Tube Wall Thickness	et	3.0480 mm.
Tube Outside Diameter	D	19.0500 mm.
Total Straight Tube Length	Lt	3658.00 mm.
Straight Tube Length (bet. inner tubsht faces) L		3478.00 mm.
Design Temperature of the Tubes		21.11 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		92.40 N./mm ²
Tube Allowable Stress At Ambient		92.40 N./mm ²
Tube Yield Stress At design Temperature	Syt	180.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.3990 mm.
Tube Layout Pattern		Square
Fillet Weld Leg	af	2.4130 mm.
Groove Weld Leg	ag	2.4130 mm.
Tube-Tubesheet Joint Weld Type		Full Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	354.223 mm.
Largest Center-to-Center Tube Distance	Ul	0.0000 mm.
Length of Expanded Portion of Tube	ltx	0.0000 mm.
Tube-side pass partition groove depth	hg	0.0000 mm.
Area of Tube Layout	Ap	3323.4 cm ²
Perimeter of Tube Layout	Cp	2394.5 mm.

Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature T 21.11 °C
 Tubesheet Material SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		K03506
Tubesheet Allowable Stress at Temperature	S	138.00 N./mm ²
Tubesheet Allowable Stress at Ambient	Tt	138.00 N./mm ²
Thickness of Tubesheet	h	180.0000 mm.
Tubesheet Corr. Allowance (Shell side)	Cats	0.0000 mm.
Tubesheet Corr. Allowance (Channel side)	Catc	0.0000 mm.
Tubesheet Outside Diameter	A	928.000 mm.
Area of the Untubed Lanes	AL	0.0 cm ²
Junction Stress Reduction option	Perform Plastic Calculation	
Perform Differential Pressure Design	No	
Run Multiple Load Cases	Yes	
Tubesheet Integral with	Both	
Tubesheet Extended as Flange	No	

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

Tubes - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Tubesht. Design Temp.	21.1 °C	0.20270E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20270E+09 KPa.

TubeSheet - TM-1 Carbon Steels with C> 0.3%

Elastic Mod. at Design Temperature	21.1 °C	0.20132E+09 KPa.
Elastic Mod. at Ambient Temperature	21.1 °C	0.20132E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (200 \cdot (19/2 - 0)) / (92.4 \cdot 1 + 0.4 \cdot 200)$
 $= 1.8938 + 0.0000 = 1.8938 \text{ mm.}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart	CS-1	at	21.11 °C
Elastic Modulus for Material			199943392.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
3.0480	19.05	3832.22	6.25	50.0000	0.0281600	95.84

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 252 \text{ bars}$

Results for Req'd Thickness for Ext. Pressure (Tca):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
3.1281	19.05	3832.22	6.09	50.0000	0.0296597	95.84

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 261 \text{ bars}$



Summary of Tube Required Thickness Results:

<i>Total Required Thickness including Corrosion all.</i>	<i>3.1281 mm.</i>
Allowable Internal Pressure at Corroded thickness	339.06 bars
Required Internal Design Pressure	199.57 bars
<i>Allowable External Pressure at Corroded thickness</i>	<i>252.45 bars</i>
Required External Design Pressure	261.15 bars
Required Thickness due to Shell Side pressure	3.1281 mm.

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:

$$\begin{aligned} &= 2 \cdot r_o + dt \\ &= 2 \cdot 354 + 19 \\ &= 727.496 \text{ mm.} \end{aligned}$$

Basic ligament efficiency for shear [μ]:

$$\begin{aligned} &= (p - dt) / p \\ &= (25.4 - 19) / 25.4 \\ &= 0.250 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:

$$\begin{aligned} &= D_s / D_o \text{ (Configurations a, b, c)} \\ &= 780 / 727 \\ &= 1.0722 \end{aligned}$$

Ratio [pc]:

$$\begin{aligned} &= D_c / D_o \text{ (Configurations a, e, f)} \\ &= 780 / 727 \\ &= 1.0722 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [ρ]:

$$\begin{aligned} &= l_{tx} / h \\ &= 0 / 180 \\ &= 0.000 \text{ (must be } 0 \leq \rho \leq 1 \text{)} \end{aligned}$$

Effective tube hole Diameter [d^*]:

$$\begin{aligned} &= \max(dt - 2t_t(E_t T / E) \cdot (S_t T / S) \cdot (\rho); dt - 2t_t) \\ &= \max(19 - 2 \cdot 3.05 (202701232 / 201322320) \cdot \\ &\quad (92.4 / 138) \cdot (0); 19 - 2 \cdot 3.05) \\ &= 19.0500 \text{ mm.} \end{aligned}$$

Effective tube pitch [p^*]:

$$\begin{aligned} &= p / \sqrt{ (1 - 4 \cdot \min(A_l; 4 \cdot D_o \cdot \rho) / (\pi \cdot D_o^2)) } \\ &= 25.4 / \sqrt{ (1 - 4 \cdot \min(0; 4 \cdot 727 \cdot 25.4) / (\pi \cdot 727^2)) } \\ &= 25.3990 \text{ mm.} \end{aligned}$$

Effective ligament efficiency for bending [μ^*]:

$$\begin{aligned} &= (p^* - d^*) / p^* \\ &= (25.4 - 19) / 25.4 \\ &= 0.2500 \end{aligned}$$

$h/p = 7.086893$; $\mu^* = 0.249970$; $E^* = 57855828 \text{ KPa.}$

$E^*/E = 0.287379$; $v^* = 0.332864$

Note: As h/p (7.087) is > 2 , data values for $h/p = 2$ were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$= \left(\left(12(1 - v_s^2) \right)^{1/4} / \sqrt{(D_s + t_s)t_s} \right)$$

$$= \left(\left(12(1 - 0.3^2) \right)^{1/4} / \sqrt{(780 + 74)74} \right)$$

$$= 0.0072 \text{ 1/mm.}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - v_s^2))$$

$$= 0.0072 \cdot 2027012 \cdot 74^3 / (6(1 - 0.3^2))$$

$$= 1087851776.0000 \text{ bars} \cdot \text{mm.}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2)$$

$$= 6(780 + 74) \cdot 1087851776 / 180^3 \cdot (1 + 180 \cdot 0.0072 + (180 \cdot 0.0072)^2 / 2)$$

$$= 3009508.2 \text{ bars}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot v_s / (2(D_s + t_s)))$$

$$= 780^2 / (4 \cdot 202701232 \cdot 74) \cdot (1 - 780 \cdot 0.3 / (2(780 + 74)))$$

$$= 0.87504\text{E-}02 \text{ mm./N./mm}^2$$

Parameter [ω_s]:

$$= \rho_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s)$$

$$= 1.07 \cdot 1087851776 \cdot 0.0072 \cdot 0.008750352 (1 + 180 \cdot 0.0072)$$

$$= 16987.4590 \text{ mm.}^2$$

Channel coefficient [β_c]:

$$= \left(\left(12(1 - v_c^2) \right)^{1/4} / \sqrt{(D_c + t_c)t_c} \right)$$

$$= \left(\left(12(1 - 0.3^2) \right)^{1/4} / \sqrt{(780 + 56)56} \right)$$

$$= 0.0084 \text{ 1/mm.}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - v_s^2))$$

$$= 0.0084 \cdot 2027012 \cdot 56^3 / (6(1 - 0.3^2))$$

$$= 547754944.0000 \text{ bars} \cdot \text{mm.}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta)$$

$$= 6(780 + 56 / \cos(0)) \cdot 547754944 / 180^3 \cdot (1 + 180 \cdot 0.0084 \cdot (1 + 1) / 2 + (180 \cdot 0.0084)^2 / 2 \cdot 1)$$

$$= 1722283.2 \text{ bars}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot v_c / (2(D_c + t_c)))$$

$$= 780^2 / (4 \cdot 202701232 \cdot 56) \cdot (1 - 780 \cdot 0.3 / (2(780 + 56)))$$

$$= 0.11523\text{E-}01 \text{ mm./N./mm}^2$$

Channel coefficient [zetac]:

$$zetac = 0$$

Parameter [ω_c]:

$$= \rho_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)]$$

$$= 1.07 \cdot 547754944 [0.01152344 (1 + 0.0084 \cdot 1 \cdot 180) - 0.000 (1/180 + 1 \cdot 0.0084/2)]$$

$$= 14285.1064 \text{ mm.}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o$$

$$= 928/727$$

$$= 1.2756$$

Coefficient [F]:

$$= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K))$$



$$= (1 - 0.33) / 57855828 \cdot (3009508 + 1722283 + 201322320 \cdot \ln(1.28)) \\ = 6.0213$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$= Do^2/16 \cdot [(ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt] \\ = 727^2/16 \cdot [(1.07 - 1) \cdot (1.07^2 + 1) \cdot 261 - (1.07 - 1) \cdot (1.07^2 + 1) \cdot 200] \\ = 315998.1250 \text{ bars} \cdot \text{mm}^2$$

Moment acting on unperforated tubesheet rim [M*]:

$$= Mts + \omega c \cdot Pt - \omega s \cdot Ps \\ = 315998 + 14285 \cdot 200 - 16987 \cdot 261 \\ = -1269383.2500 \text{ bars} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$= (M^*) - Do^2/32 \cdot F(Ps - Pt) / (1 + F) \\ = (-1269383) - 727^2/32 \cdot 6.02(261 - 200) / (1 + 6.02) \\ = -1054199.2500 \text{ bars} \cdot \text{mm}^2$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$= Mp + Do^2/64 \cdot (3 + \nu^*) \cdot (Ps - Pt) \\ = -1054199 + 727^2/64 \cdot (3 + 0.33) \cdot (261 - 200) \\ = 642998.8750 \text{ bars} \cdot \text{mm}^2$$

Maximum bending moment acting on tubesheet [M]:

$$= \max(|Mp|; |Mo|) \\ = \max(|-1054199|; |642999|) \\ = 1054199.2500 \text{ bars} \cdot \text{mm}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$= 6 \cdot M / (\mu^* (h - hg')^2) \\ = 6 \cdot 1054199 / (0.25 (180 - 0)^2) \\ = 78.1026 \text{ N./mm}^2$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$= 2 \cdot \beta T \cdot Sy \\ = 2 \cdot 0.95 \cdot 250 \\ = 474.99 \text{ N./mm}^2$$

Tubesheet bending stress at final thickness [σ]:

$$= 6 \cdot M / (\mu^* (h - hg')^2) \\ = 6 \cdot 1013528 / (0.25 (71.6 - 0)^2) \\ = 474.9913 \text{ N./mm}^2$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$= h + Cats + Catc \\ = 71.6 + 0 + 0 \\ = 71.5680 \text{ mm.}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$= \max(HreqB; HreqS) \\ = \max(71.6; 18) \\ = 71.5680 \text{ mm.}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [|Ps - Pt|]:

$$= |Ps - Pt| \\ = |261 - 200| \\ = 61.579 \text{ bars}$$

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h/Do \cdot 0.8 \cdot \beta T \cdot Sy$$



$$= 4 \cdot 0.25 \cdot 180/727 \cdot 0.8 \cdot 0.95 \cdot 250$$
$$= 47.004 \text{ N./mm}^2$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [HreqS]:

$$= 1/(4 \cdot \mu \cdot (0.8 \cdot \beta T \cdot S_y)) \cdot (4 \cdot A_p/C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc}$$
$$= 1/(4 \cdot 0.25 \cdot 190) \cdot (4 \cdot 3323/2395) \cdot |261 - 200| + 0$$
$$= 17.9967 \text{ mm.}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 74.000mm.
Min. Shell len. adjacent to the tubesheet is: 432.450 mm.

Note: For a given Channel thickness of: 56.000 mm.
Min. Channel len. adjacent to the tubesheet is: 376.196mm.

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$
$$= \pi (19 - 3.05) 3.05$$
$$= 1.532 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$
$$= \pi \cdot 3.05 (19 - 3.05) 171$$
$$= 2671.768 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$
$$= 0.85 \cdot \pi \cdot 2.41 (19 + 0.67 \cdot 2.41) 171$$
$$= 2321.9729 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$
$$= 171/171$$
$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$
$$= 0.55 \cdot \pi \cdot 2.41 (19 + 0.67 \cdot 2.41) 171$$
$$= 1502.4530 \text{ Kg}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o \right)$$
$$= 2 \left(\sqrt{((0.75 \cdot 19)^2 + 1.07 \cdot 3.05 \cdot (19 - 3.05) 1 \cdot 1)} - 0.75 \cdot 19 \right)$$
$$= 3.445 \text{ mm.}$$

Minimum Required Fillet Weld Leg [afr]: 1.8288 mm.
Minimum Required Groove Weld Leg [agr]: 1.8288 mm.

Tube-Tubesheet Jt allowable, 2672 is $\geq$ tube strength 2672 Kg

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	78.1 $\leq$	475.0 N./mm ²	Ok
Tubesheet shear stress	19.0 $\leq$	190.0 N./mm ²	Ok



Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm.)	Required	Actual	P/F
Tubesheet Thickness :	71.568	180.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.829	2.413	Ok
Tube-Tubesheet Groove Weld Leg :	1.829	2.413	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 3 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Stresses Bend Allwd Shear Allwd	Case Type	Pass/ Fail
D1uc	126.201 ...	229	475 62 190	Ps+Pt D1	Ok
D2uc	144.824 ...	307	475 81 190	Ps+Pt D2	Ok
D3uc	71.568 ...	78	475 19 190	Ps+Pt D3	Ok
Max:	144.8239 ... mm.	0.647	0.424 (Str. Ratio)		

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Req'd. Thk. Shell Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc		...	313	...	313	Ok
D2uc		...	313	...	313	Ok
D3uc		...	313	...	313	Ok
Max		...	...	...	...	

Note:
 Because there was no net pressure differential across the tubesheet due to
 the specified pressures psd,min and ptd,min; load case D4 was not processed as
 the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 3 Load Cases:

Thickness (mm.)	Required	Actual	P/F
Tubesheet thickness :	144.824	180.000	Ok
Tubesheet thickness flanged extension :	0.000	1.000	Ok
Tube thickness :	3.128	3.048	Fail
Tube-tubesheet fillet weld leg :	1.829	2.413	Ok
Tube-tubesheet groove weld leg :	1.829	2.413	Ok

Given Shell Thickness: 74.0000 mm.
 Given Channel Thickness: 56.0000 mm.

Min Shell length of thk, (74.000) adj. to tubesheet: 432.450 mm.

Min Channel length of thk, (56.000) adj. to tubesheet: 376.196 mm.

PV Elite 28 Licensee: SL User
FileName : Syngas Cooler-AK
ASME TS Calc: Pressure Test

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Case: 1 3:29p Sep 3, 2026



HEXAGON

Note: This is a full strength Tube to Tubesheet Joint.

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External Pressure Calculation Results Summary:

ASME Code, Section VIII Division 1, 2025

External Pressure Calculations:

From	To	Section Length mm.	Outside Diameter mm.	Corroded Thickness mm.	Factor A	Factor B N./mm ²
10	20	No Calc	...	257	No Calc	No Calc
20	30	No Calc	...	242	No Calc	No Calc
30	Tsht	1256.97	892	53	0.013994	122.731
Tsht	40	No Calc	...	53	No Calc	No Calc
40	50	3965.83	928	71	0.0077352	111.986
50	60	No Calc	916	65	0.0098557	114.22

External Pressure Calculations:

From	To	External Actual T. mm.	External Required T. mm.	External Design Pressure bars	External M.A.W.P. bars
10	20	260	213.503	...	No Calc
20	30	245	238.049	...	No Calc
30	Tsht	56	No Calc	...	97.2
Tsht	40	56	No Calc	...	No Calc
40	50	74	No Calc	...	114
50	60	68	5.41026	1.03419	90.1

Minimum

90.1

External Pressure Calculations:

From	To	Actual Length Bet. Stiffeners mm.	Allowable Length Bet. Stiffeners mm.	Ring Inertia Required cm ⁴	Ring Inertia Available cm ⁴
10	20	No Calc	No Calc	No Calc	No Calc
20	30	No Calc	No Calc	No Calc	No Calc
30	Tsht	1256.97	No Calc	No Calc	No Calc
Tsht	40	No Calc	No Calc	No Calc	No Calc
40	50	3965.83	No Calc	No Calc	No Calc
50	60	No Calc	No Calc	No Calc	No Calc

Elements Suitable for External Pressure.

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Element and Detail Weights:

From	To	Element Metal Wgt. kg.	Element ID Volume cm ³	Corroded Metal Wgt. kg.	Corroded ID Volume cm ³	Extra due Misc % kg.
10	20	3719.7	...	3719.7	...	371.97
20	30	2736.8	176309	2715.92	177136	273.68
30	40	1025.93	430130	974.451	436773	102.593
40	50	5924.21	1241944	5704	1270360	592.421
50	60	633.955	86408.3	610.182	88228.1	63.3955
Total		14040	1934791.00	13724	1972496.88	1404



For elements specified as shell side elements, the volume(s) shown above for those elements, reflects the displacement of the tubes.

Weight of Details:

From	Type	Weight of Detail kg.	X Offset, Dtl. Cent. mm.	Y Offset, Dtl. Cent. mm.	Z Offset, Dtl. Cent. mm.	Description
10	Liqd	...	130	...	...	Liquid: 10
20	Liqd	176.201	178.5	...	...	Liquid: 20
30	Liqd	429.867	450	...	...	Liquid: 30
30	Nozl	363.302	450	-497.95	...	T1-Tube Inlet
30	Nozl	363.302	450	497.95	...	T2-Tube outlet
40	Sadl	146.651	800	553	...	Left Saddle
40	Sadl	146.651	2650	553	...	Sd1 2 Fr40
40	Liqd	62.0593	1925	...	...	Liquid: 40
40	Insl	113.562	1925	...	...	Hot Insulation
40	Nozl	583.181	600	518.6	...	S1-Shell Inlet
40	Nozl	583.181	3200	-518.6	...	S2-Shell Outlet
50	Liqd	4.31778	115.8	...	...	Liquid: 50
50	Insl	13.1548	122.9	...	...	Hot Insulation
30	FTsh	810.515	990	...	...	
30	Tube	2796.16	2729	...	...	Tubes
40	Bafl	36.8483	796	...	...	Baffles
40	Bafl	36.8483	1106	...	...	Baffles
40	Bafl	36.8483	1416	...	...	Baffles
40	Bafl	36.8483	1726	...	...	Baffles
40	Bafl	36.8483	2036	...	...	Baffles
40	Bafl	36.8483	2346	...	...	Baffles
40	Bafl	36.8483	2656	...	...	Baffles
40	Bafl	36.8483	2966	...	...	Baffles
40	Bafl	36.8483	3276	...	...	Baffles
40	Bafl	37.1667	3586	...	...	Baffles

Total Weight of Each Detail Type:

Saddles	293.3
Liquid	672.4
Insulation	126.7
Nozzles	1893.0
Exchanger Components	3975.5
Liquid in Tubes	288.7
Sum of the Detail Weights	7249.6 kg.

Weight Summation Results: (kg.)

	Fabricated	Shop Test	Shipping	Erected	Empty	Operating
Main Elements	15444.6	15444.6	15444.6	15444.6	15444.6	15444.6
Saddles	293.3	293.3	293.3	293.3	293.3	293.3



Nozzles	1893.0	1893.0	1893.0	1893.0	1893.0	1893.0
Exchanger	3975.5	3975.5	3975.5	3975.5	3975.5	3975.5
Insulation	...	...	126.7	126.7	126.7	126.7
Ope. Liquid	...	...	...	...	...	672.4
Tube Ope Lqd	...	...	...	...	...	288.7
Test Liquid	...	1933.6	...	...	...	...
Tube Tst Lqd	...	288.7	...	...	...	...
Totals	21606.4	23828.7	21733.1	21733.1	21733.1	22694.3

Field Installation Options:

Miscellaneous Weight Percent: 10.0 %

Note that the above value for the miscellaneous weight percent has been applied to the shells/heads/flange/tubesheets/tubes etc. in the weight calculations for metallic components.

Weight Summary:

Fabricated Wt.	- Bare weight without removable internals	21606.4 kg.
Shop Test Wt.	- Fabricated weight + water (full)	23828.7 kg.
Shipping Wt.	- Fab. weight + removable intls.+ shipping app.	21733.1 kg.
Erected Wt.	- Fab. wt + or - loose items (trays,platforms etc.)	21733.1 kg.
Ope. Wt. no Liq	- Fab. weight + internals. + details + weights	21733.1 kg.
Operating Wt.	- Empty weight + operating liq. uncorroded	22694.3 kg.
Oper. Wt. + CA	- Corr wt. + operating liquid	22346.3 kg.
Field Test Wt.	- Empty weight + test medium (full)	23955.4 kg.

Exchanger Tube Data

Volume of Exchanger tubes :	288878.9 cm ³
Weight of Ope Liq in tubes :	288.7 kg.
Weight of Water in tubes :	288.7 kg.

The Corroded Weight and thickness are used in the Horizontal Vessel Analysis (Ope Case) and Earthquake Load Calculations.

Surface Areas of Elements:

From	To	Outside Surface Area cm ²	Inside Surface Area cm ²
10	20	30621.3	...
20	30	28466.9	8944.12
30	40	25220.7	22054
40	50	112243	94342
50	60	11078.3	7839.79

Total	207630.0	133179.9 cm ²
Total	223.5	143.4 ft ²

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Nozzle Flange MAWP Results: (bars & °C)

Nozzle Description	Flange Rating		Des. Temp	Class	Grade/Group	Equiv. Press	Max Pressure		
	Ope.	Ambient					UG-44	50%	DNV
T1-Tube Inlet	235.55	255.10	93	1500	GR 1.1	...	...	...	...
T2-Tube outlet	235.55	255.10	93	1500	GR 1.1	...	...	...	...
S1-Shell Inlet	235.55	255.10	93	1500	GR 1.1	...	...	...	...
S2-Shell Outle	235.55	255.10	93	1500	GR 1.1	...	...	...	...

Shellside Flange Rating

Lowest Flange Pressure Rating was (Ope)[ShellSide]: 235.554 bars
Lowest Flange Pressure Rating was (Amb)[ShellSide]: 255.100 bars

Channelside Flange Rating

Lowest Flange Pressure Rating was (Ope)[TubeSide]: 235.554 bars
Lowest Flange Pressure Rating was (Amb)[TubeSide]: 255.100 bars

Flange Pressure Ratings are per ASME B16.5 2025 Metric Edition

Warning:

There are nozzles in this model, but no flange MAWPs were de-rated. Be sure this is what you intended. There is a check box in the nozzle dialog that instructs PV Elite to perform the de-rating for each nozzle flange. See ASME VIII-1, UG-44(b) for more information.

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Wind Load Calculation Results:

Input Values:

Wind Design Code	ASCE/SEI 7-22
Wind Load Reduction Scale Factor	1.100
Basic Wind Speed	[V] 180 Km/hr
Surface Roughness Category	D: Flat, unobstructed
Type of Surface	Moderately Smooth
Base Elevation	0 mm.
Percent Wind for Hydrotest	33.0
Using User defined Wind Press. Vs Elev.	N
Height of Hill or Escarpment H or Hh	0 mm.
Distance Upwind of Crest Lh	0 mm.
Distance from Crest to the Vessel x	0 mm.
Type of Terrain (Hill, Escarpment)	Flat
Damping Factor (Beta) for Wind (Ope)	0.0100
Damping Factor (Beta) for Wind (Empty)	0.0000
Damping Factor (Beta) for Wind (Filled)	0.0000
Site Elevation above Sea Level	0 mm.

Wind Analysis Results

The importance factor is not used in the calculation of wind pressure vs. elevation for the selected wind code.

Static Gust-Effect Factor, Operating Case [G]:
 $= \min(0.85, 0.925((1 + 1.7 \cdot g_Q \cdot I_{zbar} \cdot Q) / (1 + 1.7 \cdot g_V \cdot I_{zbar})))$
 $= \min(0.85, 0.925((1 + 1.7 \cdot 3.4 \cdot 0.19 \cdot 0.99) / (1 + 1.7 \cdot 3.4 \cdot 0.19)))$
 $= \min(0.85, 0.92)$
 $= 0.850$

Natural Frequency of Vessel (Operating)	33.000 Hz
Natural Frequency of Vessel (Empty)	33.000 Hz
Natural Frequency of Vessel (Test)	33.000 Hz

User Entered Importance Factor is	1.000
Force Coefficient	[Cf] 0.585
Structure Height to Diameter ratio	6.101

This is classified as a rigid structure. Static analysis performed.

Sample Calculation for the First Element:

The ASCE code performs calculations in Imperial Units. The wind pressure is therefore computed in these units.

Value of $[\alpha]$ and $[Z_g]$:
Exposure Category: D from Table 26.11-1
 $\alpha = 11.5$: $Z_g = 589788$ mm.

Effective Height $[z]$:
 $=$ Centroid Height + Vessel Base Elevation
 $= 800 + 0 = 800$ mm.
 $= 2.62$ ft. Imperial Units

Velocity Pressure coefficient evaluated at height z $[K_z]$:
Because z (2.62 ft.) < 15 ft.
 $= 2.41(15 / Z_g)^{2/\alpha}$
 $= 2.41(15/1935)^{2/11.5}$
 $= 1.04$

Type of Hill: No Hill

Wind Directionality Factor $[K_d]$:
 $= 1$, per Table 26.6-1

As there is No Hill Present: $[K_{zt}]$:
 $K_1 = 0$, $K_2 = 0$, $K_3 = 0$

Topographical Factor [Kzt]:

$$= (1 + K1 \cdot K2 \cdot K3)^2$$

$$= (1 + 0 \cdot 0 \cdot 0)^2$$

$$= 1$$

Velocity Pressure evaluated at height z, Imperial Units [qz]:

$$= \max(16, 0.00256 \cdot Kz \cdot Kzt \cdot Kd \cdot Ke \cdot V(\text{mph})^2)$$

$$= \max(16, 0.00256 \cdot 1.04 \cdot 1 \cdot 1 \cdot 1 \cdot 112^2)$$

$$= 33.1 \text{ psf } [162 \text{ Kgs/m}^2]$$

Force on the first element [F]:

$$= qz \cdot G \cdot Cf \cdot \text{WindArea}$$

$$= 33.1 \cdot 0.85 \cdot 0.59 \cdot 5.07$$

$$= 83.6 \text{ lbs. } [37.9 \text{ Kgf}]$$

Element	Hgt (z) mm.	K1	K2	K3	Kz	Kzt	qz Kgs/m ²
Blind Flange	800.0	0.000	0.000	0.000	1.035	1.000	161.849
Ch Flg	800.0	0.000	0.000	0.000	1.035	1.000	161.849
Ch Shell	800.0	0.000	0.000	0.000	1.035	1.000	161.849
Main Shell	800.0	0.000	0.000	0.000	1.035	1.000	161.849
Dish End	800.0	0.000	0.000	0.000	1.035	1.000	161.849

Wind Load Calculation:

From	To	Wind Height mm.	Wind Diameter mm.	Wind Area cm ²	Wind Pressure Kgs/m ²	Element Wind Load Kgf
10	20	800	1812	4711.2	161.849	41.7079
20	30	800	936	3341.52	161.849	29.5823
30	40	800	1070.4	9633.6	161.849	85.2856
40	50	800	1233.6	47493.6	161.849	420.458
50	60	800	1219.2	3116.98	161.849	27.5944

The Wind Loads calculated and printed in the Wind Load calculation report have been factored by the input scalar/load reduction factor of: 1.100.

Be sure the wind speed is in accordance with the specified wind design code.

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Earthquake Load Calculation Results:

Earthquake Load Calculation:

Input Values:

Seismic Design Code		ASCE/SEI 7-22	
Importance Factor		1.500	
Force Modification Factor R		3.000	
Site Class		D	
Component Elevation Ratio	z/h	0.000	
Amplification Factor	Ap	0.000	
Force Factor		0.000	
Consider Vertical Acceleration		Yes	
Minimum Acceleration Multiplier		0.000	
User Value of Sds (used if > 0)		0.000	
User Value of Sd1 (used if > 0)		0.000	
Component resonance duct. fact.	[CAR]	0.000	
Component strength factor	[Rpo]	0.000	
Overstrength factor	[Omega0]	0.000	
Response Acceleration Factor	[SMS]	0.326	
Response Acceleration Factor	[SM1]	0.139	
Building Importance Factor	[Ie]	1.000	
Site Latitude		0.000	deg
Site Longitude		0.000	deg

Seismic Analysis Results:

$$SDS = 2/3 * Sms = 2/3 * 0.33 = 0.22$$

$$SD1 = 2/3 * Sm1 = 2/3 * 0.14 = 0.093$$

Check Approximate Fundamental Period from 12.8-7 [Ta]:
 = $C_t \cdot h_{nx}$ where $C_t = 0.020$, $x = 0.75$ and h_n = Structural Height (ft.)
 = $0.020(5.10.75)$
 = 0.068 seconds

The Coefficient C_u from Table 12.8-1 is : 1.700

Fundamental Period (1/Frequency) [T]:
 = $(1/\text{Natural Frequency}) = (1/33)$
 = 0.030

Check the Value of T which is the smaller of $C_u \cdot T_a$ and T:
 = Minimum Value of $(1.7 \cdot 0.068, 0.03)$ per 12.8.2
 = 0.030
 As the time period is < 0.06 second, use section 15.4.2.

Base Shear per equation 15.4-5, [V]:
 = $0.3 \cdot Sds \cdot W \cdot I$
 = $0.3 \cdot 0.22 \cdot 22053 \cdot 1.5$
 = 2156.784 Kgf

Vertical load per 12.4-4, [Ev]:
 = $0.2 \cdot Sds \cdot W$
 = $0.2 \cdot 0.22 \cdot 22053$
 = 958.6 Kgf

Earthquake Load Calculation:

From	To	Earthquake Height mm.	Earthquake Weight Kgf	Element Ope Load Kgf
10	20	755	3150.43	526.13
20	30	390	3150.43	271.776
30	40	390	3150.43	271.776
40	Sad1	390	3150.43	271.776
Sad1	50	390	3150.43	271.776
40	50	390	3150.43	271.776



50	60	390	3150.43	271.776
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The Earthquake Weight does not include the weight of the saddles.

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Shop/Field Installation Options :

Insulation is installed in the Shop.

Note : The CG is computed from the first Element From Node

Center of Gravity of the Saddles	3425.175 mm.
Center of Gravity of the Liquid	1170.797 mm.
Center of Gravity of the Insulation	3837.773 mm.
Center of Gravity of the Nozzles	2629.049 mm.
Center of Gravity of the Tubesheet(s)	1610.175 mm.
Center of Gravity of the Tubes	3349.175 mm.
Center of Gravity of the Baffles	3892.379 mm.
Center of Gravity of Bare Shell New and Cold	1984.643 mm.
Center of Gravity of Bare Shell Corroded	1957.734 mm.
Vessel CG in the Operating Condition	2232.251 mm.
Vessel CG in the Fabricated (Shop/Empty) Condition	2264.984 mm.
Vessel CG in the Test Condition	2325.907 mm.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Operating Case

Input and Calculated Values:

Vessel Mean Radius	Rm	428.50	mm.
Shell Thickness used in this Case	t	74.000	mm.
Stiffened Vessel Length per 4.15.6	L	3900.80	mm.
Distance from Saddle to Vessel tangent	a1 or a	800.00	mm.
Saddle Width	b1 or b	200.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	480.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm.
Wear Plate Allowable Stress	fw or Sr	118.00	N./mm ²
Shell Allowable Stress used in Calculation		118.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	800.00	mm.
Coefficient of Friction	mu	0.60	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		29067.66	Kgf
Pressure used in Analysis	P	191.002	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	56.26	118.00
Long. Stress at Bottom of Midspan	59.02	118.00
Long. Stress at Top of Saddles	66.82	118.00
Long. Stress at Bottom of Saddles	52.55	118.00
Tangential Shear in Shell	6.47	94.40
Circ. Stress at Horn of Saddle	5.31	147.50
Circ. Compressive Stress in Shell	0.57	118.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (605/2 + 0) \cdot 800/821$
 $= 883.8 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(173, 0) \cdot 800/1850$
 $= 75.0 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l, \text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(2157, 0) \cdot 800/1850$
 $= 932.7 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (2157/2 + 0) \cdot 800/821$
 $= 3152.5 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:

$$= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$$

$$= 25915 + \max(75, 884, 933, 3152)$$

$$= 29067.7 \text{ Kg}$$

Longitudinal Wind Force [FI]:

$$= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$$

$$= 1.1 \cdot 1587(0 + \pi/4(0.93 \cdot 1.2)^2)$$

$$= 1700.463 \text{ N}$$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	29214.31	Kgf
Transverse Shear Load	1078.39	Kgf
Longitudinal Shear Load	2156.78	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot \text{Rm})$$

$$= \min(2(780/2 + 74 + 10)\sin(120/2), 2 \cdot 428)$$

$$= 820.992 \text{ mm.}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm)/3L)]$$

$$= -29068 \cdot 800 [1 - (1 - 800/3901 + (428^2 - 0^2)/(2 \cdot 800 \cdot 3901))/(1 + (4 \cdot 0)/(3 \cdot 3901))]$$

$$= -4085.1 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 29068 \cdot 3901/4((1 + 2(428^2 - 0^2)/(3901^2))/(1 + (4 \cdot 0)/(3 \cdot 3901)) - 4 \cdot 800/3901)$$

$$= 5776.9 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 191 \cdot 428/(2 \cdot 71) - 5777/(\pi \cdot 428^2 \cdot 71)$$

$$= 56.26 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$$

$$= 191 \cdot 428/(2 \cdot 71) + 5777/(\pi \cdot 428^2 \cdot 71)$$

$$= 59.02 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 191 \cdot 428/(2 \cdot 71) - 4085/(0.11 \cdot \pi \cdot 428^2 \cdot 71)$$

$$= 66.82 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm/(2t) + M1/(K1^* \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 191 \cdot 428/(2 \cdot 71) + 4085/(0.19 \cdot \pi \cdot 428^2 \cdot 71)$$

$$= 52.55 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot hm/3))$$

$$= 29068(3901 - 2 \cdot 800)/(3901 + (4 \cdot 0/3))$$

$$= 17144.9 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.17 \cdot 17145 / (428 \cdot 71)$$

$$= 6.47 \text{ N./mm}^2$$

Decay Length (4.15.20) [x1,x2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(428 \cdot 71)}$$

$$= 136.050 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(200 + 1.56 \cdot \sqrt{(428 \cdot 71)}, 2 \cdot 800)$$

$$= 472.10 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:

$$= \min(Sr/S, 1)$$

$$= \min(118/118, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ6,r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 29068 \cdot 0.1 / (472(71 + 1 \cdot 10))$$

$$= -0.57 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, L ≥ 8Rm (4.15.28) [σ7,r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -29068 / (4(71 + 1 \cdot 10)472) -$$

$$3 \cdot 0.053 \cdot 29068 / (2(71 + 1 \cdot 10)^2)$$

$$= -5.31 \text{ N./mm}^2$$

Distance between Saddle Supports [Ls]:

$$= 1850.0 \text{ mm.}$$

Free Un-Restrained Thermal Expansion between the Saddles [Exp]:

$$= \alpha \cdot Ls(\text{Design Temperature} - \text{Ambient Temperature})$$

$$= 0.12750\text{E-}04 \cdot 1850(200 - 21.1)$$

$$= 4.219 \text{ mm.}$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	850.0001	mm.
Baseplate Thickness	Bpthk	24.0000	mm.
Baseplate Width	Bpwid	254.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm.
Web Thickness	Webtk	10.0000	mm.
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	189.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.600	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(393 \cdot 71)} + 480)71^3) / 12$$

$$= 2208.862 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 2209 + 526(80.3 - 35.5)^2$$

$$= 12763.956 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	740.6	71.0	35.5	525.8	1866645.6	0.128E+05
Wearplate	480.0	10.0	76.0	48.0	364800.0	12.9
Web	10.0	302.0	232.0	30.2	700639.8	0.924E+04
BasePlate	254.0	24.0	395.0	61.0	2407919.8	0.604E+05



Totals	...	...	...	665.0	5340005.0	0.824E+05
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Distance to Centroid [C1]:

$$= AY / A$$

$$= 5340005/665$$

$$= 80.304 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 29068$$

$$= 5915.8984 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 4.1690 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 113.3992 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 800 - 393 \cdot \sin(120/2) / 1.05$$

$$= 474.992 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 2810.0576 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 2.6849 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 125.9991 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{(3(29068 + 147) \cdot 254) / (4 \cdot 850 \cdot 126)}$$

$$= 22.574 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (390 + 74 + 10)$$

$$= 820.992 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 821 / (3 - 1)$$

$$= 410.496 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 410 \cdot 254 / 2$$

$$= 521.330 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 29214 / (850 \cdot 254)$$

$$= 13.531 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 521 \cdot 13.5$$

$$= 7054.328 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 19 + 20.5$$



$$= 39.525 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 7054/39.5$$

$$= 17.503 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	200.0	...	20.0	...	667.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 200 - 10$$

$$= 190.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/39.5$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200/2$$

$$= 100.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(667/39.5)}$$

$$= 41.070 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 800 - \cos(120/2) (390 + 74 + 10) - 24$$

$$= 539.000 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008/189)}$$

$$= 144.511$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 539/41.1$$

$$= 13.124$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 2157 / (2 \cdot 850) \cdot 410 \cdot 539/2$$

$$= 140.357 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($13.1 < 145$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (13.1)^2 / (2 \cdot 145^2)) 189 /$$

$$(5/3 + 3 \cdot (13.1) / (8 \cdot 145) - (13.1^3) / (8 \cdot 145^3))$$

$$= 110.7 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 17.5/111 + (140 \cdot 100/6666666) / 126$$

$$= 0.322$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
--	---	---	---	---	----	----



Rib	10.0	190.0	0.0	19.0	0.0	667.
Web	410.5	10.0	0.0	41.0	0.0	3.42
Totals	...	...	...	60.0	...	670.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/60$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200/2$$

$$= 100.000 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 800 - \cos(474/2)(24 + 0 + 0) - 0$$

$$= 302.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(670/60)}$$

$$= 33.403 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 302/33.4$$

$$= 9.041$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 2157 / (2 \cdot 850)$$

$$= 1.269 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.27 \cdot 410 \cdot 302$$

$$= 157.283 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 157/67000$$

$$= 23.021 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($9.04 < 145$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.04)^2 / (2 \cdot 145^2)) 189 /$$

$$(5/3 + 3 \cdot (9.04) / (8 \cdot 145) - (9.04^3) / (8 \cdot 145^3))$$

$$= 111.6 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 22/112 + (157 \cdot 100/670) / 126$$

$$= 0.380$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	149.95 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
User Defined Area in a Single Bolt	UserBoltArea	2.4000 cm ²
BasePlate Allowable Stress	S	118.00 N./mm ²
Area Available in a Single Bolt	BltArea	1.6469 cm ²
Saddle Load Q0 (Weight)	Q0	26061.9 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	932.7 Kg

Maximum Transverse Force	Ft	1078.4 Kgf
Maximum Longitudinal Force	F1	2156.8 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [rb,l]:
= $F1 / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $2157 / (1.65 \cdot 4)$
= 32.1 N./mm². Must be less than 99.9 N./mm².

Shear Stress in a Single Bolt, Transverse Direction [rb,t]:
= $Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $1078 / (1.65 \cdot 4)$
= 16.1 N./mm². Must be less than 99.9 N./mm².

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F1 / (\text{BoltShearAllowable} \cdot \text{Nbols})$
= $2157 / (99.9 \cdot 4)$
= 0.5291 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot Ft + \text{Sum of X Moments}$
= $800 \cdot 1078 + 0$
= 862.73 Kg-m.

Eccentricity (e):
= $Rmom / QO$
= $863 / 26062$
= 33.10 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
= $\max[0, 0.53, 0]$
= 0.5291 cm²

MDMT Calculations for: Left Saddle, Saddle to Wearplate Junction, Curve: C

Govrn. thk, tg = 10, c = 3 mm., E* = 1
Thickness Ratio = $tr \cdot E* / (tg - c) = 1$, Temp. Reduction = 0 °C
Attachment governing, Attachment stress = Shell stress per Code

Min Metal Temp. w/o impact per UCS-66, Curve C	-44 °C
Min Metal Temp per UCS-66 and UCS-68(c),PWHT credit	-48 °C

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	428.50	mm.
Shell Thickness used in this Case	t	74.000	mm.
Stiffened Vessel Length per 4.15.6	L	3900.80	mm.
Distance from Saddle to Vessel tangent	a1 or a	800.00	mm.
Saddle Width	b1 or b	200.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	480.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm.
Wear Plate Allowable Stress	fw or Sr	118.00	N./mm ²

Inside or Mean Depth of Head	Hi or hm	230.50	mm.
Shell Allowable Stress used in Calculation		118.00	N./mm ²
Head Allowable Stress used in Calculation		118.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	800.00	mm.
Coefficient of Friction	mu	0.06	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Operating Case		6721.36	Kgf
Pressure used in Analysis	P	191.002	bars

Horizontal Vessel Analysis Results:		Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan		57.45	118.00
Long. Stress at Bottom of Midspan		57.83	118.00
Long. Stress at Top of Saddles		60.58	118.00
Long. Stress at Bottom of Saddles		56.01	118.00
Tangential Shear in Shell		1.39	94.40
Circ. Stress at Horn of Saddle		1.23	147.50
Circ. Compressive Stress in Shell		0.13	118.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (605/2 + 0) \cdot 800/821$
 $= 883.8\ Kg$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(Fl,wind; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(173, 0) \cdot 800/1850$
 $= 75.0\ Kg$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(Fl,eq; Sum\ of\ X\ Forces) \cdot B / Ls$
 $= \max(2157, 0) \cdot 800/1850$
 $= 932.7\ Kg$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= Ftr(Ft/Num\ of\ Saddles + Z\ Force\ Load) \cdot B / E$
 $= 3 (2157/2 + 0) \cdot 800/821$
 $= 3152.5\ Kg$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= Saddle\ Load + \max(Fwl, Fwt, Fsl, Fst)$
 $= 3569 + \max(75, 884, 933, 3152)$
 $= 6721.4\ Kg$

Longitudinal Wind Force [Fl]:
 $= WindScalar \cdot WindPress(Platform\ Area + (\pi/4(OD \cdot WindDiaMult)^2))$
 $= 1.1 \cdot 1587 (0 + \pi/4(0.93 \cdot 1.2)^2)$
 $= 1700.463\ N$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	6868.01	Kgf
Transverse Shear Load	1078.39	Kgf
Longitudinal Shear Load	2156.78	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 $= \min(2(ShellID/2 + t + WearPadThickness) \sin(\theta/2), 2 \cdot Rm)$



$$= \min(2(780/2 + 74 + 10) \sin(120/2), 2 \cdot 428)$$

$$= 820.992 \text{ mm.}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L)) / (1 + (4hm^2)/(3L))]$$

$$= -6721 \cdot 800 [1 - (1 - 800/3901 + (428^2 - 230^2) / (2 \cdot 800 \cdot 3901)) / (1 + (4 \cdot 230)/(3 \cdot 3901))]$$

$$= -1310.8 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 ((1 + 2(Rm^2 - hm^2)/(L^2)) / (1 + (4hm^2)/(3L)) - 4a/L)$$

$$= 6721 \cdot 3901 / 4 ((1 + 2(428^2 - 230^2)/(3901^2)) / (1 + (4 \cdot 230)/(3 \cdot 3901)) - 4 \cdot 800/3901)$$

$$= 803.1 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot Rm / (2t) - M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 191 \cdot 428 / (2 \cdot 71) - 803 / (\pi \cdot 428^2 \cdot 71)$$

$$= 57.45 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot Rm / (2t) + M2 / (\pi \cdot Rm^2 \cdot t)$$

$$= 191 \cdot 428 / (2 \cdot 71) + 803 / (\pi \cdot 428^2 \cdot 71)$$

$$= 57.83 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot Rm / (2t) - M1 / (K1 \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 191 \cdot 428 / (2 \cdot 71) - 1311 / (0.11 \cdot \pi \cdot 428^2 \cdot 71)$$

$$= 60.58 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot Rm / (2t) + M1 / (K1^* \cdot \pi \cdot Rm^2 \cdot t)$$

$$= 191 \cdot 428 / (2 \cdot 71) + 1311 / (0.19 \cdot \pi \cdot 428^2 \cdot 71)$$

$$= 56.01 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot hm/3))$$

$$= 6721 (3901 - 2 \cdot 800) / (3901 + (4 \cdot 230/3))$$

$$= 3674.9 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (Rm \cdot t)$$

$$= 1.17 \cdot 3675 / (428 \cdot 71)$$

$$= 1.39 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(Rm \cdot t)}$$

$$= 0.78 \cdot \sqrt{(428 \cdot 71)}$$

$$= 136.050 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(200 + 1.56 \cdot \sqrt{(428 \cdot 71)}, 2 \cdot 800)$$

$$= 472.10 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(Sr/S, 1)$$

$$= \min(118/118, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 6721 \cdot 0.1 / (472(71 + 1 \cdot 10))$$

$$= -0.13 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -6721 / (4(71 + 1 \cdot 10)472) - 3 \cdot 0.053 \cdot 6721 / (2(71 + 1 \cdot 10)^2)$$

$$= -1.23 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	850.0001	mm.
Baseplate Thickness	Bpthk	24.0000	mm.
Baseplate Width	Bpwid	254.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm.
Web Thickness	Webtk	10.0000	mm.
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	189.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.060	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(393 \cdot 71)} + 480)71^3) / 12$$

$$= 2208.862 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 2209 + 526(80.3 - 35.5)^2$$

$$= 12763.956 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	740.6	71.0	35.5	525.8	1866645.6	0.128E+05
Wearplate	480.0	10.0	76.0	48.0	364800.0	12.9
Web	10.0	302.0	232.0	30.2	700639.8	0.924E+04
BasePlate	254.0	24.0	395.0	61.0	2407919.8	0.604E+05
Totals	...	...	...	665.0	5340005.0	0.824E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 5340005 / 665$$

$$= 80.304 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 6721$$

$$= 1367.9420 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.9640 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 113.3992 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2)/(\theta/2 \text{ in radians})$$

$$= 800 - 393 \cdot \sin(120/2)/1.05$$

$$= 474.992 \text{ mm.}$$

$$\text{Bending Moment, } M = F_h \cdot d = 649.7738 \text{ Kg-m.}$$

$$\text{Bending Stress, } S_b = (M \cdot C_1 / I) = 0.6208 \text{ N./mm}^2$$

$$\text{Allowed Stress, } S_a = 2/3 \cdot \text{Yield Str} = 125.9991 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{\frac{3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}}{4 \cdot \text{BasePlateLength} \cdot \text{AllStress}}}$$

$$= \sqrt{\frac{3(6721 + 147) \cdot 254}{4 \cdot 850 \cdot 126}}$$

$$= 10.945 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (390 + 74 + 10)$$

$$= 820.992 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 821 / (3 - 1)$$

$$= 410.496 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 410 \cdot 254 / 2$$

$$= 521.330 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 6868 / (850 \cdot 254)$$

$$= 3.181 \text{ Kg/cm}^2$$

Axial Load [P]:

$$= A_p \cdot B_p$$

$$= 521 \cdot 3.18$$

$$= 1658.406 \text{ Kg}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 19 + 20.5$$

$$= 39.525 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P / A_r$$

$$= 1658 / 39.5$$

$$= 4.115 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	200.0	...	20.0	...	667.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 200 - 10$$

$$= 190.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 39.5$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:



$$= \text{Saddle Width} / 2$$

$$= 200/2$$

$$= 100.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(667/39.5)}$$

$$= 41.070 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 800 - \cos(120/2) (390 + 74 + 10) - 24$$

$$= 539.000 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008/189)}$$

$$= 144.511$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 539/41.1$$

$$= 13.124$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 2157 / (2 \cdot 850) \cdot 410 \cdot 539/2$$

$$= 140.357 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($13.1 < 145$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (13.1)^2 / (2 \cdot 145^2)) 189 / (5/3 + 3 \cdot (13.1) / (8 \cdot 145) - (13.1)^3 / (8 \cdot 145^3))$$

$$= 110.7 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 4.11/111 + (140 \cdot 100/6666666) / 126$$

$$= 0.201$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	190.0	0.0	19.0	0.0	667.
Web	410.5	10.0	0.0	41.0	0.0	3.42
Totals	...	...	...	60.0	...	670.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/60$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200/2$$

$$= 100.000 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 800 - \cos(474/2) (24 + 0 + 0) - 0$$

$$= 302.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(670/60)}$$

$$= 33.403 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 302/33.4$$

$$= 9.041$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 2157/(2 \cdot 850)$$

$$= 1.269 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.27 \cdot 410 \cdot 302$$

$$= 157.283 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 157/67000$$

$$= 23.021 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($9.04 < 145$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.04)^2 / (2 \cdot 145^2)) 189 / (5/3 + 3 \cdot (9.04) / (8 \cdot 145) - (9.04^3) / (8 \cdot 145^3))$$

$$= 111.6 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 5.17/112 + (157 \cdot 100/670) / 126$$

$$= 0.229$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Design Temperature		21.1 °C
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	149.95 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
User Defined Area in a Single Bolt	UserBoltArea	2.4000 cm ²
BasePlate Allowable Stress	S	118.00 N./mm ²
Area Available in a Single Bolt	BlArea	1.6469 cm ²
Saddle Load QO (Weight)	QO	3715.6 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	932.7 Kg
Maximum Transverse Force	Ft	1078.4 Kg
Maximum Longitudinal Force	Fl	2156.8 Kg
Saddle Bolted to Steel Foundation	No	

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 2157/(1.65 \cdot 4)$$

$$= 32.1 \text{ N./mm}^2. \text{ Must be less than } 99.9 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 1078/(1.65 \cdot 4)$$

$$= 16.1 \text{ N./mm}^2. \text{ Must be less than } 99.9 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} QO > QL \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= FI / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 2157/(99.9 \cdot 4)$$

$$= 0.5291 \text{ cm}^2$$

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

$$\begin{aligned} &= B \cdot Ft + \text{Sum of X Moments} \\ &= 800 \cdot 1078 + 0 \\ &= 862.73 \text{ Kg-m.} \end{aligned}$$

Eccentricity (e):

$$\begin{aligned} &= Rmom / QO \\ &= 863/3716 \\ &= 232.19 \text{ mm.} > Bplen/6 \rightarrow \text{Uplift in Transverse direction} \end{aligned}$$

f = Bplen / 2 - Edgedis

$$\begin{aligned} &= 850/2 - 50.8 \\ &= 374.20 \text{ mm.} \end{aligned}$$

Modular Ratio Of Steel/Concrete (n1):

$$\begin{aligned} &= ES / EC \\ &= 203402/21526 \\ &= 9.45 \end{aligned}$$

K1 = 3 (e - 0.5 • Bplen)

$$\begin{aligned} &= 3 (232 - 0.5 \cdot 850) \\ &= -578.43 \text{ mm.} \end{aligned}$$

K2 = 6 • n1 • At / Bpwid • (f + e)

$$\begin{aligned} &= 6 \cdot 9.45 \cdot 3.29/254 \cdot (374 + 232) \\ &= 44580.46 \text{ mm.}^2 \end{aligned}$$

K3 = -K2 • (0.5 • Bplen + f)

$$\begin{aligned} &= -44580 \cdot (0.5 \cdot 850 + 374) \\ &= -35628707.95 \text{ mm.}^3 \end{aligned}$$

Iteratively Solving for the Effective Bearing Length:

$$\begin{aligned} Y^3 + K1 \cdot Y^2 + K2 \cdot Y + K3 &= 0 \\ Y^3 + -578 \cdot Y^2 + 44580 \cdot Y + -35628708 &= 0 \\ Y &= 602.57 \text{ mm.} \end{aligned}$$

Num = (Bplen / 2 - Y / 3 - e)

$$\begin{aligned} &= (850/2 - 603/3 - 232) \\ &= -8.05 \end{aligned}$$

Denom = (Bplen / 2 - Y / 3 + f)

$$\begin{aligned} &= (850/2 - 603/3 + 374) \\ &= 598.34 \end{aligned}$$

Total Bolt Tension Force [Tforce]:

$$\begin{aligned} &= - QO \cdot Num / Denom \\ &= - 3716 \cdot -8.05/598 \\ &= 49.97 \text{ Kg} \end{aligned}$$

Bolt Area Required due to Transverse Load [Bltareart]:

$$\begin{aligned} &= Tforce / (Stba \cdot Nbt) \\ &= 50 / (150 \cdot 2) \\ &= 0.0163 \text{ cm}^2 \end{aligned}$$

Required Area of a Single Bolt [Bltarear]:

$$\begin{aligned} &= \max[Bltarearl, Bltarears, Bltareart] \\ &= \max[0, 0.53, 0.016] \\ &= 0.5291 \text{ cm}^2 \end{aligned}$$

Baseplate Thickness Calculation per D. Moss:

Bearing Pressure (fc)

$$\begin{aligned} &= 2(QO + Tforce) / (Y \cdot Bpwid) \\ &= 2(3716 + 50) / (603 \cdot 254) \\ &= 4.83 \text{ bars} \end{aligned}$$

Distance from Baseplate Edge to the Web [ADIST]:

$$\begin{aligned} &= (Bplen - WebIngh) / 2 \\ &= (850 - 799) / 2 \end{aligned}$$


$$= 25.4000 \text{ mm.}$$

Overturning Moment due To Bolt Tension [Mt]:

$$= T_{\text{force}} \cdot A_{\text{dist}}$$
$$= 50 \cdot 25.4$$
$$= 1.27 \text{ Kg-m.}$$

Equivalent Bearing Pressure (f1):

$$= fc \cdot (Y - Adist) / Y$$
$$= 4.83 \cdot (603 - 25.4) / 603$$
$$= 4.62 \text{ bars}$$

Overturning Moment due to Bearing Pressure [Mc]:

$$= (Adist^2 \cdot Bpwid / 6) \cdot (f1 + 2 \cdot fc)$$
$$= (25.4^2 \cdot 254/6) \cdot (4.62 + 2 \cdot 4.83)$$
$$= 3.98 \text{ Kg-m.}$$

Baseplate Required Thickness [Treq]:

$$= \sqrt{(6 \cdot \max(M_t, M_c) / (B_{pwid} \cdot S_{ba}))}$$
$$= \sqrt{(6 \cdot \max(1.27, 3.98)) / (254 \cdot 177)}$$
$$= 2.2809 \text{ mm.}$$

MDMT Calculations for: Sdl 2 Fr40, Saddle to Wearplate Junction, Curve: C

Govrn. thk, $t_g = 10$, $c = 3$ mm., $E^* = 1$

Thickness Ratio = $tr * E^* / (tg - c) = 1$, Temp. Reduction = 0 °C

Attachment governing, Attachment stress = Shell stress per Code

-44 °C

-48 °C

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : Test Case

Input and Calculated Values:

Vessel Mean Radius	Rm	427.00	mm.
Shell Thickness used in this Case	t	74.000	mm.
Stiffened Vessel Length per 4.15.6	L	3900.80	mm.
Distance from Saddle to Vessel tangent	a1 or a	800.00	mm.
Saddle Width	b1 or b	200.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	480.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm.
Wear Plate Allowable Stress	fw or Sr	118.00	N./mm ²
Shell Allowable Stress used in Calculation		198.00	N./mm ²
Head Allowable Stress used in Calculation		198.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	800.00	mm.
Coefficient of Friction	mu	0.60	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		2405.02	Kgf
Pressure used in Analysis	P	261.191	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	75.25	198.00
Long. Stress at Bottom of Midspan	75.47	198.00
Long. Stress at Top of Saddles	76.10	198.00
Long. Stress at Bottom of Saddles	74.95	198.00
Tangential Shear in Shell	0.52	117.26
Circ. Stress at Horn of Saddle	0.41	296.99
Circ. Compressive Stress in Shell	0.04	198.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr} (F_t / \text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (200/2 + 0) \cdot 800/821$
 $= 291.6 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l, \text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(57.2, 0) \cdot 800/1850$
 $= 24.7 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 2113 + \max(24.7, 292, 0, 0)$
 $= 2405.0 \text{ Kgf}$

Longitudinal Wind Force [FI]:
 $= \text{WindScalar} \cdot \text{WindPress} (\text{Platform Area} + (\pi/4 (OD \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1587 (0 + \pi/4 (0.93 \cdot 1.2)^2)$
 $= 1700.463 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	2551.67	Kgf
Transverse Shear Load	99.76	Kgf
Longitudinal Shear Load	57.22	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:

$$= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$$

$$= \min(2(780/2 + 74 + 10)\sin(120/2), 2 \cdot 427)$$

$$= 820.992 \text{ mm.}$$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $R_m/2$.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a [1 - (1 - a/L + (R_m^2 - h_m^2)/(2a \cdot L))/(1 + (4h_m)/3L)]$$

$$= -2405 \cdot 800 [1 - (1 - 800/3901 + (427^2 - 0^2)/(2 \cdot 800 \cdot 3901))/(1 + (4 \cdot 0)/(3 \cdot 3901))]$$

$$= -338.4 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L/4 [(1 + 2(R_m^2 - h_m^2)/(L^2))/(1 + (4h_m^2)/(3L)) - 4a/L]$$

$$= 2405 \cdot 3901/4 [(1 + 2(427^2 - 0^2)/(3901^2))/(1 + (4 \cdot 0)/(3 \cdot 3901)) - 4 \cdot 800/3901]$$

$$= 477.6 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m/(2t) - M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 261 \cdot 427/(2 \cdot 74) - 478/(\pi \cdot 427^2 \cdot 74)$$

$$= 75.25 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m/(2t) + M2/(\pi \cdot R_m^2 \cdot t)$$

$$= 261 \cdot 427/(2 \cdot 74) + 478/(\pi \cdot 427^2 \cdot 74)$$

$$= 75.47 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_{*3}]:

$$= P \cdot R_m/(2t) - M1/(K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 261 \cdot 427/(2 \cdot 74) - 338/(0.11 \cdot \pi \cdot 427^2 \cdot 74)$$

$$= 76.10 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_{*4}]:

$$= P \cdot R_m/(2t) + M1/(K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 261 \cdot 427/(2 \cdot 74) + 338/(0.19 \cdot \pi \cdot 427^2 \cdot 74)$$

$$= 74.95 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a)/(L + (4 \cdot h_m/3))$$

$$= 2405(3901 - 2 \cdot 800)/(3901 + (4 \cdot 0/3))$$

$$= 1418.5 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 1419 / (427 \cdot 74)$$

$$= 0.52 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(427 \cdot 74)}$$

$$= 138.651 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(Rm \cdot t)}, 2a)$$

$$= \min(200 + 1.56 \cdot \sqrt{(427 \cdot 74)}, 2 \cdot 800)$$

$$= 477.30 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [eta]:
 = 1.0000 Materials are the same, test case

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 2405 \cdot 0.1 / (477 (74 + 1 \cdot 10))$$

$$= -0.04 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8Rm$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr) b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -2405 / (4(74 + 1 \cdot 10) 477) -$$

$$3 \cdot 0.053 \cdot 2405 / (2(74 + 1 \cdot 10)^2)$$

$$= -0.41 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	850.0001	mm.
Baseplate Thickness	Bpthk	24.0000	mm.
Baseplate Width	Bpwid	254.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm.
Web Thickness	Webtk	10.0000	mm.
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	220.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.600	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth}) t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(390 \cdot 74)} + 480) 74^3) / 12$$

$$= 2515.819 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}} (C1 - Y)^2$$

$$= 2516 + 551 (80.5 - 37)^2$$

$$= 12924.834 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	745.0	74.0	37.0	551.3	2039855.0	0.129E+05
Wearplate	480.0	10.0	79.0	48.0	379200.0	5.01
Web	10.0	302.0	235.0	30.2	709699.8	0.951E+04
BasePlate	254.0	24.0	398.0	61.0	2426207.8	0.615E+05
Totals	...	...	...	690.5	5554962.0	0.839E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 5554962 / 690$$

$$= 80.452 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 2405$$

$$= 489.4736 \text{ Kgf}$$

$$\begin{aligned} \text{Tension Stress, } St &= (Fh/As) = 0.3449 \text{ N./mm}^2 \\ \text{Allowed Stress, } Sa &= 0.6 * \text{Yield Str} = 131.9976 \text{ N./mm}^2 \end{aligned}$$

Saddle Splitting Dimension [d]:
 $= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$
 $= 800 - 390 \cdot \sin(120/2) / 1.05$
 $= 477.473 \text{ mm.}$

$$\text{Bending Moment, } M = Fh * d = 233.7149 \text{ Kg-m.}$$

$$\begin{aligned} \text{Bending Stress, } Sb &= (M * C1 / I) = 0.2197 \text{ N./mm}^2 \\ \text{Allowed Stress, } Sa &= 2/3 * \text{Yield Str} = 146.6640 \text{ N./mm}^2 \end{aligned}$$

Minimum Thickness of Baseplate per Moss:

$$\begin{aligned} &= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 * \text{BasePlateLength} * \text{AllStress})} \\ &= \sqrt{3(2405 + 147)254 / (4 * 850 * 147)} \\ &= 6.184 \text{ mm.} \end{aligned}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 $= 2 * \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$
 $= 2 * \cos(90 - 120/2) (390 + 74 + 10)$
 $= 820.992 \text{ mm.}$

Distance between Ribs [e]:
 $= \text{Web Length} / (\text{Nr ribs} - 1)$
 $= 821 / (3 - 1)$
 $= 410.496 \text{ mm.}$

Baseplate Pressure Area [Ap]:
 $= e * \text{Bpwid} / 2$
 $= 410 * 254 / 2$
 $= 521.330 \text{ cm}^2$

Bearing Pressure [Bp]:
 $= (Q + \text{SaddleWt}) / (\text{BasePlateLength} * \text{BasePlateWidth})$
 $= 2552 / (850 * 254)$
 $= 1.182 \text{ Kg/cm}^2$

Axial Load [P]:
 $= Ap * Bp$
 $= 521 * 1.18$
 $= 616.147 \text{ Kg}$

Area of the Rib and Web [Ar]:
 $= \text{Rib Area} + \text{Web Area}$
 $= 19 + 20.5$
 $= 39.525 \text{ cm}^2$

Compressive Stress [Sc]:
 $= P / Ar$
 $= 616 / 39.5$
 $= 1.529 \text{ N./mm}^2$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	200.0	...	20.0	...	667.

Rib dimension [D]:
 $= \text{Saddle Width} - \text{Web Thickness}$
 $= 200 - 10$
 $= 190.000 \text{ mm.}$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/39.5$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200/2$$

$$= 100.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(667/39.5)}$$

$$= 41.070 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\text{theta}/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 800 - \cos(120/2) (390 + 74 + 10) - 24$$

$$= 539.000 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008/220)}$$

$$= 133.944$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 539/41.1$$

$$= 13.124$$

Bending Moment [Rm]:

$$= FI / (2 \cdot \text{Bplen}) \cdot e \cdot L / 2$$

$$= 57.2 / (2 \cdot 850) \cdot 410 \cdot 539/2$$

$$= 3.724 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < Cc$ ($13.1 < 134$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$$

$$= (1 - (13.1)^2 / (2 \cdot 134^2)) 220 / (5/3 + 3 \cdot (13.1) / (8 \cdot 134) - (13.1^3) / (8 \cdot 134^3))$$

$$= 128.5 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= Sc/Sca + (Rm \cdot C1 / I) / Sba$$

$$= 1.53/129 + (3.72 \cdot 100/6666666) / 147$$

$$= 0.016$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	190.0	0.0	19.0	0.0	667.
Web	410.5	10.0	0.0	41.0	0.0	3.42
Totals	...	...	...	60.0	...	670.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/60$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200/2$$

$$= 100.000 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 800 - \cos(474/2) (24 + 0 + 0) - 0$$

$$= 302.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{\text{Total Inertia} / \text{Total Area}}$$

$$= \sqrt{670/60}$$

$$= 33.403 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 302/33.4$$

$$= 9.041$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 57.2 / (2 \cdot 850)$$

$$= 0.034 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.034 \cdot 410 \cdot 302$$

$$= 4.173 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 4.17/67000$$

$$= 0.611 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($9.04 < 134$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.04)^2 / (2 \cdot 134^2)) 220 /$$

$$(5/3 + 3 \cdot (9.04) / (8 \cdot 134) - (9.04^3) / (8 \cdot 134^3))$$

$$= 129.7 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (M_{base,c} \cdot C_1 / I) / S_{ba}$$

$$= 1.92/130 + (4.17 \cdot 100/670) / 147$$

$$= 0.019$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
User Defined Area in a Single Bolt	UserBoltArea	2.4000 cm ²
BasePlate Allowable Stress	S	118.00 N./mm ²
Area Available in a Single Bolt	BltArea	1.6469 cm ²
Saddle Load Q0 (Weight)	Q0	2260.0 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	24.7 Kg
Maximum Transverse Force	Ft	99.8 Kg
Maximum Longitudinal Force	Fl	57.2 Kg
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [τ_b ,l]:

$$= FI / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 57.2 / (1.65 \cdot 4)$$

$$= 0.9 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Shear Stress in a Single Bolt, Transverse Direction [τ_b ,t]:

$$= Ft / (\text{Bolt Area} \cdot \text{Number of Bolts})$$

$$= 99.8 / (1.65 \cdot 4)$$

$$= 1.5 \text{ N./mm}^2. \text{ Must be less than } 103.2 \text{ N./mm}^2.$$

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:

$$= 0.0 \text{ (} Q_0 > Q_L \text{ --> No Uplift in Longitudinal direction)}$$

Bolt Area due to Shear Load [Bltarears]:

$$= FI / (\text{BoltShearAllowable} \cdot \text{Nbolts})$$

$$= 57.2 / (103 \cdot 4)$$

= 0.0136 cm²

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:

= B • Ft + Sum of X Moments

= 800 • 99.8 + 0

= 79.81 Kg-m.

Eccentricity (e):

= Rmom / QO

= 79.8/2260

= 35.31 mm. < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:

= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:

= max[Bltarearl, Bltarears, Bltareart]

= max[0, 0.014, 0]

= 0.0136 cm²

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	427.00	mm.
Shell Thickness used in this Case	t	74.000	mm.
Stiffened Vessel Length per 4.15.6	L	3900.80	mm.
Distance from Saddle to Vessel tangent	a1 or a	800.00	mm.
Saddle Width	b1 or b	200.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	480.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm.
Wear Plate Allowable Stress	fw or Sr	118.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	229.00	mm.
Shell Allowable Stress used in Calculation		198.00	N./mm ²
Head Allowable Stress used in Calculation		198.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	800.00	mm.
Coefficient of Friction	mu	0.06	

Results for Horizontal Vessel Analysis:

Saddle Force Q, Test Case, no Ext. Forces		2401.56	Kgf
Pressure used in Analysis	P	261.191	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	75.29	198.00
Long. Stress at Bottom of Midspan	75.43	198.00
Long. Stress at Top of Saddles	76.38	198.00
Long. Stress at Bottom of Saddles	74.80	198.00
Tangential Shear in Shell	0.48	117.26
Circ. Stress at Horn of Saddle	0.41	296.99
Circ. Compressive Stress in Shell	0.04	198.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= \text{Ftr}(\text{Ft/Num of Saddles} + \text{Z Force Load}) \cdot B / E$
 $= 3 (200/2 + 0) \cdot 800/821$
 $= 291.6 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(\text{Fl,wind}; \text{Sum of X Forces}) \cdot B / Ls$
 $= \max(57.2, 0) \cdot 800/1850$
 $= 24.7 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(\text{Fwl}, \text{Fwt}, \text{Fsl}, \text{Fst})$
 $= 2110 + \max(24.7, 292, 0, 0)$
 $= 2401.6 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1587 (0 + \pi/4(0.93 \cdot 1.2)^2)$
 $= 1700.463 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	2548.21	Kgf
Transverse Shear Load	99.76	Kgf
Longitudinal Shear Load	57.22	Kgf

Note: Wear Plate is Welded to the Shell, $k = 0.1$

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot Rm)$
 $= \min(2(780/2 + 74 + 10)\sin(120/2), 2 \cdot 427)$
 $= 820.992 \text{ mm}$

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to $Rm/2$.

Moment per Equation 4.15.1 [M1]:
 $= -Q \cdot a [1 - (1 - a/L + (Rm^2 - hm^2)/(2a \cdot L))/(1 + (4hm^2)/3L)]$
 $= -2402 \cdot 800 [1 - (1 - 800/3901 + (427^2 - 229^2)/(2 \cdot 800 \cdot 3901))/(1 + (4 \cdot 229)/(3 \cdot 3901))]$
 $= -467.8 \text{ Kg-m}$

Moment per Equation 4.15.2 [M2]:
 $= Q \cdot L/4((1 + 2(Rm^2 - hm^2)/(L^2))/(1 + (4hm^2)/(3L)) - 4a/L)$
 $= 2402 \cdot 3901/4((1 + 2(427^2 - 229^2)/(3901^2))/(1 + (4 \cdot 229)/(3 \cdot 3901)) - 4 \cdot 800/3901)$
 $= 287.8 \text{ Kg-m}$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:
 $= P \cdot Rm/(2t) - M2/(\pi \cdot Rm^2 \cdot t)$
 $= 261 \cdot 427/(2 \cdot 74) - 288/(\pi \cdot 427^2 \cdot 74)$
 $= 75.29 \text{ N./mm}^2$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:
 $= P \cdot Rm/(2t) + M2/(\pi \cdot Rm^2 \cdot t)$
 $= 261 \cdot 427/(2 \cdot 74) + 288/(\pi \cdot 427^2 \cdot 74)$
 $= 75.43 \text{ N./mm}^2$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:
 $= P \cdot Rm/(2t) - M1/(K1 \cdot \pi \cdot Rm^2 \cdot t)$
 $= 261 \cdot 427/(2 \cdot 74) - 468/(0.11 \cdot \pi \cdot 427^2 \cdot 74)$
 $= 76.38 \text{ N./mm}^2$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M1 / (K1 \cdot \pi \cdot R_m^2 \cdot t)$$

$$= 261 \cdot 427 / (2 \cdot 74) + 468 / (0.19 \cdot \pi \cdot 427^2 \cdot 74)$$

$$= 74.80 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L-2a) / (L + (4 \cdot h_m / 3))$$

$$= 2402 (3901 - 2 \cdot 800) / (3901 + (4 \cdot 229 / 3))$$

$$= 1313.7 \text{ Kgf}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 1314 / (427 \cdot 74)$$

$$= 0.48 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(427 \cdot 74)}$$

$$= 138.651 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(200 + 1.56 \cdot \sqrt{(427 \cdot 74)}, 2 \cdot 800)$$

$$= 477.30 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= 1.0000 \text{ Materials are the same, test case}$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K5 \cdot Q \cdot k / (B1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 2402 \cdot 0.1 / (477(74 + 1 \cdot 10))$$

$$= -0.04 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b1) - 3 \cdot K7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -2402 / (4(74 + 1 \cdot 10)477) - 3 \cdot 0.053 \cdot 2402 / (2(74 + 1 \cdot 10)^2)$$

$$= -0.41 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	850.0001	mm.
Baseplate Thickness	Bpthk	24.0000	mm.
Baseplate Width	Bpwid	254.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm.
Web Thickness	Webtk	10.0000	mm.
Web Location	Webloc	Center	
Saddle Yield Stress	Sy	220.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.060	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(390 \cdot 74)} + 480)74^3) / 12$$

$$= 2515.819 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C1 - Y)^2$$

$$= 2516 + 551(80.5 - 37)^2$$

$$= 12924.834 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	745.0	74.0	37.0	551.3	2039855.0	0.129E+05
Wearplate	480.0	10.0	79.0	48.0	379200.0	5.01
Web	10.0	302.0	235.0	30.2	709699.8	0.951E+04
BasePlate	254.0	24.0	398.0	61.0	2426207.8	0.615E+05



Totals	...	...	...	690.5	5554962.0	0.839E+05
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Distance to Centroid [C1]:

$$= AY / A$$

$$= 5554962/690$$

$$= 80.452 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle}/2$$

$$= 180 - 120/2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K1 \cdot Q$$

$$= 0.2 \cdot 2402$$

$$= 488.7692 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.3444 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 131.9976 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 800 - 390 \cdot \sin(120/2) / 1.05$$

$$= 477.473 \text{ mm.}$$

$$\text{Bending Moment, } M = Fh \cdot d = 233.3785 \text{ Kg-m.}$$

$$\text{Bending Stress, } Sb = (M \cdot C1 / I) = 0.2194 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 2/3 \cdot \text{Yield Str} = 146.6640 \text{ N./mm}^2$$

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{ 3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}) }$$

$$= \sqrt{ 3(2402 + 147) 254 / (4 \cdot 850 \cdot 147) }$$

$$= 6.180 \text{ mm.}$$

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$$

$$= 2 \cdot \cos(90 - 120/2) (390 + 74 + 10)$$

$$= 820.992 \text{ mm.}$$

Distance between Ribs [e]:

$$= \text{Web Length} / (\text{Nr ribs} - 1)$$

$$= 821 / (3 - 1)$$

$$= 410.496 \text{ mm.}$$

Baseplate Pressure Area [Ap]:

$$= e \cdot \text{Bpwid} / 2$$

$$= 410 \cdot 254 / 2$$

$$= 521.330 \text{ cm}^2$$

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 2548 / (850 \cdot 254)$$

$$= 1.180 \text{ Kgf/cm}^2$$

Axial Load [P]:

$$= Ap \cdot Bp$$

$$= 521 \cdot 1.18$$

$$= 615.312 \text{ Kgf}$$

Area of the Rib and Web [Ar]:

$$= \text{Rib Area} + \text{Web Area}$$

$$= 19 + 20.5$$



$$= 39.525 \text{ cm}^2$$

Compressive Stress [Sc]:

$$= P/A_r$$

$$= 615/39.5$$

$$= 1.527 \text{ N./mm}^2$$

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	200.0	...	20.0	...	667.

Rib dimension [D]:

$$= \text{Saddle Width} - \text{Web Thickness}$$

$$= 200 - 10$$

$$= 190.000 \text{ mm.}$$

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/39.5$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200/2$$

$$= 100.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(667/39.5)}$$

$$= 41.070 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 800 - \cos(120/2) (390 + 74 + 10) - 24$$

$$= 539.000 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{(2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress})}$$

$$= \sqrt{(2 \cdot \pi^2 \cdot 199943008/220)}$$

$$= 133.944$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 539/41.1$$

$$= 13.124$$

Bending Moment [Rm]:

$$= FI / (2 \cdot B_{plen}) \cdot e \cdot L / 2$$

$$= 57.2 / (2 \cdot 850) \cdot 410 \cdot 539/2$$

$$= 3.724 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($13.1 < 134$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (13.1)^2 / (2 \cdot 134^2)) 220 /$$

$$(5/3 + 3 \cdot (13.1) / (8 \cdot 134) - (13.1^3) / (8 \cdot 134^3))$$

$$= 128.5 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c/S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 1.53/129 + (3.72 \cdot 100/6666666) / 147$$

$$= 0.016$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
--	---	---	---	---	----	----

Rib	10.0	190.0	0.0	19.0	0.0	667.
Web	410.5	10.0	0.0	41.0	0.0	3.42
Totals	...	...	...	60.0	...	670.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/60$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200/2$$

$$= 100.000 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 800 - \cos(474/2) (24 + 0 + 0) - 0$$

$$= 302.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(670/60)}$$

$$= 33.403 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 302/33.4$$

$$= 9.041$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 57.2/(2 \cdot 850)$$

$$= 0.034 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 0.034 \cdot 410 \cdot 302$$

$$= 4.173 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 4.17/67000$$

$$= 0.611 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($9.04 < 134$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.04)^2 / (2 \cdot 134^2)) 220 /$$

$$(5/3 + 3 \cdot (9.04) / (8 \cdot 134) - (9.04^3) / (8 \cdot 134^3))$$

$$= 129.7 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C1/I) / S_{ba}$$

$$= 1.92/130 + (4.17 \cdot 100/670) / 147$$

$$= 0.019$$

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	4
Total Number of Bolts in Tension/Baseplate	Nbt	2
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	172.00 N./mm ²
Bolt Corrosion Allowance	Bca	1.5 mm.
Distance from Bolts to Edge	Edgedis	50.8 mm.
User Defined Area in a Single Bolt	UserBoltArea	2.4000 cm ²
BasePlate Allowable Stress	S	118.00 N./mm ²
Area Available in a Single Bolt	BltArea	1.6469 cm ²
Saddle Load Q0 (Weight)	Q0	2256.6 Kg
Saddle Load QL (Wind/Seismic contribution)	QL	24.7 Kg
Maximum Transverse Force	Ft	99.8 Kg



Maximum Longitudinal Force	F1	57.2 Kgf
Saddle Bolted to Steel Foundation		No

Shear Stress in a Single Bolt, Longitudinal Direction [tb,l]:
= $F_l / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $57.2 / (1.65 \cdot 4)$
= 0.9 N./mm^2 . Must be less than 103.2 N./mm^2 .

Shear Stress in a Single Bolt, Transverse Direction [tb,t]:
= $F_t / (\text{Bolt Area} \cdot \text{Number of Bolts})$
= $99.8 / (1.65 \cdot 4)$
= 1.5 N./mm^2 . Must be less than 103.2 N./mm^2 .

Bolt Area Calculation per Dennis R. Moss

Bolt Area Requirement Due to Longitudinal Load [Bltarearl]:
= 0.0 (QO > QL --> No Uplift in Longitudinal direction)

Bolt Area due to Shear Load [Bltarears]:
= $F_l / (\text{BoltShearAllowable} \cdot \text{Nbols})$
= $57.2 / (103 \cdot 4)$
= 0.0136 cm^2

Bolt Area due to Transverse Load:

Moment on Baseplate Due to Transverse Load [Rmom]:
= $B \cdot F_t + \text{Sum of X Moments}$
= $800 \cdot 99.8 + 0$
= 79.81 Kg-m .

Eccentricity (e):
= R_{mom} / QO
= $79.8 / 2257$
= 35.37 mm . < Bplen/6 --> No Uplift in Transverse direction

Bolt Area due to Transverse Load [Bltareart]:
= 0 (No Uplift)

Required Area of a Single Bolt [Bltarear]:
= $\max[\text{Bltarearl}, \text{Bltarears}, \text{Bltareart}]$
= $\max[0, 0.014, 0]$
= 0.0136 cm^2

Review notes about nozzle loadings and supports in the Warnings report.

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ASME VIII Division 2 Horizontal Vessel Analysis, Left Saddle:

Horizontal Vessel Stress Calculations : External Pressure

Input and Calculated Values:

Vessel Mean Radius	Rm	428.50	mm.
Shell Thickness used in this Case	t	74.000	mm.
Stiffened Vessel Length per 4.15.6	L	3900.80	mm.
Distance from Saddle to Vessel tangent	a1 or a	800.00	mm.
Saddle Width	b1 or b	200.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	480.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm.
Wear Plate Allowable Stress	fw or Sr	118.00	N./mm ²
Shell Allowable Stress used in Calculation		118.00	N./mm ²
Head Allowable Stress used in Calculation		138.00	N./mm ²
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	800.00	mm.
Coefficient of Friction	mu	0.60	

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case		29067.66	Kgf
Pressure used in Analysis	P	-0.000	bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-1.38	-182.10
Long. Stress at Top of Midspan	1.38	118.00
Long. Stress at Bottom of Midspan	1.38	118.00
Long. Stress at Top of Saddles	9.18	118.00
Long. Stress at Bottom of Saddles	-5.09	-182.10
Long. Stress at Bottom of Saddles	5.09	118.00
Tangential Shear in Shell	6.47	94.40
Circ. Stress at Horn of Saddle	5.31	147.50
Circ. Compressive Stress in Shell	0.57	118.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (605/2 + 0) \cdot 800/821$
 $= 883.8 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,\text{wind}}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(173, 0) \cdot 800/1850$
 $= 75.0 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l,\text{eq}}; \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(2157, 0) \cdot 800/1850$
 $= 932.7 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (2157/2 + 0) \cdot 800/821$

= 3152.5 Kgf

Load Combination Results for Q + Wind or Seismic [Q]:

= Saddle Load + max(Fwl, Fwt, Fsl, Fst)
 = 25915 + max(75, 884, 933, 3152)
 = 29067.7 Kgf

Longitudinal Wind Force [FI]:

= WindScalar • WindPress(Platform Area + ($\pi/4$ (OD • WindDiaMult)²))
 = 1.1 • 1587 (0 + $\pi/4$ (0.93 • 1.2)²)
 = 1700.463 N

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight)	29214.31	Kgf
Transverse Shear Load	1078.39	Kgf
Longitudinal Shear Load	2156.78	Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:

= min(2(ShellID/2 + t + WearPadThickness)sin($\theta/2$), 2•Rm)
 = min(2(780/2 + 74 + 10)sin(120/2), 2•428)
 = 820.992 mm.

The Computed K values from Table 4.15.1:

K1 = 0.1066	K2 = 1.1707	K3 = 0.8799	K4 = 0.4011
K5 = 0.7603	K6 = 0.0529	K7 = 0.0529	K8 = 0.3405
K9 = 0.2711	K10 = 0.0581	K1* = 0.1923	K6p = 0.0434
K7p = 0.0434			

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

= -Q•a [1 - (1 - a/L + (Rm²-hm²)/(2a•L))/(1+(4hm)/3L)]
 = -29068•800 [1 - (1 - 800/3901 + (428²-0²)/(2•800•3901))/(1+(4•0)/(3•3901))]
 = -4085.1 Kg-m.

Moment per Equation 4.15.2 [M2]:

= Q•L/4 ((1+2(Rm²-hm²)/(L²))/(1+(4hm²)/(3L)) - 4a/L)
 = 29068•3901/4 ((1+2(428²-0²)/(3901²))/(1+(4•0)/(3•3901)) - 4•800/3901)
 = 5776.9 Kg-m.

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

= P • Rm/(2t) - M2/(π •Rm²t)
 = -0 • 428/(2•71) - 5777/(π •428²•71)
 = -1.38 N./mm²

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

= P • Rm/(2t) + M2/(π • Rm² • t)
 = -0 • 428/(2 • 71) + 5777/(π • 428² • 71)
 = 1.38 N./mm²

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ_3]:

= P • Rm/(2t) - M1/(K1• π •Rm²t)
 = -0•428/(2•71)--4085/(0.11• π •428²•71)
 = 9.18 N./mm²

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ_4]:

= P • Rm/(2t) + M1/(K1* • π • Rm² • t)
 = -0•428/(2•71)+-4085/(0.19• π •428²•71)
 = -5.09 N./mm²

Maximum Shear Force in the Saddle (4.15.3) [T]:

= Q(L-2a)/(L+(4•hm/3))
 = 29068 (3901 - 2 • 800)/(3901 + (4 • 0/3))
 = 17144.9 Kgf

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 17145 / (428 \cdot 71)$$

$$= 6.47 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(428 \cdot 71)}$$

$$= 136.050 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(200 + 1.56 \cdot \sqrt{(428 \cdot 71)}, 2 \cdot 800)$$

$$= 472.10 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r/S, 1)$$

$$= \min(118/118, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot tr))$$

$$= -0.76 \cdot 29068 \cdot 0.1 / (472(71 + 1 \cdot 10))$$

$$= -0.57 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) [σ_7, r]:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -29068 / (4(71 + 1 \cdot 10)472) -$$

$$3 \cdot 0.053 \cdot 29068 / (2(71 + 1 \cdot 10)^2)$$

$$= -5.31 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	850.0001	mm.
Baseplate Thickness	Bpthk	24.0000	mm.
Baseplate Width	Bpwid	254.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm.
Web Thickness	Webtk	10.0000	mm.
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	189.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.600	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(393 \cdot 71)} + 480)71^3) / 12$$

$$= 2208.862 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 2209 + 526(80.3 - 35.5)^2$$

$$= 12763.956 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	740.6	71.0	35.5	525.8	1866645.6	0.128E+05
Wearplate	480.0	10.0	76.0	48.0	364800.0	12.9
Web	10.0	302.0	232.0	30.2	700639.8	0.924E+04
BasePlate	254.0	24.0	395.0	61.0	2407919.8	0.604E+05
Totals	...	...	...	665.0	5340005.0	0.824E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 5340005/665$$

$$= 80.304 \text{ mm.}$$

Angle [β]:
 = 180 - Saddle Angle/2
 = 180 - 120/2
 = 120.0

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

 = 0.2035

Saddle Splitting Force [Fh]:
 = K1 • Q
 = 0.2 • 29068
 = 5915.8984 Kgf

Tension Stress, St = (Fh/As) = 4.1690 N./mm²
 Allowed Stress, Sa = 0.6 • Yield Str = 113.3992 N./mm²

Saddle Splitting Dimension [d]:
 = B - R • sin(theta/2)/(theta/2 in radians)
 = 800 - 393 • sin(120/2)/1.05
 = 474.992 mm.

Bending Moment, M = Fh • d = 2810.0576 Kg-m.

Bending Stress, Sb = (M • C1 / I) = 2.6849 N./mm²
 Allowed Stress, Sa = 2/3 • Yield Str = 125.9991 N./mm²

Minimum Thickness of Baseplate per Moss:

$$= \sqrt{3(Q + \text{SaddleWt}) \text{BasePlateWidth} / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress})}$$

$$= \sqrt{3(29068 + 147) 254 / (4 \cdot 850 \cdot 126)}$$

 = 22.574 mm.

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:
 = 2 • cos(90 - Saddle Angle/2)(Inside Radius + Shell Thk + Wear Plate Thk)
 = 2 • cos(90 - 120/2)(390 + 74 + 10)
 = 820.992 mm.

Distance between Ribs [e]:
 = Web Length / (Nr ribs - 1)
 = 821/(3 - 1)
 = 410.496 mm.

Baseplate Pressure Area [Ap]:
 = e • Bpwid / 2
 = 410 • 254/2
 = 521.330 cm²

Bearing Pressure [Bp]:

$$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$$

$$= 29214 / (850 \cdot 254)$$

 = 13.531 Kgf/cm²

Axial Load [P]:
 = Ap • Bp
 = 521 • 13.5
 = 7054.328 Kgf

Area of the Rib and Web [Ar]:
 = Rib Area + Web Area
 = 19 + 20.5
 = 39.525 cm²

Compressive Stress [Sc]:
 = P/Ar
 = 7054/39.5
 = 17.503 N./mm²



Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	200.0	...	20.0	...	667.

Rib dimension [D]:
 = Saddle Width - Web Thickness
 = 200 - 10
 = 190.000 mm.

Distance to Centroid from Datum [ytot]:
 = AY / A
 = 0/39.5
 = 0.000 mm.

Distance to Centroid [C1]:
 = Saddle Width / 2
 = 200/2
 = 100.000 mm.

Radius of Gyration [r]:
 = $\sqrt{\text{Total Inertia} / \text{Total Area}}$
 = $\sqrt{667/39.5}$
 = 41.070 mm.

Length of Outer Rib [L]:
 = Saddle Height - cos(theta/2)(radius + shlthk + wpdthk) - bpthk
 = 800 - cos(120/2)(390 + 74 + 10) - 24
 = 539.000 mm.

Intermediate Term [Cc]:
 = $\sqrt{2 \cdot \pi^2 \cdot \text{Elastic Modulus} / \text{Yield Stress}}$
 = $\sqrt{2 \cdot \pi^2 \cdot 199943008/189}$
 = 144.511

Slenderness ratio [KL/r]:
 = KL/r
 = 1 • 539/41.1
 = 13.124

Bending Moment [Rm]:
 = FI / (2 • Bplen) • e • L / 2
 = 2157 / (2 • 850) • 410 • 539/2
 = 140.357 Kg-m.

Compressive Allowable, KL/r < Cc (13.1 < 145) per AISC E2-1 [Sca]:
 = $(1 - (KL/r)^2 / (2 \cdot Cc^2)) Fy / (5/3 + 3 \cdot (KL/r) / (8 \cdot Cc) - (KL/r)^3 / (8 \cdot Cc^3))$
 = $(1 - (13.1)^2 / (2 \cdot 145^2)) 189 / (5/3 + 3 \cdot (13.1) / (8 \cdot 145) - (13.1^3) / (8 \cdot 145^3))$
 = 110.7 N./mm²

AISC Unity Check of Outside Ribs (must be ≤ 1)
 = Sc/Sca + (Rm • C1 / I) / Sba
 = 17.5/111 + (140 • 100/6666666) / 126
 = 0.322

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	190.0	0.0	19.0	0.0	667.
Web	410.5	10.0	0.0	41.0	0.0	3.42
Totals	...	...	...	60.0	...	670.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0/60$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200/2$$

$$= 100.000 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 800 - \cos(474/2)(24 + 0 + 0) - 0$$

$$= 302.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{(\text{Total Inertia} / \text{Total Area})}$$

$$= \sqrt{(670/60)}$$

$$= 33.403 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= KL/r$$

$$= 1 \cdot 302/33.4$$

$$= 9.041$$

Unit Force [Force,u]:

$$= FI / (2 \cdot \text{Baseplate Length})$$

$$= 2157 / (2 \cdot 850)$$

$$= 1.269 \text{ Kg/mm.}$$

Moment at base of inner Rib [Mbase,c]:

$$= \text{Unit Force} \cdot e \cdot L$$

$$= 1.27 \cdot 410 \cdot 302$$

$$= 157.283 \text{ Kg-m.}$$

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:

$$= \text{Bending Moment} / \text{Section Modulus}$$

$$= 157/67000$$

$$= 23.021 \text{ N./mm}^2$$

Compressive Allowable, $KL/r < C_c$ ($9.04 < 145$) per AISC E2-1 [Sca]:

$$= (1 - (KL/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (KL/r) / (8 \cdot C_c) - (KL/r)^3 / (8 \cdot C_c^3))$$

$$= (1 - (9.04)^2 / (2 \cdot 145^2)) 189 / (5/3 + 3 \cdot (9.04) / (8 \cdot 145) - (9.04)^3 / (8 \cdot 145^3))$$

$$= 111.6 \text{ N./mm}^2$$

AISC Unity Check of Inside Ribs (must be ≤ 1)

$$= Sc/Sca + (M_{base,c} \cdot C_1/I) / S_{ba}$$

$$= 22/112 + (157 \cdot 100/670) / 126$$

$$= 0.380$$

ASME VIII Division 2 Horizontal Vessel Analysis, Right Saddle:

Input and Calculated Values:

Vessel Mean Radius	Rm	428.50	mm.
Shell Thickness used in this Case	t	74.000	mm.
Stiffened Vessel Length per 4.15.6	L	3900.80	mm.
Distance from Saddle to Vessel tangent	a1 or a	800.00	mm.
Saddle Width	b1 or b	200.00	mm.
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	480.00	mm.
Wear Plate Bearing Angle	delta2 or theta1	132.00	degrees
Wear Plate Thickness	e2 or tr	10.0	mm.
Wear Plate Allowable Stress	fw or Sr	118.00	N./mm ²
Inside or Mean Depth of Head	Hi or hm	230.50	mm.
Shell Allowable Stress used in Calculation		118.00	N./mm ²

Head Allowable Stress used in Calculation 118.00 N./mm²
 Circumferential Efficiency in Plane of Saddle 1.00
 Circumferential Efficiency at Mid-Span 1.00
 Distance from Saddle Base to Centerline B 800.00 mm.
 Coefficient of Friction mu 0.06

Results for Horizontal Vessel Analysis:

Saddle Force Q, External Pressure Case 6721.36 Kgf
 Pressure used in Analysis P -0.000 bars

Horizontal Vessel Analysis Results:	Actual N./mm ²	Allowable N./mm ²
Long. Stress at Top of Midspan	-0.19	-182.10
Long. Stress at Top of Midspan	0.19	118.00
Long. Stress at Bottom of Midspan	0.19	118.00
Long. Stress at Top of Saddles	2.94	118.00
Long. Stress at Bottom of Saddles	-1.63	-182.10
Long. Stress at Bottom of Saddles	1.63	118.00
Tangential Shear in Shell	1.39	94.40
Circ. Stress at Horn of Saddle	1.23	147.50
Circ. Compressive Stress in Shell	0.13	118.00

Intermediate Results: Saddle Reaction Q due to Wind or Seismic:

Transverse Saddle Reaction Force [Fwt]:
 $= F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (605/2 + 0) \cdot 800/821$
 $= 883.8 \text{ Kgf}$

Longitudinal Saddle Reaction Force [Fwl]:
 $= \max(F_{l,\text{wind}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(173, 0) \cdot 800/1850$
 $= 75.0 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Fl [Fsl]:
 $= \max(F_{l,\text{eq}}, \text{Sum of X Forces}) \cdot B / L_s$
 $= \max(2157, 0) \cdot 800/1850$
 $= 932.7 \text{ Kgf}$

Saddle Reaction Force due to Earthquake Ft [Fst]:
 $= F_{tr}(F_t/\text{Num of Saddles} + Z \text{ Force Load}) \cdot B / E$
 $= 3 (2157/2 + 0) \cdot 800/821$
 $= 3152.5 \text{ Kgf}$

Load Combination Results for Q + Wind or Seismic [Q]:
 $= \text{Saddle Load} + \max(F_{wl}, F_{wt}, F_{sl}, F_{st})$
 $= 3569 + \max(75, 884, 933, 3152)$
 $= 6721.4 \text{ Kgf}$

Longitudinal Wind Force [Fl]:
 $= \text{WindScalar} \cdot \text{WindPress}(\text{Platform Area} + (\pi/4(\text{OD} \cdot \text{WindDiaMult})^2))$
 $= 1.1 \cdot 1587 (0 + \pi/4(0.93 \cdot 1.2)^2)$
 $= 1700.463 \text{ N}$

Summary of Loads at the Base of this Saddle:

Vertical Load (including saddle weight) 6868.01 Kgf
 Transverse Shear Load 1078.39 Kgf
 Longitudinal Shear Load 2156.78 Kgf

Note: Wear Plate is Welded to the Shell, k = 0.1

Saddle Dimension [E]:
 $= \min(2(\text{ShellID}/2 + t + \text{WearPadThickness})\sin(\theta/2), 2 \cdot R_m)$
 $= \min(2(780/2 + 74 + 10)\sin(120/2), 2 \cdot 428)$
 $= 820.992 \text{ mm.}$

The Computed K values from Table 4.15.1:

K1 = 0.1066 K2 = 1.1707 K3 = 0.8799 K4 = 0.4011
 K5 = 0.7603 K6 = 0.0529 K7 = 0.0529 K8 = 0.3405
 K9 = 0.2711 K10 = 0.0581 K1* = 0.1923 K6p = 0.0434
 K7p = 0.0434

The suffix 'p' denotes the values for a wear plate if it exists.

Note: Dimension a is greater than or equal to Rm/2.

Moment per Equation 4.15.1 [M1]:

$$= -Q \cdot a \left[1 - \left(\frac{a}{L} + \frac{(R_m^2 - h_m^2)}{(2a \cdot L)} \right) / \left(1 + \frac{(4h_m)}{3L} \right) \right]$$

$$= -6721 \cdot 800 \left[1 - \left(\frac{1 - 800/3901 + (428^2 - 230^2)}{(2 \cdot 800 \cdot 3901)} \right) / \left(1 + \frac{(4 \cdot 230)}{(3 \cdot 3901)} \right) \right]$$

$$= -1310.8 \text{ Kg-m.}$$

Moment per Equation 4.15.2 [M2]:

$$= Q \cdot L / 4 \left(\frac{(1 + 2(R_m^2 - h_m^2)/(L^2)) / (1 + (4h_m^2)/(3L)) - 4a/L}{(3 \cdot 3901)} \right) - 4 \cdot 800 / 3901$$

$$= 6721 \cdot 3901 / 4 \left(\frac{(1 + 2(428^2 - 230^2)/(3901^2)) / (1 + (4 \cdot 230)/(3 \cdot 3901)) - 4 \cdot 800 / 3901}{(3 \cdot 3901)} \right) - 4 \cdot 800 / 3901$$

$$= 803.1 \text{ Kg-m.}$$

Longitudinal Stress at Top of Shell (4.15.4) [σ_1]:

$$= P \cdot R_m / (2t) - M_2 / (\pi \cdot R_m^2 t)$$

$$= -0 \cdot 428 / (2 \cdot 71) - 803 / (\pi \cdot 428^2 \cdot 71)$$

$$= -0.19 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell (4.15.5) [σ_2]:

$$= P \cdot R_m / (2t) + M_2 / (\pi \cdot R_m^2 \cdot t)$$

$$= -0 \cdot 428 / (2 \cdot 71) + 803 / (\pi \cdot 428^2 \cdot 71)$$

$$= 0.19 \text{ N./mm}^2$$

Longitudinal Stress at Top of Shell at Support (4.15.8) [σ^*3]:

$$= P \cdot R_m / (2t) - M_1 / (K_1 \cdot \pi \cdot R_m^2 t)$$

$$= -0 \cdot 428 / (2 \cdot 71) - 1311 / (0.11 \cdot \pi \cdot 428^2 \cdot 71)$$

$$= 2.94 \text{ N./mm}^2$$

Longitudinal Stress at Bottom of Shell at Support (4.15.9) [σ^*4]:

$$= P \cdot R_m / (2t) + M_1 / (K_1^* \cdot \pi \cdot R_m^2 \cdot t)$$

$$= -0 \cdot 428 / (2 \cdot 71) + 1311 / (0.19 \cdot \pi \cdot 428^2 \cdot 71)$$

$$= -1.63 \text{ N./mm}^2$$

Maximum Shear Force in the Saddle (4.15.3) [T]:

$$= Q(L - 2a) / (L + (4 \cdot h_m / 3))$$

$$= 6721 (3901 - 2 \cdot 800) / (3901 + (4 \cdot 230 / 3))$$

$$= 3674.9 \text{ Kg}$$

Shear Stress in the shell no rings, not stiffened (4.15.12) [τ_2]:

$$= K_2 \cdot T / (R_m \cdot t)$$

$$= 1.17 \cdot 3675 / (428 \cdot 71)$$

$$= 1.39 \text{ N./mm}^2$$

Decay Length (4.15.20) [x_1, x_2]:

$$= 0.78 \cdot \sqrt{(R_m \cdot t)}$$

$$= 0.78 \cdot \sqrt{(428 \cdot 71)}$$

$$= 136.050 \text{ mm.}$$

Effective reinforcing plate width (4.15.24) [B1]:

$$= \min(b + 1.56 \cdot \sqrt{(R_m \cdot t)}, 2a)$$

$$= \min(200 + 1.56 \cdot \sqrt{(428 \cdot 71)}, 2 \cdot 800)$$

$$= 472.10 \text{ mm.}$$

Wear Plate/Shell Stress ratio (4.15.26) [η]:

$$= \min(S_r / S, 1)$$

$$= \min(118 / 118, 1)$$

$$= 1.0000$$

Circumferential Stress at Saddle Base with Wear Plate (4.15.25) [σ_6, r]:

$$= -K_5 \cdot Q \cdot k / (B_1(t + \eta \cdot t_r))$$

$$= -0.76 \cdot 6721 \cdot 0.1 / (472 (71 + 1 \cdot 10))$$

$$= -0.13 \text{ N./mm}^2$$

Circ. Comp. Stress at Horn of Saddle, $L \geq 8R_m$ (4.15.28) $[\sigma_{7,r}]$:

$$= -Q / (4(t + \eta \cdot tr)b_1) - 3 \cdot K_7 \cdot Q / (2(t + \eta \cdot tr)^2)$$

$$= -6721 / (4(71 + 1 \cdot 10)472) -$$

$$3 \cdot 0.053 \cdot 6721 / (2(71 + 1 \cdot 10)^2)$$

$$= -1.23 \text{ N./mm}^2$$

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	850.0001	mm.
Baseplate Thickness	Bpthk	24.0000	mm.
Baseplate Width	Bpwid	254.0000	mm.
Number of Ribs (inc. outside ribs)	Nribs	3	
Rib Thickness	Ribtk	10.0000	mm.
Web Thickness	Webtk	10.0000	mm.
Web Location	Webloc	Center	
Saddle Design Temperature		21.1	°C
Saddle Yield Stress	Sy	189.0	N./mm ²
Height of Web at Center	Hw,c	306.0	mm.
Friction Coefficient	mu	0.060	

Moment of Inertia of Saddle - Transverse Direction (90 degrees to long axis)

Inertia of Shell [ShellInertia]:

$$= ((1.56 \cdot \sqrt{(R \cdot t)} + \text{WearPlateWidth})t^3) / 12$$

$$= ((1.56 \cdot \sqrt{(393 \cdot 71)} + 480)71^3) / 12$$

$$= 2208.862 \text{ cm}^4$$

Resolved Inertia for the Shell [I,shell]:

$$= \text{ShellInertia} + A_{\text{Shell}}(C_1 - Y)^2$$

$$= 2209 + 526 (80.3 - 35.5)^2$$

$$= 12763.956 \text{ cm}^4$$

	B mm.	D mm.	Y mm.	A cm ²	AY mm. ³	I + AD ⁽²⁾ cm ⁴
Shell	740.6	71.0	35.5	525.8	1866645.6	0.128E+05
Wearplate	480.0	10.0	76.0	48.0	364800.0	12.9
Web	10.0	302.0	232.0	30.2	700639.8	0.924E+04
BasePlate	254.0	24.0	395.0	61.0	2407919.8	0.604E+05
Totals	...	...	...	665.0	5340005.0	0.824E+05

Distance to Centroid [C1]:

$$= AY / A$$

$$= 5340005 / 665$$

$$= 80.304 \text{ mm.}$$

Angle [β]:

$$= 180 - \text{Saddle Angle} / 2$$

$$= 180 - 120 / 2$$

$$= 120.0$$

Saddle Splitting Coefficient [K1]:

$$= (1 + \cos(\beta) - 0.5 \cdot \sin(\beta)^2) / (\pi - \beta + \sin(\beta) \cos(\beta))$$

$$= (1 + \cos(120) - 0.5 \cdot \sin(120)^2) / (\pi - 2.09 + \sin(120) \cos(120))$$

$$= 0.2035$$

Saddle Splitting Force [Fh]:

$$= K_1 \cdot Q$$

$$= 0.2 \cdot 6721$$

$$= 1367.9420 \text{ Kgf}$$

$$\text{Tension Stress, } St = (Fh / As) = 0.9640 \text{ N./mm}^2$$

$$\text{Allowed Stress, } Sa = 0.6 \cdot \text{Yield Str} = 113.3992 \text{ N./mm}^2$$

Saddle Splitting Dimension [d]:

$$= B - R \cdot \sin(\theta/2) / (\theta/2 \text{ in radians})$$

$$= 800 - 393 \cdot \sin(120/2) / 1.05$$

= 474.992 mm.

Bending Moment, $M = F_h \cdot d = 649.7738 \text{ Kg-m.}$

Bending Stress, $S_b = (M \cdot C_1 / I) = 0.6208 \text{ N./mm}^2$

Allowed Stress, $S_a = 2/3 \cdot \text{Yield Str} = 125.9991 \text{ N./mm}^2$

Minimum Thickness of Baseplate per Moss:

$= \sqrt{(3(Q + \text{SaddleWt}) \cdot \text{BasePlateWidth}) / (4 \cdot \text{BasePlateLength} \cdot \text{AllStress}))}$

$= \sqrt{(3(6721 + 147) \cdot 254 / (4 \cdot 850 \cdot 126))}$

= 10.945 mm.

Calculation of Axial Load, Intermediate Values and Compressive Stress:

Web Length Dimension [Web Length]:

$= 2 \cdot \cos(90 - \text{Saddle Angle}/2) (\text{Inside Radius} + \text{Shell Thk} + \text{Wear Plate Thk})$

$= 2 \cdot \cos(90 - 120/2) (390 + 74 + 10)$

= 820.992 mm.

Distance between Ribs [e]:

$= \text{Web Length} / (\text{Nribs} - 1)$

$= 821 / (3 - 1)$

= 410.496 mm.

Baseplate Pressure Area [Ap]:

$= e \cdot \text{Bpwid} / 2$

$= 410 \cdot 254 / 2$

= 521.330 cm²

Bearing Pressure [Bp]:

$= (Q + \text{SaddleWt}) / (\text{BasePlateLength} \cdot \text{BasePlateWidth})$

$= 6868 / (850 \cdot 254)$

= 3.181 Kg/cm²

Axial Load [P]:

$= A_p \cdot B_p$

$= 521 \cdot 3.18$

= 1658.406 Kg

Area of the Rib and Web [Ar]:

$= \text{Rib Area} + \text{Web Area}$

$= 19 + 20.5$

= 39.525 cm²

Compressive Stress [Sc]:

$= P / A_r$

$= 1658 / 39.5$

= 4.115 N./mm²

Check of Outside Ribs:

Inertia of Saddle, Outer Ribs - Longitudinal Direction

	B	D	Y	A	AY	Io
Rib+Web	10.0	200.0	...	20.0	...	667.

Rib dimension [D]:

$= \text{Saddle Width} - \text{Web Thickness}$

$= 200 - 10$

= 190.000 mm.

Distance to Centroid from Datum [ytot]:

$= AY / A$

$= 0 / 39.5$

= 0.000 mm.

Distance to Centroid [C1]:

$= \text{Saddle Width} / 2$

= 200/2



= 100.000 mm.

Radius of Gyration [r]:

$$= \sqrt{\frac{\text{Total Inertia}}{\text{Total Area}}}$$

$$= \sqrt{\frac{667}{39.5}}$$

$$= 41.070 \text{ mm.}$$

Length of Outer Rib [L]:

$$= \text{Saddle Height} - \cos(\theta/2) (\text{radius} + \text{shlthk} + \text{wpdthk}) - \text{bpthk}$$

$$= 800 - \cos(120/2) (390 + 74 + 10) - 24$$

$$= 539.000 \text{ mm.}$$

Intermediate Term [Cc]:

$$= \sqrt{\frac{2 \cdot \pi^2 \cdot \text{Elastic Modulus}}{\text{Yield Stress}}}$$

$$= \sqrt{\frac{2 \cdot \pi^2 \cdot 199943008}{189}}$$

$$= 144.511$$

Slenderness ratio [KL/r]:

$$= \frac{KL}{r}$$

$$= 1 \cdot 539 / 41.1$$

$$= 13.124$$

Bending Moment [Rm]:

$$= \frac{F_l}{(2 \cdot \text{Bplen})} \cdot e \cdot L / 2$$

$$= 2157 / (2 \cdot 850) \cdot 410 \cdot 539 / 2$$

$$= 140.357 \text{ Kg-m.}$$

Compressive Allowable, $KL/r < C_c$ ($13.1 < 145$) per AISC E2-1 [Sca]:

$$= \left(1 - \frac{(KL/r)^2}{(2 \cdot C_c^2)} \right) F_y / \left(\frac{5}{3} + 3 \cdot \frac{(KL/r)}{(8 \cdot C_c)} - \frac{(KL/r)^3}{(8 \cdot C_c^3)} \right)$$

$$= \left(1 - \frac{(13.1)^2}{(2 \cdot 145^2)} \right) 189 / \left(\frac{5}{3} + 3 \cdot \frac{(13.1)}{(8 \cdot 145)} - \frac{(13.1^3)}{(8 \cdot 145^3)} \right)$$

$$= 110.7 \text{ N./mm}^2$$

AISC Unity Check of Outside Ribs (must be ≤ 1)

$$= S_c / S_{ca} + (R_m \cdot C_1 / I) / S_{ba}$$

$$= 4.11 / 111 + (140 \cdot 100 / 6666666) / 126$$

$$= 0.201$$

Check of Inside Ribs:

Inertia of Saddle, Inner Ribs - Axial Direction

	B	D	Y	A	AY	Io
Rib	10.0	190.0	0.0	19.0	0.0	667.
Web	410.5	10.0	0.0	41.0	0.0	3.42
Totals	...	...	...	60.0	...	670.

Distance to Centroid from Datum [ytot]:

$$= AY / A$$

$$= 0 / 60$$

$$= 0.000 \text{ mm.}$$

Distance to Centroid [C1]:

$$= \text{Saddle Width} / 2$$

$$= 200 / 2$$

$$= 100.000 \text{ mm.}$$

Length of Inner Rib [L]:

$$= \text{Saddle Height} - \text{Outside Radius} - \text{Bpthk}$$

$$= 800 - \cos(474/2) (24 + 0 + 0) - 0$$

$$= 302.000 \text{ mm.}$$

Radius of Gyration [r]:

$$= \sqrt{\frac{\text{Total Inertia}}{\text{Total Area}}}$$

$$= \sqrt{\frac{670}{60}}$$

$$= 33.403 \text{ mm.}$$

Slenderness ratio [KL/r]:

$$= \frac{KL}{r}$$

$$= 1 \cdot 302 / 33.4$$

= 9.041

Unit Force [Force,u]:
= $F_l / (2 \cdot \text{Baseplate Length})$
= $2157 / (2 \cdot 850)$
= 1.269 Kgf/mm.

Moment at base of inner Rib [Mbase,c]:
= Unit Force $\cdot e \cdot L$
= $1.27 \cdot 410 \cdot 302$
= 157.283 Kg-m.

Bending Stress due to Transverse Force and Weight Load [σ_B ,base,c]:
= Bending Moment / Section Modulus
= $157/67000$
= 23.021 N./mm²

Compressive Allowable, $KL/r < C_c$ ($9.04 < 145$) per AISC E2-1 [Sca]:
= $(1 - (Kl/r)^2 / (2 \cdot C_c^2)) F_y / (5/3 + 3 \cdot (Kl/r) / (8 \cdot C_c) - (Kl/r)^3 / (8 \cdot C_c^3))$
= $(1 - (9.04)^2 / (2 \cdot 145^2)) 189 /$
 $(5/3 + 3 \cdot (9.04) / (8 \cdot 145) - (9.04^3) / (8 \cdot 145^3))$
= 111.6 N./mm²

AISC Unity Check of Inside Ribs (must be ≤ 1)
= $S_c / S_{ca} + (M_{base,c} \cdot C_{1/I}) / S_{ba}$
= $5.17/112 + (157 \cdot 100/670) / 126$
= 0.229

Review notes about nozzle loadings and supports in the Warnings report.

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Input, Nozzle Desc: T1-Tube Inlet

From: 30

Pressure for Reinforcement Calculations	P	147.076	bars
Temperature for Internal Pressure	Temp	93	°C
Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	118.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	780.00	mm.
Shell Finished (Minimum) Thickness	t	56.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1070.17	mm.
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		250.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	55.4637	mm.
Height of Beveled Transition	L'	35.4637	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		90.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		56.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: T1-Tube Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	215.900	mm.
Actual Thickness Used in Calculation	28.575	mm.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 147 \cdot 393 / (118 \cdot 1 - 0.6 \cdot 147)$$

$$= 52.9476 \text{ mm.}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 147 \cdot 111 / (138 \cdot 1 - 0.6 \cdot 147)$$

$$= 12.6336 \text{ mm.}$$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

$$= 61.4 / 0.5773$$

$$= 106.3913 \text{ mm.}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	443.8000	mm.
Parallel to Vessel Wall, opening length	d	221.9000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	132.5000	mm.

Weld Strength Reduction Factor [fr1]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 138 / 118)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr2]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 138 / 118)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(1, 1)$$

$$= 1.000$$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	117.491	NA	NA
Area in Shell	A1	0.116	NA	NA
Area in Nozzle Wall	A2	34.295	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	89.921	NA	NA
TOTAL AREA AVAILABLE	Atot	125.332	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (222 \cdot 52.9 \cdot 1 + 2 \cdot 87 \cdot 52.9 \cdot 1 \cdot (1 - 1))$$

$$= 117.491 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 222 (1 \cdot 53 - 1 \cdot 52.9) - 2 \cdot 87$$

$$(1 \cdot 53 - 1 \cdot 52.9) \cdot (1 - 1)$$

$$= 0.116 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np}) (t_n - t_{rn}) f_{r2}$$

$$= (2 \cdot 133) (25.6 - 12.6) \cdot 1$$

$$= 34.295 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot f_{r2} + (W_i - c_n / 0.707)^2 \cdot f_{r2}$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(133, 200, 55.5)) \cdot (90 - 28.6) \cdot 1$$

$$= 89.921 \text{ cm}^2 \text{ Includes Bevel Area of } 21.784 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 15.6336 \text{ mm.}$
Wall Thickness per UG16(b),	$t_{r16b} = 5.5000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure	$t_{rb1} = 55.9476 \text{ mm.}$
Wall Thickness	$t_{b1} = \max(t_{rb1}, t_{r16b}) = 55.9476 \text{ mm.}$
Wall Thickness	$t_{b2} = \max(t_{rb2}, t_{r16b}) = 5.5000 \text{ mm.}$
Wall Thickness per table UG-45	$t_{b3} = 11.1026 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[t_{b3}, \max(t_{b1}, t_{b2})]$$

$$= \min[11.1, \max(55.9, 5.5)]$$

$$= 11.1026 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(15.6, 11.1)$$

$$= 15.6336 \text{ mm.}$$

Available Nozzle Neck Thickness = $0.875 \cdot 28.575 = 25.003 \text{ mm.} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 24.6,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 320.3 N./mm ²	Passed
Occasional	: 28.6,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 11.2,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govern. thk, $t_g = 25$, $t_r = 12.6$, $c = 3 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.57$, Temp. Reduction = 25°C

Min Metal Temp. w/o impact per UCS-66, Curve A	19 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-6 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

Govern. thk, $t_g = 56$, $t_r = 52.9$, $c = 3 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 1$, Temp. Reduction = 0°C

Min Metal Temp. w/o impact per UCS-66, Curve B	19 °C
--	-------

Gov. MDMT of the nozzle to shell joint welded assembly : 19 °C

Flange MDMT including Temperature reduction per UCS-66.1:



Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -42 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
Design Pressure/Ambient Rating = 147.08/255.10 = 0.577

Weld Size Calculations, Description: T1-Tube Inlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
= max(0, (A-A1+2•tn•fr1•(E1•t-tr))Sv)
= max(0, (117 - 0.12 + 2 • 87 • 1 •
(1 • 53 - 52.9))118)
= 141337.28 Kgf

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
= (A2+A4-(Wi-Can/.707)²•fr2)•Sv
= (124 + 1 - 0 • 1) • 118
= 150661.98 Kgf

Weld Load [W2]:
= (A2 + A3 + A4 + (2 • tn • t • fr1)) • Sv
= (124 + 0 + 1 + (92.2)) • 118
= 261623.06 Kgf

Weld Load [W3]:
= (A2+A3+A4+A5+(2•tn•t•fr1))•S
= (124 + 0 + 1 + 0 + (92.2)) • 118
= 261623.06 Kgf

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
= (π/2) • Dlo • Wo • 0.49 • Snw
= (3.14/2.0) • 396 • 10 • 0.49 • 118
= 36665. Kgf

Shear, Nozzle Wall [Snw]:
= (π • (Dlr + Dlo)/4) • (Thk - Can) • 0.7 • Sn
= (3.14 • 154) • (90 - 3) • 0.7 • 138
= 415813. Kgf

Tension, Shell Groove Weld [Tngw]:
= (π/2) • Dlo • (T - cas) • 0.74 • Sng
= (3.14/2.0) • 396 • (56 - 3) • 0.74 • 138
= 343207. Kgf

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
= (SONW + SNW) = (36665 + 415813) = 452477 Kgf

Failure Path [Path 2-2]:
= (Sonw + Tpgw + Tngw + Sinw)
= (36665 + 0 + 343207 + 0) = 379871 Kgf

Failure Path [Path 3-3]:
 = (Sonw + Tngw + Sinw)
 = (36665 + 343207 + 0) = 379871 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 452477 Kgf, must exceed W = 141337 Kgf or W1 = 150661 Kgf
 Path 2-2 = 379871 Kgf, must exceed W = 141337 Kgf or W2 = 261623 Kgf
 Path 3-3 = 379871 Kgf, must exceed W = 141337 Kgf or W3 = 261623 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 147 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 53.9705 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 309.9706 mm.

Input Echo, WRC297 Item 1, Description: T1-Tube Inlet :

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	780.000	mm.
Shell Thickness	56.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 60	
Vessel Cold S.I. Allowable	Smc 118.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	215.900	mm.
Nozzle Thickness	90.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSained loads.

Design Internal Pressure	Dp	147.08	bars
Radial Load	P	-8158.27	Kgf
Circumferential Shear	Vc	8158.27	Kgf
Longitudinal Shear	Vl	6118.70	Kgf
Circumferential Moment	Mc	5.30	Kg-m.
Longitudinal Moment	Ml	5.30	Kg-m.
Torsional Moment	Mt	6.12	Kg-m.
Include Axial Pressure Thrust		No	
Include Pressure Stress Indices per Div. 2		No	
Local Loads applied at end of Nozzle/Attachment		No	

Warning - The ratio, Dv/Tv (15.929) is < 20
 check the limitations of WRC bulletin 297.
 Warning - The ratio, Dn/Tn (4.399) is < 10
 check the limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 = NozzleOD + 2 • 1.65 • √(Rmean(t - ca))
 = 395.9 + 2 • 1.65 • √(419.5(56.0 - 3.0))

= 887.960 mm.

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 839.000 mm.
 Nozzle Outside Diameter (d) = 395.900 mm.
 Vessel Thickness used (T) = 53.000 mm.
 Nozzle Thickness used (t) = 87.000 mm.
 T / t = 0.609
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.877$$

Nr/P = 0.075
 Mr/P = 0.142
 M0/P = 0.042
 N0/P = 0.102
 MrD/Mc = 0.279
 NrDL/Mc = 0.109
 M0d/Mc = 0.089
 N0DL/Mc = 0.064
 MrD/Ml = 0.119
 NrDL/Ml = 0.070
 M0D/Ml = 0.040
 N0DL/Ml = 0.127

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	102	116	102	116
Outplane Stress Summary	112	112	112	112

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	24	-24	24	-24
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	51	51	51	51
Inplane Stress Summary	77	28	77	28

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	2	2	-2	-2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Top	Al Top	Bu Bottom	Bl Bottom
---	-----------	-----------	--------------	--------------



	Outside	Inside	Outside	Inside
Two * Max Shear Stress	112	112	112	112

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	51	51	51	51
Outplane Stress Summary	61	46	61	46

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	24	-24	24	-24
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	102	116	102	116
Inplane Stress Summary	128	94	128	94

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-1	-1	1	1

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	128	94	128	94

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	16	33	16	33
Hoop Stress Summary	19	36	19	36

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to	Au Top	Al Top	Bu Bottom	Bl Bottom
---	-----------	-----------	--------------	--------------



pipe cross-section)	Outside	Inside	Outside	Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	6	-6	6	-6
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	8	8	8	8
Axial Stress Summary	15	2	15	2

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	1	1	-1	-1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	19	36	19	36

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	16	33	16	33
Hoop Stress Summary	19	36	19	36

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	6	-6	6	-6
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	8	8	8	8
Axial Stress Summary	15	2	15	2

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-1	-1	1	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE	Cu	Cl	Du	Dl
-----------------------	----	----	----	----

(Stress Intensities Outside Inside Outside Inside

Two * Max Shear Stress 19 36 19 36

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	102	116	102	116	102	116	102	116
Circ. Pl (SUS)	2	2	2	2	2	2	2	2
Circ. Q (SUS)	7	-7	7	-7	24	-24	24	-24
Long. Pm (SUS)	51	51	51	51	51	51	51	51
Long. Pl (SUS)	2	2	2	2	2	2	2	2
Long. Q (SUS)	24	-24	24	-24	7	-7	7	-7
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-1	-1	1	1
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	102.0	116.0	102.0	116.0	102.0	116.0	102.0	116.0
Pm+Pl (SUS)	104.1	118.1	104.1	118.1	104.0	118.0	104.0	118.0
Pm+Pl+Q (Total)	111.1	111.0	111.1	111.0	128.0	94.0	128.0	94.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	116.00	118.00	Passed
Pm+Pl (SUS)	118.06	177.00	Passed
Pm+Pl+Q (TOTAL)	128.01	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	16	33	16	33	16	33	16	33
Circ. Pl (SUS)	2	2	2	2	2	2	2	2
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	8	8	8	8	8	8	8	8
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	6	-6	6	-6	6	-6	6	-6
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	16.0	33.0	16.0	33.0	16.0	33.0	16.0	33.0
Pm+Pl (SUS)	18.1	35.0	18.1	35.0	18.1	35.0	18.1	35.0
Pm+Pl+Q (Total)	18.2	35.0	18.2	35.0	18.2	35.0	18.2	35.0

Nozzle Stress Summation Comparison (N./mm²):



Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	33.00	138.00	Passed
Pm+Pl (SUS)	35.04	207.00	Passed
Pm+Pl+Q (TOTAL)	35.03	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: T2-Tube outlet
From: 30

Pressure for Reinforcement Calculations	P	147.000	bars
Temperature for Internal Pressure	Temp	93	°C
Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	118.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	780.00	mm.
Shell Finished (Minimum) Thickness	t	56.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		1070.17	mm.
User Entered Minimum Design Metal Temperature		0.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		250.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	55.4637	mm.
Height of Beveled Transition	L'	35.4637	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		90.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel Wgnv		56.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: T2-Tube outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	215.900	mm.
Actual Thickness Used in Calculation	28.575	mm.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 147 \cdot 393 / (118 \cdot 1 - 0.6 \cdot 147)$$

$$= 52.9179 \text{ mm.}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 147 \cdot 111 / (138 \cdot 1 - 0.6 \cdot 147)$$

$$= 12.6265 \text{ mm.}$$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

$$= 61.4 / 0.5773$$

$$= 106.3913 \text{ mm.}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	443.8000	mm.
Parallel to Vessel Wall, opening length	d	221.9000	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	132.5000	mm.

Weld Strength Reduction Factor [fr1]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 138 / 118)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr2]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 138 / 118)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(1, 1)$$

$$= 1.000$$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	117.425	NA	NA
Area in Shell	A1	0.182	NA	NA
Area in Nozzle Wall	A2	34.313	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	89.921	NA	NA
TOTAL AREA AVAILABLE	Atot	125.416	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (222 \cdot 52.9 \cdot 1 + 2 \cdot 87 \cdot 52.9 \cdot 1 \cdot (1 - 1))$$

$$= 117.425 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$

$$= 222 (1 \cdot 53 - 1 \cdot 52.9) - 2 \cdot 87$$

$$(1 \cdot 53 - 1 \cdot 52.9) \cdot (1 - 1)$$

$$= 0.182 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np}) (t_n - t_{rn}) f_{r2}$$

$$= (2 \cdot 133) (25.6 - 12.6) \cdot 1$$

$$= 34.313 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot f_{r2} + (W_i - c_n / 0.707)^2 \cdot f_{r2}$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(133, 200, 55.5)) \cdot (90 - 28.6) \cdot 1$$

$$= 89.921 \text{ cm}^2 \text{ Includes Bevel Area of } 21.784 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 15.6265 \text{ mm.}$
Wall Thickness per UG16(b),	$t_{r16b} = 5.5000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure	$t_{rb1} = 55.9179 \text{ mm.}$
Wall Thickness	$t_{b1} = \max(t_{rb1}, t_{r16b}) = 55.9179 \text{ mm.}$
Wall Thickness	$t_{b2} = \max(t_{rb2}, t_{r16b}) = 5.5000 \text{ mm.}$
Wall Thickness per table UG-45	$t_{b3} = 11.1026 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[t_{b3}, \max(t_{b1}, t_{b2})]$$

$$= \min[11.1, \max(55.9, 5.5)]$$

$$= 11.1026 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(15.6, 11.1)$$

$$= 15.6265 \text{ mm.}$$

Available Nozzle Neck Thickness = $0.875 \cdot 28.575 = 25.003 \text{ mm.} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 24.6,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 320.4 N./mm ²	Passed
Occasional	: 28.6,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 11.2,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govern. thk, $t_g = 25$, $t_r = 12.6$, $c = 3 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.57$, Temp. Reduction = 25°C

Min Metal Temp. w/o impact per UCS-66, Curve A	19 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-6 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

Govern. thk, $t_g = 56$, $t_r = 52.9$, $c = 3 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 1$, Temp. Reduction = 0°C

Min Metal Temp. w/o impact per UCS-66, Curve B	19 °C
Min Metal Temp. at Required thickness (UCS 66.1)	19 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 19 °C



Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -42 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 147.00/255.10 = 0.576

Weld Size Calculations, Description: T2-Tube outlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (117 - 0.18 + 2 \cdot 87 \cdot 1 \cdot (1 \cdot 53 - 52.9)) \cdot 118)$
 $= 141240.70 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i - \text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (124 + 1 - 0 \cdot 1) \cdot 118$
 $= 150684.39 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (124 + 0 + 1 + (92.2)) \cdot 118$
 $= 261645.44 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (124 + 0 + 1 + 0 + (92.2)) \cdot 118$
 $= 261645.44 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 396 \cdot 10 \cdot 0.49 \cdot 118$
 $= 36665. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 154) \cdot (90 - 3) \cdot 0.7 \cdot 138$
 $= 415813. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 396 \cdot (56 - 3) \cdot 0.74 \cdot 138$
 $= 343207. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (36665 + 415813) = 452477 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (36665 + 0 + 343207 + 0) = 379871 \text{ Kgf}$

Failure Path [Path 3-3]:
 = (Sonw + Tngw + Sinw)
 = (36665 + 343207 + 0) = 379871 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 452477 Kgf, must exceed W = 141240 Kgf or W1 = 150684 Kgf
 Path 2-2 = 379871 Kgf, must exceed W = 141240 Kgf or W2 = 261645 Kgf
 Path 3-3 = 379871 Kgf, must exceed W = 141240 Kgf or W3 = 261645 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 147 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 53.9705 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 309.9706 mm.

Input Echo, WRC297 Item 1, Description: T2-Tube outlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	780.000	mm.
Shell Thickness	56.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 60	
Vessel Cold S.I. Allowable	Smc 118.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	215.900	mm.
Nozzle Thickness	90.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	147.00	bars
Radial Load	P	-8158.27	Kgf
Circumferential Shear	Vc	8158.27	Kgf
Longitudinal Shear	Vl	6118.70	Kgf
Circumferential Moment	Mc	5.30	Kg-m.
Longitudinal Moment	Ml	5.30	Kg-m.
Torsional Moment	Mt	6.12	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dv/Tv (15.929) is < 20
 check the Limitations of WRC bulletin 297.

Warning - The ratio, Dn/Tn (4.399) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:
 = NozzleOD + 2 • 1.65 • √(Rmean(t - ca))

$$= 395.9 + 2 \cdot 1.65 \cdot \sqrt{(419.5(56.0 - 3.0))}$$

$$= 887.960 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 839.000 mm.
 Nozzle Outside Diameter (d) = 395.900 mm.
 Vessel Thickness used (T) = 53.000 mm.
 Nozzle Thickness used (t) = 87.000 mm.
 T / t = 0.609
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.877$$

Nr/P = 0.075
 Mr/P = 0.142
 M0/P = 0.042
 N0/P = 0.102
 MrD/Mc = 0.279
 NrDL/Mc = 0.109
 M0d/Mc = 0.089
 N0DL/Mc = 0.064
 MrD/M1 = 0.119
 NrDL/M1 = 0.070
 M0D/M1 = 0.040
 N0DL/M1 = 0.127

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	102	116	102	116
Outplane Stress Summary	112	112	112	112

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	24	-24	24	-24
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	51	51	51	51
Inplane Stress Summary	77	28	77	28

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	2	2	-2	-2
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	2	2	-2	-2

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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(Stress Intensities	Top Outside	Top Inside	Bottom Outside	Bottom Inside
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Two * Max Shear Stress	112	112	112	112
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
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Outplane Membrane (P)	2	2	2	2
Outplane Bending (P)	7	-7	7	-7
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	51	51	51	51

Outplane Stress Summary	61	46	61	46
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Inplane Membrane (P)	2	2	2	2
Inplane Bending (P)	24	-24	24	-24
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	102	116	102	116

Inplane Stress Summary	128	94	128	94
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	0	0	0	0

Shear Stress Summary	-1	-1	1	1
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Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
--	-----------------------	----------------------	------------------------	-----------------------

Two * Max Shear Stress	128	94	128	94
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
---	----------------------	---------------------	-------------------------	------------------------

Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	16	33	16	33

Hoop Stress Summary	19	36	19	36
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE	Au	Al	Bu	Bl
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(Stresses Normal to pipe cross-section)	Top Outside	Top Inside	Bottom Outside	Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	6	-6	6	-6
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	8	8	8	8

Axial Stress Summary	15	2	15	2
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Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	1	1	-1	-1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	19	36	19	36

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	2	2	2	2
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	16	33	16	33
Hoop Stress Summary	19	36	19	36

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	6	-6	6	-6
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	8	8	8	8
Axial Stress Summary	15	2	15	2

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	-1	-1	1	1
Shear due to Torsion	0	0	0	0
Shear Stress Summary	-1	-1	1	1

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities	Cu Outside	Cl Inside	Du Outside	Dl Inside
Two * Max Shear Stress	19	36	19	36

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	102	116	102	116	102	116	102	116
Circ. Pl (SUS)	2	2	2	2	2	2	2	2
Circ. Q (SUS)	7	-7	7	-7	24	-24	24	-24
Long. Pm (SUS)	51	51	51	51	51	51	51	51
Long. Pl (SUS)	2	2	2	2	2	2	2	2
Long. Q (SUS)	24	-24	24	-24	7	-7	7	-7
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	2	2	-2	-2	-1	-1	1	1
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	102.0	116.0	102.0	116.0	102.0	116.0	102.0	116.0
Pm+Pl (SUS)	104.1	118.1	104.1	118.1	104.0	118.0	104.0	118.0
Pm+Pl+Q (Total)	111.1	111.0	111.1	111.0	128.0	94.0	128.0	94.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	116.00	118.00	Passed
Pm+Pl (SUS)	118.06	177.00	Passed
Pm+Pl+Q (TOTAL)	128.01	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	16	33	16	33	16	33	16	33
Circ. Pl (SUS)	2	2	2	2	2	2	2	2
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	8	8	8	8	8	8	8	8
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	6	-6	6	-6	6	-6	6	-6
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	16.0	33.0	16.0	33.0	16.0	33.0	16.0	33.0
Pm+Pl (SUS)	18.1	35.0	18.1	35.0	18.1	35.0	18.1	35.0
Pm+Pl+Q (Total)	18.2	35.0	18.2	35.0	18.2	35.0	18.2	35.0

Nozzle Stress Summation Comparison (N./mm²):



Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	33.00	138.00	Passed
Pm+Pl (SUS)	35.04	207.00	Passed
Pm+Pl+Q (TOTAL)	35.03	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: S1-Shell Inlet

From: 40

Pressure for Reinforcement Calculations	P	191.000	bars
Temperature for Internal Pressure	Temp	93	°C
Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	118.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	780.00	mm.
Shell Finished (Minimum) Thickness	t	74.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		2300.17	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		90.00	deg
Diameter		300.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	75.0000	mm.
Height of Beveled Transition	L'	45.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		110.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	74.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: S1-Shell Inlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	257.200	mm.
Actual Thickness Used in Calculation	33.325	mm.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 191 \cdot 393 / (118 \cdot 1 - 0.6 \cdot 191)$$

$$= 70.4612 \text{ mm.}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 191 \cdot 132 / (138 \cdot 1 - 0.6 \cdot 191)$$

$$= 19.8654 \text{ mm.}$$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

$$= 76.7 / 0.5773$$

$$= 132.8054 \text{ mm.}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	Dl	526.4008	mm.
Parallel to Vessel Wall, opening length	d	263.2004	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	177.5000	mm.

Weld Strength Reduction Factor [fr1]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 138 / 118)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr2]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 138 / 118)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(1, 1)$$

$$= 1.000$$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	185.454	NA	NA
Area in Shell	A1	1.418	NA	NA
Area in Nozzle Wall	A2	37.131	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	149.517	NA	NA
TOTAL AREA AVAILABLE	Atot	189.066	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (263 \cdot 70.5 \cdot 1 + 2 \cdot 107 \cdot 70.5 \cdot 1 \cdot (1 - 1))$$

$$= 185.454 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$



$$= 263 (1 \cdot 71 - 1 \cdot 70.5) - 2 \cdot 107$$

$$(1 \cdot 71 - 1 \cdot 70.5) \cdot (1 - 1)$$

$$= 1.418 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np}) (t_n - t_{rn}) f_{r2}$$

$$= (2 \cdot 178) (30.3 - 19.9) 1$$

$$= 37.131 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot f_{r2} + (W_i - c_{an}/0.707)^2 \cdot f_{r2}$$

$$= 10^2 \cdot 1 + (0)^2 \cdot 1$$

$$= 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2}$$

$$= (2 \cdot \min(178, 200, 75)) \cdot (110 - 33.3) \cdot 1$$

$$= 149.517 \text{ cm}^2 \text{ Includes Bevel Area of } 34.504 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 22.8654 \text{ mm.}$
Wall Thickness per UG16(b),	$tr_{16b} = 4.5000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure	$tr_{b1} = 73.4612 \text{ mm.}$
Wall Thickness	$tb_1 = \max(tr_{b1}, tr_{16b}) = 73.4612 \text{ mm.}$
Wall Thickness	$tb_2 = \max(tr_{b2}, tr_{16b}) = 4.5000 \text{ mm.}$
Wall Thickness per table UG-45	$tb_3 = 11.3312 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[tb_3, \max(tb_1, tb_2)]$$

$$= \min[11.3, \max(73.5, 4.5)]$$

$$= 11.3312 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b)$$

$$= \max(22.9, 11.3)$$

$$= 22.8654 \text{ mm.}$$

Available Nozzle Neck Thickness = $0.875 \cdot 33.325 = 29.159 \text{ mm.} \rightarrow \text{OK}$

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 33.8,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 311.2 N./mm ²	Passed
Occasional	: 37.2,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 9.6,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 29.2$, $t_r = 19.9$, $c = 3 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.76$, Temp. Reduction = $13 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A	23 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	10 $^\circ\text{C}$

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

Govrn. thk, $t_g = 74$, $t_r = 70.5$, $c = 3 \text{ mm.}$, $E^* = 1$
 Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.99$, Temp. Reduction = $0 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B	25 $^\circ\text{C}$
Min Metal Temp. at Required thickness (UCS 66.1)	25 $^\circ\text{C}$

Gov. MDMT of the nozzle to shell joint welded assembly : 25 $^\circ\text{C}$



Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -32 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 191.00/255.10 = 0.749

Weld Size Calculations, Description: S1-Shell Inlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_r1 \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (185 - 1.42 + 2 \cdot 107 \cdot 1 \cdot (1 \cdot 71 - 70.5)) \cdot 118)$
 $= 222823.31 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i - \text{Can})/0.707)^2 \cdot f_r2 \cdot S_v$
 $= (187 + 1 - 0 \cdot 1) \cdot 118$
 $= 225781.30 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_r1)) \cdot S_v$
 $= (187 + 0 + 1 + (152)) \cdot 118$
 $= 408598.72 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_r1)) \cdot S$
 $= (187 + 0 + 1 + 0 + (152)) \cdot 118$
 $= 408598.72 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 477 \cdot 10 \cdot 0.49 \cdot 118$
 $= 44194. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 185) \cdot (110 - 3) \cdot 0.7 \cdot 138$
 $= 612888. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 477 \cdot (74 - 3) \cdot 0.74 \cdot 138$
 $= 554183. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (44194 + 612888) = 657082 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (44194 + 0 + 554183 + 0) = 598377 \text{ Kgf}$

Failure Path [Path 3-3]:
 = (Sonw + Tngw + Sinw)
 = (44194 + 554183 + 0) = 598377 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 657081 Kgf, must exceed W = 222823 Kgf or W1 = 225781 Kgf
 Path 2-2 = 598377 Kgf, must exceed W = 222823 Kgf or W2 = 408598 Kgf
 Path 3-3 = 598377 Kgf, must exceed W = 222823 Kgf or W3 = 408598 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 192 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 81.5037 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 355.5037 mm.

Input Echo, WRC297 Item 1, Description: S1-Shell Inlet:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	780.000	mm.
Shell Thickness	74.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 60	
Vessel Cold S.I. Allowable	Smc 118.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	257.200	mm.
Nozzle Thickness	110.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	191.00	bars
Radial Load	P	-9789.91	Kgf
Circumferential Shear	Vc	9789.91	Kgf
Longitudinal Shear	Vl	7342.47	Kgf
Circumferential Moment	Mc	7.63	Kg-m.
Longitudinal Moment	ML	7.63	Kg-m.
Torsional Moment	Mt	8.81	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dv/Tv (12.541) is < 20
 check the Limitations of WRC bulletin 297.
 Warning - The ratio, Dn/Dv (.514) is > 0.5
 check the Limitations of WRC bulletin 297.
 Warning - The ratio, Dn/Tn (4.338) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:



Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 477.2 + 2 \cdot 1.65 \cdot \sqrt{(428.5(74.0 - 3.0))}$$

$$= 1052.797 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 857.000 mm.
 Nozzle Outside Diameter (d) = 477.200 mm.
 Vessel Thickness used (T) = 71.000 mm.
 Nozzle Thickness used (t) = 107.000 mm.
 T / t = 0.664
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.935$$

Nr/P = 0.075
 Mr/P = 0.136
 M0/P = 0.041
 N0/P = 0.101
 MrD/Mc = 0.273
 NrDL/Mc = 0.114
 M0d/Mc = 0.089
 N0DL/Mc = 0.064
 MrD/Ml = 0.116
 NrDL/Ml = 0.069
 M0D/Ml = 0.039
 N0DL/Ml = 0.127

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	96	116	96	116
Outplane Stress Summary	103	113	103	113

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	48	48	48	48
Inplane Stress Summary	65	34	65	34

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	1	1	-1	-1



Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	103	113	103	113

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	48	48	48	48
Outplane Stress Summary	55	45	55	45

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	96	116	96	116
Inplane Stress Summary	113	101	114	101

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-1	-1	1	1

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	113	102	114	101

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	20	42	20	42
Hoop Stress Summary	21	44	21	44

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	4	-4	4	-4
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	10	10	10	10
Axial Stress Summary	15	5	15	5

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	1	1	-1	-1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	22	44	22	44

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	20	42	20	42
Hoop Stress Summary	21	44	21	44

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	4	-4	4	-4
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	10	10	10	10
Axial Stress Summary	15	5	15	5

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0



Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Stress Intensities Cu Cl Du Dl
 Outside Inside Outside Inside

Two * Max Shear Stress 22 44 22 44

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	96	116	96	116	96	116	96	116
Circ. Pl (SUS)	1	1	1	1	1	1	1	1
Circ. Q (SUS)	4	-4	4	-4	15	-15	15	-15
Long. Pm (SUS)	48	48	48	48	48	48	48	48
Long. Pl (SUS)	1	1	1	1	1	1	1	1
Long. Q (SUS)	15	-15	15	-15	4	-4	4	-4
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	96.0	116.0	96.0	116.0	96.0	116.0	96.0	116.0
Pm+Pl (SUS)	97.0	117.0	97.0	117.0	97.0	117.0	97.0	117.0
Pm+Pl+Q (Total)	101.0	113.0	101.0	113.0	112.0	102.0	112.0	102.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	116.00	118.00	Passed
Pm+Pl (SUS)	117.01	177.00	Passed
Pm+Pl+Q (TOTAL)	113.01	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	20	42	20	42	20	42	20	42
Circ. Pl (SUS)	1	1	1	1	1	1	1	1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	10	10	10	10	10	10	10	10
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	4	-4	4	-4	4	-4	4	-4
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	20.0	42.0	20.0	42.0	20.0	42.0	20.0	42.0
Pm+Pl (SUS)	21.1	43.0	21.1	43.0	21.0	43.0	21.0	43.0
Pm+Pl+Q (Total)	21.1	43.0	21.1	43.0	21.0	43.0	21.0	43.0



Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	42.00	138.00	Passed
Pm+Pl (SUS)	43.03	207.00	Passed
Pm+Pl+Q (TOTAL)	43.03	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input, Nozzle Desc: S2-Shell Outlet

From: 40

Pressure for Reinforcement Calculations	P	191.004	bars
Temperature for Internal Pressure	Temp	93	°C
Parent Material		SA-516 60	
Parent Allowable Stress at Temperature	Sv	118.00	N./mm ²
Parent Allowable Stress At Ambient	Sva	118.00	N./mm ²
Inside Diameter of Cylindrical Shell	D	780.00	mm.
Shell Finished (Minimum) Thickness	t	74.0000	mm.
Shell Internal Corrosion Allowance	c	3.0000	mm.
Shell External Corrosion Allowance	co	0.0000	mm.
Distance from Bottom/Left Tangent		4900.18	mm.
User Entered Minimum Design Metal Temperature		-10.00	°C

Type of Element Connected to the Parent : Nozzle

Material		SA-266 2	
Material UNS Number		K03506	
Material Specification/Type		Forgings	
Allowable Stress at Temperature	Sn	138.00	N./mm ²
Allowable Stress At Ambient	Sna	138.00	N./mm ²
Diameter Basis (for tr calc only)		Inside	
Layout Angle		270.00	deg
Diameter		300.0000	mm
Size and Thickness Basis		Nominal	
Nominal Thickness		160	
Flange Material		SA-105	
Hub Height of Integral Nozzle	h	75.0000	mm.
Height of Beveled Transition	L'	45.0000	mm.
Hub Thickness of Integral Nozzle (tn or x+tp)		110.0000	mm.
Corrosion Allowance	can	3.0000	mm.
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
Outside Projection	ho	200.0000	mm.
Weld leg size between Nozzle and Pad/Shell	Wo	10.0000	mm.
Groove weld depth between Nozzle and Vessel	Wgnv	74.0000	mm.
Inside Projection	h	0.0000	mm.
Weld leg size, Inside Element to Shell	Wi	0.0000	mm.
Flange Class		1500	
Flange Grade		GR 1.1	

The Pressure Design option was Design Pressure + static head.

Reinforcement CALCULATION, Description: S2-Shell Outlet

ASME Code, Section VIII, Div. 1, UG-37 to UG-45

Actual Inside Diameter Used in Calculation	257.200	mm.
Actual Thickness Used in Calculation	33.325	mm.

Note:

Post Weld Heat Treatment is required for this nozzle and it was specified as being heat treated.

Nozzle input data check completed without errors.

Reqd thk per UG-37(a) of Cylindrical Shell, tr [Int. Press]

$$= P \cdot R / (S_v \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 191 \cdot 393 / (118 \cdot 1 - 0.6 \cdot 191)$$

$$= 70.4627 \text{ mm.}$$

Reqd thk per UG-37(a) of Nozzle Wall, trn [Int. Press]

$$= P \cdot R / (S_n \cdot E - 0.6 \cdot P) \text{ per UG-27 (c)(1)}$$

$$= 191 \cdot 132 / (138 \cdot 1 - 0.6 \cdot 191)$$

$$= 19.8659 \text{ mm.}$$

Intermediate Hub Nozzle Calculations:

Check to determine use of Sketch (e-1) or (e-2):

Height value from sketch (e-1) [te]:

$$= (\text{Hub Thickness} - \text{Neck Thickness}) / \tan(30)$$

$$= 76.7 / 0.5773$$

$$= 132.8054 \text{ mm.}$$

Hub Height was < 2.5 times Hub Thickness, use sketch UG-40 (e-1).

UG-40, Limits of Reinforcement: [Internal Pressure]

Parallel to Vessel Wall (Diameter Limit)	D1	526.4008	mm.
Parallel to Vessel Wall, opening length	d	263.2004	mm.
Normal to Vessel Wall (Thickness Limit), no pad	Tlnp	177.5000	mm.

Weld Strength Reduction Factor [fr1]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 138 / 118)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr2]:

$$= \min(1, S_n / S_v)$$

$$= \min(1, 138 / 118)$$

$$= 1.000$$

Weld Strength Reduction Factor [fr3]:

$$= \min(fr2, fr4)$$

$$= \min(1, 1)$$

$$= 1.000$$

Results of Nozzle Reinforcement Area Calculations: (cm²)

AREA AVAILABLE, A1 to A5		Design	External	Mapnc
Area Required	Ar	185.458	NA	NA
Area in Shell	A1	1.414	NA	NA
Area in Nozzle Wall	A2	37.129	NA	NA
Area in Inward Nozzle	A3	0.000	NA	NA
Area in Welds	A41+A42+A43	1.000	NA	NA
Area in Element	A5	0.000	NA	NA
Area in Hub	A6	149.517	NA	NA
TOTAL AREA AVAILABLE	Atot	189.060	NA	NA

The Internal Pressure Case Governs the Analysis.

Nozzle Angle Used in Area Calculations 90.00 degs.

The area available without a pad is Sufficient.

Area Required [A]:

$$= (d \cdot tr \cdot F + 2 \cdot tn \cdot tr \cdot F \cdot (1 - fr1)) \text{ UG-37(c)}$$

$$= (263 \cdot 70.5 \cdot 1 + 2 \cdot 107 \cdot 70.5 \cdot 1 \cdot (1 - 1))$$

$$= 185.458 \text{ cm}^2$$

Reinforcement Areas per Figure UG-37.1

Area Available in Shell [A1]:

$$= d(E1 \cdot t - F \cdot tr) - 2 \cdot tn(E1 \cdot t - F \cdot tr) \cdot (1 - fr1)$$



$$= 263 (1 \cdot 71 - 1 \cdot 70.5) - 2 \cdot 107 \\ (1 \cdot 71 - 1 \cdot 70.5) \cdot (1 - 1) \\ = 1.414 \text{ cm}^2$$

Area Available in Nozzle Projecting Outward [A2]:

$$= (2 \cdot t_{np}) (t_n - t_{rn}) f_{r2} \\ = (2 \cdot 178) (30.3 - 19.9) 1 \\ = 37.129 \text{ cm}^2$$

Area Available in Inward Weld + Outward Weld [A41 + A43]:

$$= W_o^2 \cdot f_{r2} + (W_i - c_{an}/0.707)^2 \cdot f_{r2} \\ = 10^2 \cdot 1 + (0)^2 \cdot 1 \\ = 1.000 \text{ cm}^2$$

Area Available in the Hub Section [A6]:

$$= (2 \cdot \min(t_{np}, h_o, \text{Hubht})) \cdot (\text{Hubtk} - t_n) \cdot f_{r2} \\ = (2 \cdot \min(178, 200, 75)) \cdot (110 - 33.3) \cdot 1 \\ = 149.517 \text{ cm}^2 \text{ Includes Bevel Area of } 34.504 \text{ cm}^2$$

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 22.8659 \text{ mm.}$
Wall Thickness per UG16(b),	$t_{r16b} = 4.5000 \text{ mm.}$
Wall Thickness, shell/head, internal pressure	$t_{rb1} = 73.4627 \text{ mm.}$
Wall Thickness	$t_{b1} = \max(t_{rb1}, t_{r16b}) = 73.4627 \text{ mm.}$
Wall Thickness	$t_{b2} = \max(t_{rb2}, t_{r16b}) = 4.5000 \text{ mm.}$
Wall Thickness per table UG-45	$t_{b3} = 11.3312 \text{ mm.}$

Determine Nozzle Thickness candidate [tb]:

$$= \min[t_{b3}, \max(t_{b1}, t_{b2})] \\ = \min[11.3, \max(73.5, 4.5)] \\ = 11.3312 \text{ mm.}$$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$$= \max(t_a, t_b) \\ = \max(22.9, 11.3) \\ = 22.8659 \text{ mm.}$$

Available Nozzle Neck Thickness = $0.875 \cdot 33.325 = 29.159 \text{ mm.}$ -> OK

Stresses on Nozzle due to External and Pressure Loads per the ASME B31.3 Piping Code, (see 319.4.4 and 302.3.5):

Sustained	: 33.8,	Allowable	: 138.0 N./mm ²	Passed
Expansion	: 0.0,	Allowable	: 311.2 N./mm ²	Passed
Occasional	: 37.2,	Allowable	: 183.5 N./mm ²	Passed
Shear	: 9.6,	Allowable	: 96.6 N./mm ²	Passed

The number of cycles on this nozzle was assumed to be 7000 or less for the determination of the expansion stress allowable.

Nozzle Junction Minimum Design Metal Temperature (MDMT) Calculations:

Nozzle Neck to Flange Weld, min(Curve:B, Curve:A)

Govrn. thk, $t_g = 29.2$, $t_r = 19.9$, $c = 3 \text{ mm.}$, $E^* = 1$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.76$, Temp. Reduction = $13 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve A	23 °C
Min Metal Temp. at Required thickness (UCS 66.1)	10 °C

Nozzle-Shell/Head Weld (UCS-66(a)1(b)), min(Curve:B, Curve:C)

Govrn. thk, $t_g = 74$, $t_r = 70.5$, $c = 3 \text{ mm.}$, $E^* = 1$
Thickness Ratio = $t_r \cdot E^* / (t_g - c) = 0.99$, Temp. Reduction = $0 \text{ }^\circ\text{C}$

Min Metal Temp. w/o impact per UCS-66, Curve B	25 °C
Min Metal Temp. at Required thickness (UCS 66.1)	25 °C

Gov. MDMT of the nozzle to shell joint welded assembly : 25 °C



Flange MDMT including Temperature reduction per UCS-66.1:

Unadjusted MDMT of ASME B16.5/47 flanges per UCS-66(c) -18 °C
 Flange MDMT with Temp reduction per UCS-66(b)(1)(-b) -32 °C

Where the Stress Reduction Ratio per UCS-66(b)(1)(-b) is :
 Design Pressure/Ambient Rating = 191.00/255.10 = 0.749

Weld Size Calculations, Description: S2-Shell Outlet

Intermediate Calc. for nozzle/shell Welds Tmin 19.0000 mm.

Results Per UW-16.1:

	Required Thickness	Actual Thickness
Nozzle Weld	6.0000 = Min per Code	7.0700 = 0.7 * Wo mm.

Weld Strength and Weld Loads per UG-41.1, Sketch (a) or (b)

Weld Load [W]:
 $= \max(0, (A-A1+2 \cdot t_n \cdot f_{r1} \cdot (E1 \cdot t - t_r)) \cdot S_v)$
 $= \max(0, (185 - 1.41 + 2 \cdot 107 \cdot 1 \cdot (1 \cdot 71 - 70.5)) \cdot 118)$
 $= 222829.22 \text{ Kgf}$

For hub type nozzles, A2 includes the area of the hub.

Note: F is always set to 1.0 throughout the calculation.

Weld Load [W1]:
 $= (A2+A4-(W_i-\text{Can}/.707)^2 \cdot f_{r2}) \cdot S_v$
 $= (187 + 1 - 0 \cdot 1) \cdot 118$
 $= 225779.44 \text{ Kgf}$

Weld Load [W2]:
 $= (A2 + A3 + A4 + (2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S_v$
 $= (187 + 0 + 1 + (152)) \cdot 118$
 $= 408596.88 \text{ Kgf}$

Weld Load [W3]:
 $= (A2+A3+A4+A5+(2 \cdot t_n \cdot t \cdot f_{r1})) \cdot S$
 $= (187 + 0 + 1 + 0 + (152)) \cdot 118$
 $= 408596.88 \text{ Kgf}$

Strength of Connection Elements for Failure Path Analysis

Shear, Outward Nozzle Weld [Sonw]:
 $= (\pi/2) \cdot D_{lo} \cdot W_o \cdot 0.49 \cdot S_{nw}$
 $= (3.14/2.0) \cdot 477 \cdot 10 \cdot 0.49 \cdot 118$
 $= 44194. \text{ Kgf}$

Shear, Nozzle Wall [Snw]:
 $= (\pi \cdot (D_{lr} + D_{lo})/4) \cdot (Thk - \text{Can}) \cdot 0.7 \cdot S_n$
 $= (3.14 \cdot 185) \cdot (110 - 3) \cdot 0.7 \cdot 138$
 $= 612888. \text{ Kgf}$

Tension, Shell Groove Weld [Tngw]:
 $= (\pi/2) \cdot D_{lo} \cdot (T - \text{cas}) \cdot 0.74 \cdot S_{ng}$
 $= (3.14/2.0) \cdot 477 \cdot (74 - 3) \cdot 0.74 \cdot 138$
 $= 554183. \text{ Kgf}$

See UG-41 and UW-15 for clarification of allowable stresses.

Strength of Failure Paths:

Failure Path [Path 1-1]:
 $= (\text{SONW} + \text{SNW}) = (44194 + 612888) = 657082 \text{ Kgf}$

Failure Path [Path 2-2]:
 $= (\text{Sonw} + \text{Tpgw} + \text{Tngw} + \text{Sinw})$
 $= (44194 + 0 + 554183 + 0) = 598377 \text{ Kgf}$

Failure Path [Path 3-3]:
 = (Sonw + Tngw + Sinw)
 = (44194 + 554183 + 0) = 598377 Kgf

Summary of Failure Path Calculations:

Path 1-1 = 657081 Kgf, must exceed W = 222829 Kgf or W1 = 225779 Kgf
 Path 2-2 = 598377 Kgf, must exceed W = 222829 Kgf or W2 = 408596 Kgf
 Path 3-3 = 598377 Kgf, must exceed W = 222829 Kgf or W3 = 408596 Kgf

Maximum Allowable Pressure for this Nozzle at this Location:

Converged Maximum Allowable Pressure in the Operating case: 192 bars

Note: The MAWP of this junction was limited by the parent Shell/Head.

The Drop for this Nozzle is : 81.5037 mm.
 The Cut Length for this Nozzle is, Drop + Ho + H + T : 355.5037 mm.

Input Echo, WRC297 Item 1, Description: S2-Shell Outle:

Diameter Basis for Cylindrical Shell	ID	
Shell Internal Corrosion Allowance	3.0000	mm.
Shell External Corrosion Allowance	0.0000	mm.
Shell Diameter	780.000	mm.
Shell Thickness	74.0000	mm.
Shell Stress Concentration Factor	1.000	
Vessel Material	SA-516 60	
Vessel Cold S.I. Allowable	Smc 118.00	N./mm ²
Vessel Hot S.I. Allowable	Smh 118.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Diameter Basis for Nozzle	ID	
Nozzle Corrosion Allowance	3.0000	mm.
Nozzle Diameter	257.200	mm.
Nozzle Thickness	110.0000	mm.
Nozzle Stress Concentration Factor	1.000	
Nozzle Material	SA-266 2	
Nozzle Cold S.I. Allowable	SNmc 138.00	N./mm ²
Nozzle Hot S.I. Allowable	SNmh 138.00	N./mm ²

See ASME VIII Div. 2, 4.1.6.3, for determination of the SPS allowable.

Note:

External Forces and Moments in WRC 107/537 Convention:
 These loads are assumed to be SUSstained loads.

Design Internal Pressure	Dp	191.00	bars
Radial Load	P	-9789.91	Kgf
Circumferential Shear	Vc	9789.91	Kgf
Longitudinal Shear	Vl	7342.47	Kgf
Circumferential Moment	Mc	7.63	Kg-m.
Longitudinal Moment	ML	7.63	Kg-m.
Torsional Moment	Mt	8.81	Kg-m.

Include Axial Pressure Thrust	No
Include Pressure Stress Indices per Div. 2	No
Local Loads applied at end of Nozzle/Attachment	No

Warning - The ratio, Dv/Tv (12.541) is < 20
 check the Limitations of WRC bulletin 297.
 Warning - The ratio, Dn/Dv (.514) is > 0.5
 check the Limitations of WRC bulletin 297.
 Warning - The ratio, Dn/Tn (4.338) is < 10
 check the Limitations of WRC bulletin 297.

Stress Computations at the Edge of the Nozzle:

Stress Attenuation Diameter (for Insert Plates) per WRC 297:

$$= \text{NozzleOD} + 2 \cdot 1.65 \cdot \sqrt{(\text{Rmean}(t - ca))}$$

$$= 477.2 + 2 \cdot 1.65 \cdot \sqrt{(428.5(74.0 - 3.0))}$$

$$= 1052.797 \text{ mm.}$$

WRC 297 Curve Access Parameters:

Vessel Mean Diameter (D) = 857.000 mm.
 Nozzle Outside Diameter (d) = 477.200 mm.
 Vessel Thickness used (T) = 71.000 mm.
 Nozzle Thickness used (t) = 107.000 mm.
 T / t = 0.664
 d / t = 10.000

$$\lambda = [(d/D) \cdot \sqrt{(D/T)}] = 1.935$$

Nr/P = 0.075
 Mr/P = 0.136
 M0/P = 0.041
 N0/P = 0.101
 MrD/Mc = 0.273
 NrDL/Mc = 0.114
 M0d/Mc = 0.089
 N0DL/Mc = 0.064
 MrD/Ml = 0.116
 NrDL/Ml = 0.069
 M0D/Ml = 0.039
 N0DL/Ml = 0.127

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	96	116	96	116
Outplane Stress Summary	103	113	103	113

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses parallel to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	48	48	48	48
Inplane Stress Summary	65	34	65	34

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress normal to longitudinal plane)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Outplane Shear (Vc)	1	1	-1	-1
Outplane Shear (Vl)	0	0	0	0
Outplane Shear (Mt)	0	0	0	0
Shear Stress Summary	1	1	-1	-1

Vessel Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Two * Max Shear Stress	103	113	103	113

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Membrane (P)	1	1	1	1
Outplane Bending (P)	4	-4	4	-4
Outplane Membrane (Mc)	0	0	0	0
Outplane Bending (Mc)	0	0	0	0
Outplane Membrane (ML)	0	0	0	0
Outplane Bending (ML)	0	0	0	0
Normal Pressure Stress	48	48	48	48
Outplane Stress Summary	55	45	55	45

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses parallel to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Inplane Membrane (P)	1	1	1	1
Inplane Bending (P)	15	-15	15	-15
Inplane Membrane (Mc)	0	0	0	0
Inplane Bending (Mc)	0	0	0	0
Inplane Membrane (ML)	0	0	0	0
Inplane Bending (ML)	0	0	0	0
Inplane Pressure Stress	96	116	96	116
Inplane Stress Summary	113	101	114	101

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress normal to circumferential plane)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Outplane Shear (Vc)	0	0	0	0
Outplane Shear (Vl)	-1	-1	1	1
Torsional Shear (Mt)	0	0	0	0
Shear Stress Summary	-1	-1	1	1

Vessel Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stress Intensities)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Two * Max Shear Stress	113	102	114	101

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses in the hoop direction)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	20	42	20	42
Hoop Stress Summary	21	44	21	44

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stresses Normal to pipe cross-section)	Au Top Outside	Al Top Inside	Bu Bottom Outside	Bl Bottom Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	4	-4	4	-4
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	10	10	10	10
Axial Stress Summary	15	5	15	5

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Shear stress)	Au Outside	Al Inside	Bu Outside	Bl Inside
Shear due to (Vc)	1	1	-1	-1
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	1	1	-1	-1

Nozzle Stresses (N./mm²):

LONGITUDINAL PLANE (Stress Intensities)	Au Outside	Al Inside	Bu Outside	Bl Inside
Two * Max Shear Stress	22	44	22	44

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses in the hoop direction)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Hoop Membrane (P)	1	1	1	1
Hoop Bending (P)	0	0	0	0
Hoop Membrane (Mc)	0	0	0	0
Hoop Bending (Mc)	0	0	0	0
Hoop Membrane (ML)	0	0	0	0
Hoop Bending (ML)	0	0	0	0
Hoop Pressure Stress	20	42	20	42
Hoop Stress Summary	21	44	21	44

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Stresses Normal to pipe cross-section)	Cu Left Outside	Cl Left Inside	Du Right Outside	Dl Right Inside
Axial Membrane (P)	0	0	0	0
Axial Bending (P)	4	-4	4	-4
Axial Membrane (Mc)	0	0	0	0
Axial Bending (Mc)	0	0	0	0
Axial Membrane (ML)	0	0	0	0
Axial Bending (ML)	0	0	0	0
Axial Pressure Stress	10	10	10	10
Axial Stress Summary	15	5	15	5

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE (Shear stress)	Cu Outside	Cl Inside	Du Outside	Dl Inside
Shear due to (Vc)	0	0	0	0
Shear due to (Vl)	0	0	0	0
Shear due to Torsion	0	0	0	0
Shear Stress Summary	0	0	0	0

Nozzle Stresses (N./mm²):

CIRCUMFERENTIAL PLANE
 (Stress Intensities Outside Inside Outside Inside

Two * Max Shear Stress 22 44 22 44

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Vessel Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	96	116	96	116	96	116	96	116
Circ. Pl (SUS)	1	1	1	1	1	1	1	1
Circ. Q (SUS)	4	-4	4	-4	15	-15	15	-15
Long. Pm (SUS)	48	48	48	48	48	48	48	48
Long. Pl (SUS)	1	1	1	1	1	1	1	1
Long. Q (SUS)	15	-15	15	-15	4	-4	4	-4
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	-1	-1	1	1
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	96.0	116.0	96.0	116.0	96.0	116.0	96.0	116.0
Pm+Pl (SUS)	97.0	117.0	97.0	117.0	97.0	117.0	97.0	117.0
Pm+Pl+Q (Total)	101.0	113.0	101.0	113.0	112.0	102.0	112.0	102.0

Vessel Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	116.00	118.00	Passed
Pm+Pl (SUS)	117.01	177.00	Passed
Pm+Pl+Q (TOTAL)	113.01	353.99	Passed

Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.

WRC 297 Stress Summations per ASME Sec. VIII Div. 2:

Nozzle Stress Summation at Vessel-Nozzle Junction (N./mm²):

Type of Stress Int. Location	Stress Values at							
	Au	Al	Bu	Bl	Cu	Cl	Du	Dl
Circ. Pm (SUS)	20	42	20	42	20	42	20	42
Circ. Pl (SUS)	1	1	1	1	1	1	1	1
Circ. Q (SUS)	0	0	0	0	0	0	0	0
Long. Pm (SUS)	10	10	10	10	10	10	10	10
Long. Pl (SUS)	0	0	0	0	0	0	0	0
Long. Q (SUS)	4	-4	4	-4	4	-4	4	-4
Shear Pm (SUS)	0	0	0	0	0	0	0	0
Shear Pl (SUS)	1	1	-1	-1	0	0	0	0
Shear Q (SUS)	0	0	0	0	0	0	0	0
Pm (SUS)	20.0	42.0	20.0	42.0	20.0	42.0	20.0	42.0
Pm+Pl (SUS)	21.1	43.0	21.1	43.0	21.0	43.0	21.0	43.0
Pm+Pl+Q (Total)	21.1	43.0	21.1	43.0	21.0	43.0	21.0	43.0



Nozzle Stress Summation Comparison (N./mm²):

Type of Stress Int.	Max. S.I.	S.I. Allowable	Result
Pm (SUS)	42.00	138.00	Passed
Pm+Pl (SUS)	43.03	207.00	Passed
Pm+Pl+Q (TOTAL)	43.03	413.99	Passed

*Because only sustained loads were specified, the Pm+Pl+Q allowable was 3 * Smh.*

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Input Echo, Tubesheet Number 1, Description:

Shell Data:

Main Shell Description: Main Shell

Shell Maximum Design Pressure	Psd,max	191.00 bars
Shell Maximum Operating Pressure	Psox,max	191.00 bars
Shell Minimum Operating Pressure	Psox,min	0.00 bars
Shell Thickness	ts	74.0000 mm.
Shell Internal Corrosion Allowance	cas	3.0000 mm.
Shell External Corrosion Allowance	casext	0.0000 mm.
Inside Diameter of Shell	Ds	780.000 mm.
Shell Temperature for Internal Pressure	Ts	200.00 °C
Shell Material		SA-516 60

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number		K02100
Shell Allowable Stress at Temperature	Ss	118.00 N./mm ²
Shell Allowable Stress at Ambient		118.00 N./mm ²

Channel Data:

Channel Description: Ch Shell

Channel Type:		Cylinder
Channel Maximum Design Pressure	Ptd,max	147.00 bars
Channel Maximum Operating Pressure	Ptox,max	147.00 bars
Channel Minimum Operating Pressure	Ptox,min	0.00 bars
Channel Thickness	tc	56.0000 mm.
Channel Internal Corrosion Allowance	cac	3.0000 mm.
Channel External Corrosion Allowance	cacext	0.0000 mm.
Inside Diameter of Channel	Dc	780.000 mm.
Channel Design Temperature	TEMPC	85.00 °C
Channel Flange Matl		SA-516 60

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number		K02100
Channel Allowable Stress at Temperature	Sc	118.00 N./mm ²
Channel Allowable Stress at Ambient		118.00 N./mm ²

Tube Data:

Number of Tube Holes	Nt	520
Tube Wall Thickness	et	3.0480 mm.
Tube Outside Diameter	D	19.0500 mm.
Total Straight Tube Length	Lt	3658.00 mm.
Straight Tube Length (bet. inner tubsht faces) L		3478.00 mm.
Design Temperature of the Tubes		200.00 °C
Tube Material		SA-179
Tube Material UNS Number		K01200
Is this a Welded Tube		No
Tube Material Specification used	Smls. tube	
Tube Allowable Stress at Temperature		92.40 N./mm ²
Tube Allowable Stress At Ambient		92.40 N./mm ²
Tube Yield Stress At design Temperature	Syt	154.00 N./mm ²
Tube Pitch (Center to Center Spacing)	P	25.3990 mm.
Tube Layout Pattern		Square
Fillet Weld Leg	af	2.4130 mm.
Groove Weld Leg	ag	2.4130 mm.
Tube-Tubesheet Joint Weld Type	Full	Strength
Method for Tube-Tubesheet Jt. Allowable		UW-20.3
Is the Tube Inset		No
Radius to Outermost Tube Hole Center	ro	354.223 mm.
Largest Center-to-Center Tube Distance	Ul	0.0000 mm.
Length of Expanded Portion of Tube	ltx	0.0000 mm.
Tube-side pass partition groove depth	hg	0.0000 mm.
Area of Tube Layout	Ap	3323.4 cm ²
Perimeter of Tube Layout	Cp	2394.5 mm.



Tubesheet Data:

Tubesheet TYPE: U-tube, Integral both sides, Conf. a

Tubesheet Design Metal Temperature T 200.00 °C
 Tubesheet Material SA-266 2

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number K03506
 Tubesheet Allowable Stress at Temperature S 138.00 N./mm²
 Tubesheet Allowable Stress at Ambient Tt 138.00 N./mm²
 Thickness of Tubesheet h 180.0000 mm.
 Tubesheet Corr. Allowance (Shell side) Cats 6.0000 mm.
 Tubesheet Corr. Allowance (Channel side) Catc 6.0000 mm.
 Tubesheet Outside Diameter A 928.000 mm.

Area of the Untubed Lanes AL 0.0 cm²

Junction Stress Reduction option Perform Plastic Calculation
 Perform Differential Pressure Design No
 Run Multiple Load Cases Yes

Tubesheet Integral with Both
 Tubesheet Extended as Flange No

ASME TubeSheet Results per Part UHX/4.18, 2025

Elasticity/Expansion Material Properties :

Shell - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature 200.0 °C 0.19258E+09 KPa.
 Elastic Mod. at Ambient Temperature 21.1 °C 0.20270E+09 KPa.

Channel - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Design Temperature 85.0 °C 0.19904E+09 KPa.
 Elastic Mod. at Ambient Temperature 21.1 °C 0.20270E+09 KPa.

Tubes - TM-1 Carbon Steels with C<= 0.3%

Elastic Mod. at Tubsht. Design Temp. 200.0 °C 0.19258E+09 KPa.
 Elastic Mod. at Ambient Temperature 21.1 °C 0.20270E+09 KPa.

TubeSheet - TM-1 Carbon Steels with C> 0.3%

Elastic Mod. at Design Temperature 200.0 °C 0.19120E+09 KPa.
 Elastic Mod. at Ambient Temperature 21.1 °C 0.20132E+09 KPa.

Tube Required Thickness under Internal Pressure (Tubeside pressure):

Thickness Due to Internal Pressure:
 $= (P \cdot (D/2 - CAE)) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)
 $= (147 \cdot (19/2 - 0)) / (92.4 \cdot 1 + 0.4 \cdot 147)$
 $= 1.4248 + 0.0000 = 1.4248 \text{ mm.}$

Tube Required Thickness under External Pressure (Shellside pressure) :

External Pressure Chart CS-1 at 200.00 °C
 Elastic Modulus for Material 193602528.00 KPa.

Results for Max. Allowable External Pressure (Emawp):

TCA	ODCA	SLEN	D/T	L/D	Factor A	B
3.0480	19.05	3832.22	6.25	50.0000	0.0281600	93.30

EMAWP = $(2.167 / (D/T) - 0.0833) \cdot B = 246 \text{ bars}$

Results for Req'd Thickness for Ext. Pressure (Tca):



TCA	ODCA	SLEN	D/T	L/D	Factor A	B
2.5322	19.05	3832.22	7.52	50.0000	0.0194350	93.30

EMAWP = $(2.167 / (D/T) - 0.0833) * B = 191$ bars

Summary of Tube Required Thickness Results:

Total Required Thickness including Corrosion all.	2.5322 mm.
Allowable Internal Pressure at Corroded thickness	339.06 bars
Required Internal Design Pressure	147.00 bars
Allowable External Pressure at Corroded thickness	245.77 bars
Required External Design Pressure	191.00 bars
Required Thickness due to Shell Side pressure	2.5322 mm.

Detailed Results for load Case D3 un-corr. (Psd,max + Ptd,max)

Results for ASME U-tube Tubesheet Calculations for Configuration a:

Per Edition 2025, Original Thickness :

ASME Section VIII-2 Part 4.18.7.4 Step 1:

Equivalent outer tube limit circle diameter [Do]:
 $= 2 \cdot r_o + dt$
 $= 2 \cdot 354 + 19$
 $= 727.496$ mm.

Basic ligament efficiency for shear [μ]:
 $= (p - dt) / p$
 $= (25.4 - 19) / 25.4$
 $= 0.250$

ASME Section VIII-2 Part 4.18.7.4 Step 2:

Ratio [ps]:
 $= D_s / D_o$ (Configurations a, b, c)
 $= 780 / 727$
 $= 1.0722$

Ratio [pc]:
 $= D_c / D_o$ (Configurations a, e, f)
 $= 780 / 727$
 $= 1.0722$

ASME Section VIII-2 Part 4.18.7.4 Step 3:

Ratio [p]:
 $= l_{tx} / h$
 $= 0 / 180$
 $= 0.000$ (must be $0 \leq p \leq 1$)

Effective tube hole Diameter [d^*]:
 $= \max(dt - 2tt(EtT/E) \cdot (StT/S) \cdot (p); dt - 2tt)$
 $= \max(19 - 2 \cdot 3.05 (192580832 / 191201920) \cdot (92.4 / 138) \cdot (0); 19 - 2 \cdot 3.05)$
 $= 19.0500$ mm.

Effective tube pitch [p^*]:
 $= p / \sqrt{ (1 - 4 \cdot \min(A; 4 \cdot D_o \cdot p) / (\pi \cdot D_o^2)) }$
 $= 25.4 / \sqrt{ (1 - 4 \cdot \min(0; 4 \cdot 727 \cdot 25.4) / (\pi \cdot 727^2)) }$
 $= 25.3990$ mm.

Effective ligament efficiency for bending [μ^*]:
 $= (p^* - d^*) / p^*$
 $= (25.4 - 19) / 25.4$
 $= 0.2500$

$h/p = 7.086893$; $\mu^* = 0.249970$; $E^* = 54947436$ KPa.
 $E^*/E = 0.287379$; $v^* = 0.332864$



Note: As h/p (7.087) is > 2, data values for h/p = 2 were used.

ASME Section VIII-2 Part 4.18.7.4 Step 4:

Shell coefficient [β_s]:

$$= \left(\left(12(1 - \nu_s^2) \right)^{1/4} / \sqrt{(D_s + t_s)t_s} \right) \\ = \left(\left(12(1 - 0.3^2) \right)^{1/4} / \sqrt{(780 + 74)74} \right) \\ = 0.0072 \text{ 1/mm.}$$

Shell coefficient [k_s]:

$$= \beta_s \cdot E_s \cdot t_s^3 / (6(1 - \nu_s^2)) \\ = 0.0072 \cdot 1925808 \cdot 74^3 / (6(1 - 0.3^2)) \\ = 1033537856.0000 \text{ bars} \cdot \text{mm.}^2$$

Shell coefficient [λ_s]:

$$= 6(D_s + t_s) \cdot k_s / h^3 \cdot (1 + h \cdot \beta_s + (h \cdot \beta_s)^2 / 2) \\ = 6(780 + 74) \cdot 1033537856 / 180^3 \cdot (1 + 180 \cdot 0.0072 + (180 \cdot 0.0072)^2 / 2) \\ = 2859250.5 \text{ bars}$$

Shell coefficient [δ_s]:

$$= D_s^2 / (4 \cdot E_s \cdot t_s) \cdot (1 - D_s \cdot \nu_s / (2(D_s + t_s))) \\ = 780^2 / (4 \cdot 192580832 \cdot 74) \cdot (1 - 780 \cdot 0.3 / (2(780 + 74))) \\ = 0.92102\text{E-}02 \text{ mm./N./mm}^2$$

Parameter [w_s]:

$$= p_s \cdot k_s \cdot \beta_s \cdot \delta_s (1 + h \cdot \beta_s) \\ = 1.07 \cdot 1033537856 \cdot 0.0072 \cdot 0.009210195 (1 + 180 \cdot 0.0072) \\ = 16987.4590 \text{ mm.}^2$$

Channel coefficient [β_c]:

$$= \left(\left(12(1 - \nu_c^2) \right)^{1/4} / \sqrt{(D_c + t_c)t_c} \right) \\ = \left(\left(12(1 - 0.3^2) \right)^{1/4} / \sqrt{(780 + 56)56} \right) \\ = 0.0084 \text{ 1/mm.}$$

Channel coefficient [k_c]:

$$= \beta_c \cdot E_c \cdot t_c^3 / (6(1 - \nu_s^2)) \\ = 0.0084 \cdot 1990422 \cdot 56^3 / (6(1 - 0.3^2)) \\ = 537867328.0000 \text{ bars} \cdot \text{mm.}^2$$

Channel coefficients [CC, CqDelta, CqTheta, CmTheta, CmDelta]:

$$CqDelta = CqTheta = CmTheta = CmDelta = 1$$

Channel coefficient [λ_c]:

$$= 6(D_c + t_c / \cos(\theta)) \cdot k_c / h^3 \cdot (CmTheta + h \cdot \beta_c \cdot (CmDelta + CqTheta) / 2 + (h \cdot \beta_c)^2 / 2 \cdot CqDelta) \\ = 6(780 + 56 / \cos(0)) \cdot 537867328 / 180^3 \cdot (1 + 180 \cdot 0.0084 \cdot (1 + 1) / 2 + (180 \cdot 0.0084)^2 / 2 \cdot 1) \\ = 1691194.1 \text{ bars}$$

Channel coefficient [δ_c]:

$$= D_c^2 / (4 \cdot E_c \cdot t_c) \cdot (1 - D_c \cdot \nu_c / (2(D_c + t_c))) \\ = 780^2 / (4 \cdot 199042256 \cdot 56) \cdot (1 - 780 \cdot 0.3 / (2(780 + 56))) \\ = 0.11735\text{E-}01 \text{ mm./N./mm}^2$$

Channel coefficient [zetac]:

$$\text{zetac} = 0$$

Parameter [w_c]:

$$= p_c \cdot k_c [\delta_c (CmTheta \cdot \beta_c + CqDelta \cdot h \cdot \beta_c^2) - \zeta_c (CmTheta + h \cdot CqTheta \cdot h \cdot \beta_c / 2)] \\ = 1.07 \cdot 537867328 [0.011735274 (1 + 0.0084 \cdot 1 \cdot 180) - 0.000 (1/180 + 1 \cdot 0.0084/2)] \\ = 14285.1025 \text{ mm.}^2$$

ASME Section VIII-2 Part 4.18.7.4 Step 5:

Diameter ratio [K]:

$$= A / D_o \\ = 928/727$$



$$= 1.2756$$

Coefficient [F]:

$$\begin{aligned} &= (1 - v^*) / E^* \cdot (\lambda_s + \lambda_c + E \cdot \ln(K)) \\ &= (1 - 0.33) / 54947436 \cdot (2859250 + 1691194 + \\ &\quad 191201920 \cdot \ln(1.28)) \\ &= 6.0900 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 6:

Moment on tubesheet due to pressures (Ps, Pt) [Mts]:

$$\begin{aligned} &= Do^2/16 \cdot [(ps-1) \cdot (ps^2+1) \cdot Ps - (pc-1) \cdot (pc^2+1) \cdot Pt] \\ &= 727^2/16 \cdot [(1.07-1) \cdot (1.07^2+1) \cdot 191 - \\ &\quad (1.07-1) \cdot (1.07^2+1) \cdot 147] \\ &= 225789.3125 \text{ bars} \cdot \text{mm}^2 \end{aligned}$$

Moment acting on unperforated tubesheet rim [M*]:

$$\begin{aligned} &= Mts + \omega_c \cdot Pt - \omega_s \cdot Ps \\ &= 225789 + 14285 \cdot 147 - 16987 \cdot 191 \\ &= -918905.4375 \text{ bars} \cdot \text{mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 7:

Maximum bending moment acting on periphery of tubesheet [Mp]:

$$\begin{aligned} &= (M^*) - Do^2/32 \cdot F(Ps - Pt) / (1 + F) \\ &= (-918905) - 727^2/32 \cdot 6.09(191 - 147) / (1 + 6.09) \\ &= -754685.1250 \text{ bars} \cdot \text{mm}^2 \end{aligned}$$

Maximum bending moment acting on center of tubesheet [Mo]:

$$\begin{aligned} &= Mp + Do^2/64 \cdot (3 + v^*) \cdot (Ps - Pt) \\ &= -754685 + 727^2/64 \cdot (3 + 0.33) \cdot (191 - 147) \\ &= 458009.5938 \text{ bars} \cdot \text{mm}^2 \end{aligned}$$

Maximum bending moment acting on tubesheet [M]:

$$\begin{aligned} &= \max(|Mp|; |Mo|) \\ &= \max(|-754685|; |458010|) \\ &= 754685.1250 \text{ bars} \cdot \text{mm}^2 \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 8:

Tubesheet Bending Stress at original thickness:

$$\begin{aligned} &= 6 \cdot M / (\mu^* (h - hg')^2) \\ &= 6 \cdot 754685 / (0.25 (180 - 0)^2) \\ &= 55.9125 \text{ N./mm}^2 \end{aligned}$$

Allowable Tubesheet Bending Stress [σ_{All}]:

$$\begin{aligned} &= 2 \cdot S \\ &= 2 \cdot 138 \\ &= 275.99 \text{ N./mm}^2 \end{aligned}$$

Tubesheet bending stress at final thickness [σ]:

$$\begin{aligned} &= 6 \cdot M / (\mu^* (h - hg')^2) \\ &= 6 \cdot 724195 / (0.25 (79.4 - 0)^2) \\ &= 275.9950 \text{ N./mm}^2 \end{aligned}$$

Required tubesheet thickness for Bending Stress [HreqB]:

$$\begin{aligned} &= h + Cats + Catc \\ &= 79.4 + 0 + 0 \\ &= 79.3635 \text{ mm.} \end{aligned}$$

Required tubesheet thickness for given loadings (includes CA) [Hreq]:

$$\begin{aligned} &= \max(HreqB; HreqS) \\ &= \max(79.4; 22.1) \\ &= 79.3635 \text{ mm.} \end{aligned}$$

ASME Section VIII-2 Part 4.18.7.4 Step 9:

Absolute pressure differential [|Ps - Pt|]:

$$\begin{aligned} &= |Ps - Pt| \\ &= |191 - 147| \end{aligned}$$



= 44.000 bars

Shear Stress check [τ_{limit}]:

$$= 4 \cdot \mu \cdot h / D_o \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)$$

$$= 4 \cdot 0.25 \cdot 180 / 727 \cdot \min(0.8 \cdot 138; 0.533 \cdot 213)$$

$$= 27.312 \text{ N./mm}^2$$

$|P_s - P_t|$ is $\leq$ the Shear Stress limit (τ_{limit}). Therefore, the required thickness for shear is for information only.

Minimum required thickness for Shear [H_{reqS}]:

$$= 1 / (4 \cdot \mu \cdot \min(0.8 \cdot S; 0.533 \cdot S_y)) \cdot (4 \cdot A_p / C_p) \cdot |P_s - P_t| + C_{ats} + C_{atc}$$

$$= 1 / (4 \cdot 0.25 \cdot 110) \cdot (4 \cdot 3323 / 2395) \cdot |191 - 147| + 0$$

$$= 22.1307 \text{ mm.}$$

ASME Section VIII-2 Part 4.18.7.4 Step 10:

Note: For a given Shell thickness of: 74.000mm.

Min. Shell len. adjacent to the tubesheet is: 432.450 mm.

Axial Shell Membrane Stress [$\sigma_{s,m}$]:

$$= D_s^2 / (4 \cdot t_s (D_s + t_s)) \cdot P_s$$

$$= 780^2 / (4 \cdot 74 (780 + 74)) \cdot 191$$

$$= 45.9725 \text{ N./mm}^2$$

Axial Shell Bending Stress [$\sigma_{s,b}$]:

$$= 6 \cdot k_s / t_s^2 \cdot [\beta S \cdot P_s + 6(1 - \nu^*) / E^* \cdot D_o / h^3 \cdot (1 + h \cdot \beta S / 2) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 23235388 / 74^2 \cdot [0.0072 \cdot 0.17 + 6(1 - 0.33) / 54947436 \cdot 727 / 180^3 \cdot (1 + 180 \cdot 0.0072 / 2) \cdot (-754685 + 727^2 / 32 \cdot (191 - 147))]$$

$$= 139.490 \text{ N./mm}^2$$

Shell Membrane + Bending Stress [σ_s]:

$$= |\sigma_{s,m}| + |\sigma_{s,b}| \text{ (Should be } \leq 1.5 \cdot S_s)$$

$$= 185.463 \text{ N./mm}^2 \text{ (Should be } \leq 177.00)$$

Note: For a given Channel thickness of: 56.000 mm.

Min. Channel len. adjacent to the tubesheet is: 376.196mm.

Axial Channel Membrane Stress [$\sigma_{c,m}$]:

$$= D_c^2 / (4 \cdot t_c (D_c + t_c) \cdot \cos(\theta)) \cdot P_t$$

$$= 780^2 / (4 \cdot 56 (780 + 56) \cdot \cos(0)) \cdot 147$$

$$= 47.761 \text{ N./mm}^2$$

Axial Channel Bending Stress [$\sigma_{c,b}$]:

$$= 6 \cdot k_c / t_c^2 \cdot [(\delta_c \cdot C_m \Delta \cdot \beta_c - \zeta_c \cdot C_m \theta) \cdot P_t - 6((1 - \nu^*) / E^*) \cdot D_o / h^3 \cdot (C_m \theta + h \cdot \beta_c / 2 \cdot C_m \Delta) \cdot (M_p + D_o^2 / 32 \cdot (P_s - P_t))]$$

$$= 6 \cdot 12092016 / 56^2 \cdot [(0.011735274 \cdot 1 \cdot 0.0084 - 0.000 \cdot 1) \cdot 147 - 6((1 - 0.33) / 54947436) \cdot 727 / 180^3 \cdot (1 + 0.0084 \cdot 180 / 2 \cdot 1) \cdot (-754685 + 727^2 / 32 \cdot (191 - 147))]$$

$$= 153.595 \text{ N./mm}^2$$

Channel Membrane + Bending Stress [σ_c]:

$$= |\sigma_{c,m}| + |\sigma_{c,b}| \text{ (Should be } \leq 1.5 \cdot S_c)$$

$$= 201.356 \text{ N./mm}^2 \text{ (Should be } \leq 177.00)$$

Step 11, The Cylinder-to-Tubesheet Juncture is Overstressed.

Performing a simplified Elastic-Plastic calculation

(option 3) to reduce the overstress condition.

Modify E_s and/or E_c and recompute from Step 4 onwards.

Modified Elasticity [E_s]:

$$= E_s \cdot \sqrt{1.5 \cdot S_s / \sigma_s}$$

$$= 192580832 \cdot \sqrt{1.5 \cdot 118 / 185}$$

$$= 188134096.0 \text{ KPa.}$$

Modified Elasticity [E_c]:



$$= E_c \cdot \sqrt{(1.5 \cdot S_c / \sigma_c)}$$

$$= 199042256 \cdot \sqrt{(1.5 \cdot 118/201)}$$

$$= 186614576.0 \text{ KPa.}$$

Tubesheet Bending Stress at Original Thk., after the Elastic-Plastic iteration:

$$= 6 \cdot M / (\mu \cdot (h - h'g)^2)$$

$$= 6 \cdot 755502 / (0.25 \cdot (180 - 0)^2)$$

$$= 56.0 \text{ N./mm}^2 \text{ (Should be } \leq 276.0 \text{)}$$



Note: The tubesheet is Not overstressed after the Elastic-Plastic iteration.
 The design is acceptable. Recomputing tubesheet required thickness.

Required Tubesheet Thk., for Bending Stress after the Elastic-Plastic iteration [HreqB]:

$$= h + C_{ats} + C_{atc} = 79.4 + 0 + 0 = 79.4 \text{ mm.}$$

Required Tubesheet Thk. after the Elastic-Plastic iteration (includes CA) [Hreq]:

$$= \max(H_{reqB}; H_{reqS}) = \max(79.4 ; 22.1) = 79.4 \text{ mm.}$$

Tube Metal Area [At]:

$$= \pi (d_o - t) t$$

$$= \pi (19 - 3.05) 3.05$$

$$= 1.532 \text{ cm}^2$$

Tube Weld Size Results per UW-20.3:

Tube Strength [Ft]:

$$= \pi \cdot t (d_o - t) S$$

$$= \pi \cdot 3.05 (19 - 3.05) 92.4$$

$$= 1443.692 \text{ Kg}$$

Groove Weld Strength [Fg]:

$$= 0.85 \cdot \pi \cdot a_g (d_o + 0.67 \cdot a_g) S_w \text{ (but not } > F_t)$$

$$= 0.85 \cdot \pi \cdot 2.41 (19 + 0.67 \cdot 2.41) 92.4$$

$$= 1254.6801 \text{ Kg}$$

Weld Strength Factor [fw]:

$$= S / S_w \text{ where } S_w = \min(S \text{ and } S_t)$$

$$= 92.4/92.4$$

$$= 1.000$$

Design Strength Ratio [fd]:

$$= 1 \text{ for full strength welds}$$

Fillet Weld Strength [Ff]:

$$= 0.55 \cdot \pi \cdot a_f (d_o + 0.67 \cdot a_f) S_w \text{ (but not } > F_t)$$

$$= 0.55 \cdot \pi \cdot 2.41 (19 + 0.67 \cdot 2.41) 92.4$$

$$= 811.8518 \text{ Kg}$$

Minimum Length of Weld [ar]:

$$= 2 \left(\sqrt{((0.75 \cdot d_o)^2 + 1.07 \cdot t \cdot (d_o - t) f_w \cdot f_d)} - 0.75 \cdot d_o \right)$$

$$= 2 \left(\sqrt{((0.75 \cdot 19)^2 + 1.07 \cdot 3.05 \cdot (19 - 3.05) 1 \cdot 1)} - 0.75 \cdot 19 \right)$$

$$= 3.445 \text{ mm.}$$

Minimum Required Fillet Weld Leg [afr]: 1.8288 mm.

Minimum Required Groove Weld Leg [agr]: 1.8288 mm.

Tube-Tubesheet Jt allowable, 1444 is >= tube strength 1444 Kg

Note: This tube-tubesheet joint is a Full Strength joint

Stress/Force summary for loadcase D3 un-corr. (Psd,max + Ptd,max):

Stress Description	Actual	Allowable	Pass/Fail
Tubesheet bend. stress	56.0 <=	276.0 N./mm ²	Ok
Tubesheet shear stress	13.6 <=	110.4 N./mm ²	Ok



Stress in Shell at Tubesheet 185.5 <= 378.0 N./mm² Ok
 Stress in Channel at Tubesheet 201.4 <= 408.0 N./mm² Ok

Thickness results for loadcase D3 un-corr. (Psd,max + Ptd,max):

Thickness (mm.)	Required	Actual	P/F
Tubesheet Thickness :	79.364	180.000	Ok
Tube-Tubesheet Fillet Weld Leg :	1.829	2.413	Ok
Tube-Tubesheet Groove Weld Leg :	1.829	2.413	Ok

U-Tube Tubesheet results per ASME Div. 2 Part 4.18.7 2025

Results for 6 Load Cases:

Case#	--Reqd. Thk. + CA Tbsht Extnsn	----	Tubesheet Bend Allwd	Stresses Shear Allwd	Case Type	Pass/ Fail
D1uc	141.594	...	168	276	45 110	Ps+Pt D1 Ok
D2uc	162.411	...	224	276	59 110	Ps+Pt D2 Ok
D3uc	79.364	...	56	276	14 110	Ps+Pt D3 Ok
D1c	152.881	...	191	276	49 110	Ps+Pt-c D1 Ok
D2c	173.548	...	254	276	63 110	Ps+Pt-c D2 Ok
D3c	91.300	...	63	276	15 110	Ps+Pt-c D3 Ok
Max:	173.5484	... mm.	0.921	0.572	(Str. Ratio)	

Load Case Definitions:

[Ps & Pt]:
 Shell-side and Tube-side Design or Operating Pressures
 derived from Psd,min Ptd,max, Psox,min, Ptox,max etc. per the
 Load Case Tables

[c]:
 With or Without Corrosion Allowance

[D1, D2, D3]:
 Design Load Cases using the Maximum and Minimum Design Pressures

[D4]:
 Design Load Case using the Minimum (Vacuum) Pressures (if specified)

Shell and Channel Stress Summary:

Case#	-- Integral Cyl. Req'd. Thk. Shell Channel	Shell Stress	Stress Allwd	Channel Stress	Stress Allwd	Pass Fail
D1uc	...	...	26	177	222 408	Ok
D2uc	...	...	211	378	21 177	Ok
D3uc	...	...	185	378	201 408	Ok
D1c	...	...	36	177	245 408	Ok
D2c	...	...	232	378	31 177	Ok
D3c	...	...	197	378	214 408	Ok
Max	...	...	0.615	0.600		

Note:
 Because there was no net pressure differential across the tubesheet due to
 the specified pressures psd,min and ptd,min; load case D4 was not processed as
 the resulting tubesheet bending stress would be zero.

Summary of Thickness Comparisons for 6 Load Cases:

Thickness (mm.)	Required	Actual	P/F
Tubesheet thickness :	173.548	180.000	Ok
Tubesheet thickness flanged extension :	0.000	1.000	Ok
Tube thickness :	2.532	3.048	Ok



Tube-tubesheet fillet weld leg :	1.829	2.413	Ok
Tube-tubesheet groove weld leg :	1.829	2.413	Ok

Given Shell Thickness: 74.0000 mm.
 Given Channel Thickness: 56.0000 mm.

Min Shell length of thk, (74.000) adj. to tubesheet: 432.450 mm.

Min Channel length of thk, (56.000) adj. to tubesheet: 376.196 mm.

Note: This is a full strength Tube to Tubesheet Joint.

Tubesheet MAWP used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAWP	Ø shellside Stress Rat.	Shellside MAWP	Ø tubeside Stress Rat.
Tubesheet Bending Stress	211.661	1.000	206.666	1.000
Tubesheet Shear Stress	334.014	1.000	334.014	1.000
Shell Stress (Axial, Junction)	334.014	0.457	206.666	1.000
Tubesheet-Channel Junction	211.661	1.000	206.666	1.000
Tube Pressure Stress	339.056	1.000	245.766	1.000
Minimum MAWP	211.661		206.666	

Tubesheet MAPnc used to Compute Hydrotest Pressure:

Stress / Force Condition	Tubeside MAPnc	Ø shellside Stress Rat.	Shellside MAPnc	Ø tubeside Stress Rat.
Tubesheet Bending Stress	240.538	1.000	234.705	1.000
Tubesheet Shear Stress	357.872	1.000	357.872	1.000
Shell Stress (Axial, Junction)	357.872	0.361	357.872	1.525
Tubesheet-Channel Junction	240.538	1.000	357.872	1.525
Tube Pressure Stress	339.056	1.000	252.447	1.000
Minimum MAPnc	240.538		234.705	

(*) Only the design load cases were analyzed to compute the MAWP for determining the test pressure.

Tubesheet MDMT Calculations:

Shell Side MDMT calculation:

Governing thickness on the shell side per figure UCS-66.3 (e):

$$= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{shell thickness}))$$

$$= \max(180/4, \min(180, 74))$$

$$= 74.000 \text{ mm.}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):

$$= \min(\text{ps}/\text{Shellside MAPnc}, 1)$$

$$= \min(191/235, 1)$$

$$= 0.814$$

Thickness Ratio = 0.81, Temperature Reduction per Fig. UCS 66.1 = 10 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	25 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	9 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-2 °C

Channel Side MDMT calculation:

Governing Channel side thickness per figure UCS-66.3 (e):

$$= \max(\text{tubesheet thk}/4, \min(\text{tubesheet thk}, \text{channel thickness}))$$

$$= \max(180/4, \min(180, 56))$$

$$= 56.000 \text{ mm.}$$

MDMT reduction ratio per UCS 66 (b)(1)(b):



= min(pt/Tubeside MAPnc, 1)
= min(147/241, 1)
= 0.611

Thickness Ratio = 0.61, Temperature Reduction per Fig. UCS 66.1 = 22 °C

Min Metal Temp. w/o impact per UCS-66, Curve B	19 °C
Min Metal Temp per UCS-66 and UCS-68(c), PWHT credit	3 °C
Min Metal Temp. at Required thickness (UCS 66.1)	-19 °C

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Noted issues for the current model:

Listed below are the known problem areas for the current design. If one or more of the design flags are turned on, please re-run the analysis. Some of these issues may be resolved when using updated input values.

**** Warning: The net force on one of the saddles putting it in tension!**

Please review all reports carefully!

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